

 Parts, Materials, and Processes Engineering Board Approved Revision (BAR)		
Document: 20098308Rev. V	Mark.A.Romanowski@lmco.com <i>PMP Subject Matter Expert (SME)</i> <u>Not Applicable</u> <i>Revision Author</i> Amy.M.Becker@lmco.com <i>BAR Chair</i>	
Change: PDM CT 0799657		
Type: Part Drawing		
Category: Source Control Drawing (SoCD)		
CAGE: 08GY3	Externally: The Purchase Order/Subcontract cited revision is to be used. Internally: The current released revision on the LM Space intranet is to be used.	

LOW VOLTAGE TRANSFORMER EFD TYPE CORE

1 SCOPE

This document establishes the requirements for EFD style low-voltage custom transformer intended for use in the environments encountered in high reliability spacecraft applications. These transformers are rated to temperature Class S per MIL-PRF-27. A "Low voltage" device is defined as one that is not intended to be tested for partial discharge or Corona, and as such is not to be used at voltages in excess of 350V peak between windings or between a winding to case or core.

2 APPLICABLE DOCUMENTS

The following documents, of the issue in effect on the date of the purchase order, form a part of this specification to the extent specified herein.

2.1 Government Documents

The following government documents, of the latest issue, form a part of this specification to the extent specified herein

A-A-56032	Ink, Marking, Epoxy Based
A-A-59770	Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive
MIL-PRF-27	Transformers and Inductors (Audio, Power, and High-Power Pulse), General Specification for
MIL-STD-1276	Leads for Electronic Component Parts
MIL-STD-981	Design, Manufacturing and Quality Standards for Custom Electromagnetic Devices for Space Applications

2.2 Industry Documents

The following industry documents, of the latest issue, form a part of this specification to the extent specified herein.

Association Connecting Electronics Industries (IPC)

IPC J-STD-006	Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications
ASTM B152	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM D5213	Standard Specification for Polymeric Resin Film for Electrical Insulation and Dielectric Applications

National Electrical Manufacturers Association (NEMA)

NEMA MW-1000	Magnet Wire
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Lockheed Martin Space (LM Space)

20043779	Low Voltage Magnetics Screening and Data Requirements
3GPDE001	GPS Low Voltage Non-Hermetic Magnetics Screening Specification

2.3 Order of Precedence

In the event of a conflict between this document and any referenced documents, this document governs followed by the applicable screening document (20043779 or 3GPDE001). The referenced documents in effect at the time of manufacturing are the latest revision unless otherwise specified.

3 REQUIREMENTS

Unless otherwise specified in this drawing, all documentation, manufacturing and testing requirements shall be per the applicable document 20043779 or 3GPDE001.

3.1 Item Requirements

The devices specified herein shall be as follows with the exceptions and additions specified in the applicable document 20043779 or 3GPDE001.

MIL-PRF-27, Product Level "T", Grade "6", Temperature Class "S"

MIL-STD-981, Families "04", "20", "37" and "41", Product Class "S"

3.2 Build and Screening Requirements

The devices specified herein shall use the low voltage screening requirement document based on the "P number" as stated below:

P1 to P12 – Use drawing 20043779

P503, P512 & P515 – P518 – Use drawing 3GPDE001

3.2.1 Non-Flight Part Requirements and Marking

Electromagnetic devices procured for non-flight purposes shall be manufactured in accordance with para. 3.9 of 20043779/ para. 3.18 of 3GPDE001 and marked in accordance with para. 3.12 of 20043779/para. 3.6 of 3GPDE001, except the customer part number shall be PX00X for the non-flight/unscreened version of the PX device. The designated non-flight part numbers are not included with the manufacturing detail, but are form, fit and functionally identical to the flight version, excluding the testing, part marking, and do not require any applicable source inspection(s).

3.3 Outline Dimensions

Each electromagnetic device shall meet the physical dimensions as shown in the applicable figure(s) for that "P" number.

3.4 Materials

Materials used are specified for each "P" number. Materials shall meet the requirements specified in the applicable document 20043779 or 3GPDE001.

3.4.1 Deleted.

3.5 Ratings

Unless otherwise specified for the applicable "P" number, all ratings are per document 20043779 (Low Voltage Magnetics Screening and Data Requirements).

3.6 Electrical Characteristics

Electrical characteristics are specified for each "P" number.

3.7 Deleted.

3.8 Deleted.

3.9 Deleted.

3.10 Deleted.

3.11 Lead Finish (Final Lead Dressing)

Unless otherwise specified on the Winding Information Sheet or if Litz wire is used in the construction of the device, final lead dressing of flying leads (stripping and tinning) is not required prior to shipment to the Customer.

3.12 Lead/Pin Length

Final lead and pin length shall be per the applicable Winding Information Sheet.

3.13 Deleted.

3.14 Deleted.

3.15 Deleted.

3.16 Deleted.

4 Deleted.

1. Purchased by Standex. Holtsville facility no longer in business. Left for historical purposes.
2. No longer an approved source. Left for historical purposes.
3. This part is identical to 23025745P15 except the screening requirements are per 3GPDE001.

6 NOTES

1. Alternate equivalent input waveforms (i.e., sine wave vs. square wave) may be used for Burn-in and Life Test Circuits with prior approval of purchaser's Components Engineering.
2. Use of pink poly packaging material for shipment of hardware is prohibited.

6.1 Definitions

Customer/Purchaser – Lockheed Martin Space

6.2 Drawing Interpretation

Interpret dimensions and tolerances per ASME Y14.5M-1994. Interpret drawing per ASME Y14.100-2004.

6.3 Dimensions. Unless otherwise specified dimensions are in inches:

- a. Fractions: $\pm 1/16$
- b. Angles: ± 2 degrees
- c. Decimals:
 - .X = $\pm .1$
 - .XX = $\pm .03$
 - .XXX = $\pm .010$

6.4 Miscellaneous

6.4.1 Related Data

For internal users, PMPDB (<https://pmpdb-prod.us.lmco.com/>) and Warehouse (<https://edt.apps.nsrca-ocpp.ssc.lmco.com/>) provide additional resources that may be useful:

- a. Requirements: Procurement, receiving-inspection, manufacturers/distributors, internal shelf-life, shelf-life extension, "special" process, and other requirements.
- b. Recommendations: Design usage and preferences recommendations.
- c. Records: Revision history and justification records.

6.4.2 Changes

Technical changes from the previous revision are marked using change bars in the margin. For internal users, the revision history is available at:

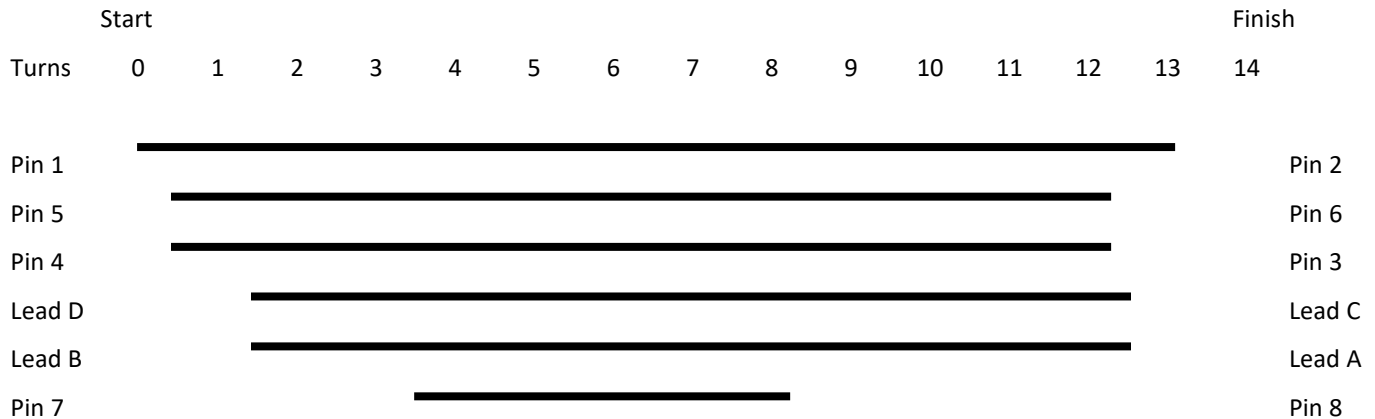
<https://edt.apps.nsrca-ocpp.ssc.lmco.com/showCT/0799657>

P1 Materials List			
Item	Material	Description	Source
1	Wire (3)	#30 AWG MW035CHX000030* or MW016CHX000030*	Elektrisola, Essex, MWS, or REA
2	Wire (3)	#26 AWG MW035CHX000026* or MW016CHX000026*	Elektrisola, Essex, MWS, or REA
3	Bobbin (4)	CSH-EFD15-1S-8P or CSH-EFD15-1S-8P-*Z	Ferroxcube
4	Core	EFD15-F3-A160-S or EFD15/8/5/3F3-A160-S	Ferroxcube
5	Tape, Kapton	P224 or K102 per ASTM D5213	Per approved manufacturer's Baseline
6	Tape	#79 Glass per A-A-59770	Per approved manufacturer's Baseline
7	Core Cement (1), (2)	Stycast 3050 w/CAT 11, Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved manufacturer's Baseline
8	RTV	RTV142 or CV-1142 or DC6-1104	Per approved manufacturer's Baseline
9	Impregnation	Scotchcast #280 or Stycast 3050 w/Cat 11	Per approved manufacturer's Baseline
10	Solder	Sn60Pb40 or Sn63Pb37 per IPC J-STD-006	Per approved manufacturer's Baseline
11	Ink	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved manufacturer's Baseline
12	Shrink Sleeving	FIT-500	Per approved manufacturer's Baseline
NOTES: - Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW. - Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only. * - Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax. - The "*" is a letter for the country of origin. Prior to winding, bobbin pins need to be solder dipped twice in their entirety using item 10.			

NOTES:

WINDING INFORMATION SHEET AND NOTES FOR P1

Winding Designation	Np (1-2)	Ns1 (D-C)	Ns2 (B-A)	Ns3 (5-6)	Ns4 (4-3)	Ns5 (7-8)
No. of Turns	14	12	12	12	12	5
Wire Size AWG single strand	30	26	26	30	30	30



C-D and A-B are 2" flying leads.

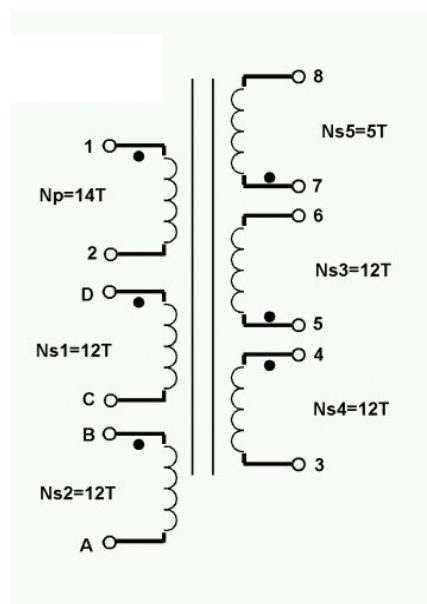
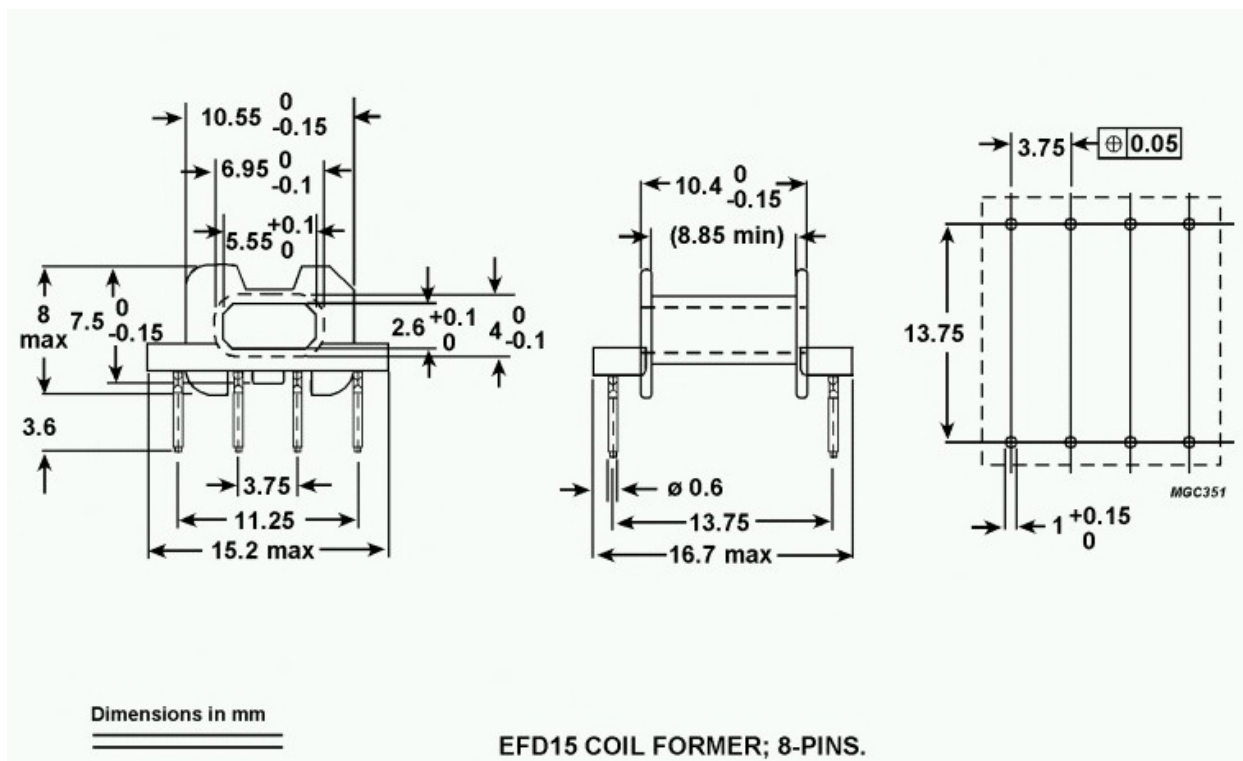
P1 Winding Instructions
Windings will be continuous. Wind as tight as possible. Minimize crossovers.
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 10 (Solder).
Observe polarity during the winding process.
Begin winding Np (1-2) using Item 1 (30 AWG Wire) onto Item 3 (Bobbin). Start on the left side of the bobbin with Pin #1 at the bottom left as viewed from the pin side of the bobbin. Wind 1 turn of Np (1-2).
Using Item 1 (30 AWG Wire), pick up windings Ns3 (5-6) and Ns4 (4-3) together with Np (1-2) in parallel and wind ½ turn. NOTE: You will be on top side of bobbin.
Using Item 2 (26 AWG Wire), pick up windings Ns1 (D-C) and Ns2 (B-A), together with Np (1-2), Ns3 (5-6) and Ns4 (4-3) in parallel and wind 2 ½ turns. NOTE: You will be on bottom (pin) side of bobbin.
Using Item 1 (30 AWG Wire), pick up winding Ns5 (7-8) together with the rest of the windings in parallel and wind 5 turns.
Break out winding Ns5 (7-8) (total of 5 turns) and route leads to appropriate lead locations. Continue to wind the remainder of the windings in parallel for 4 more turns.
Break out windings Ns3 (5-6) and Ns4 (4-3) (total of 12 turns each winding) and route leads to appropriate lead locations. Continue to wind the remainder of the windings in parallel for ½ a turn.
Break out windings Ns1 (D-C) and Ns2 (B-A) (total of 12 turns each winding) and route leads to appropriate lead locations. Continue to wind Np (1-2) for ½ more turn.
Break out winding Np (1-2) (total of 14 turns each winding) and route leads to appropriate lead locations.

P1 Winding Instructions

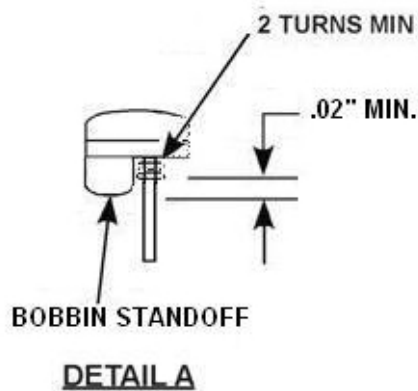
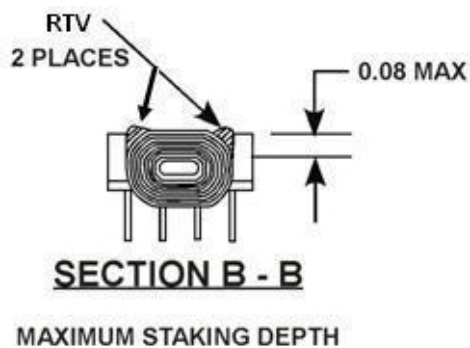
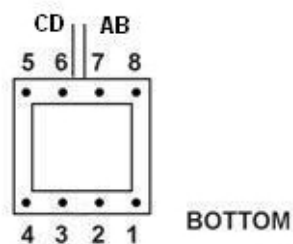
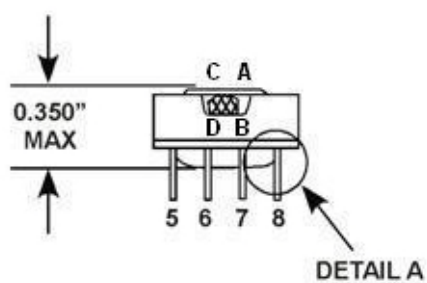
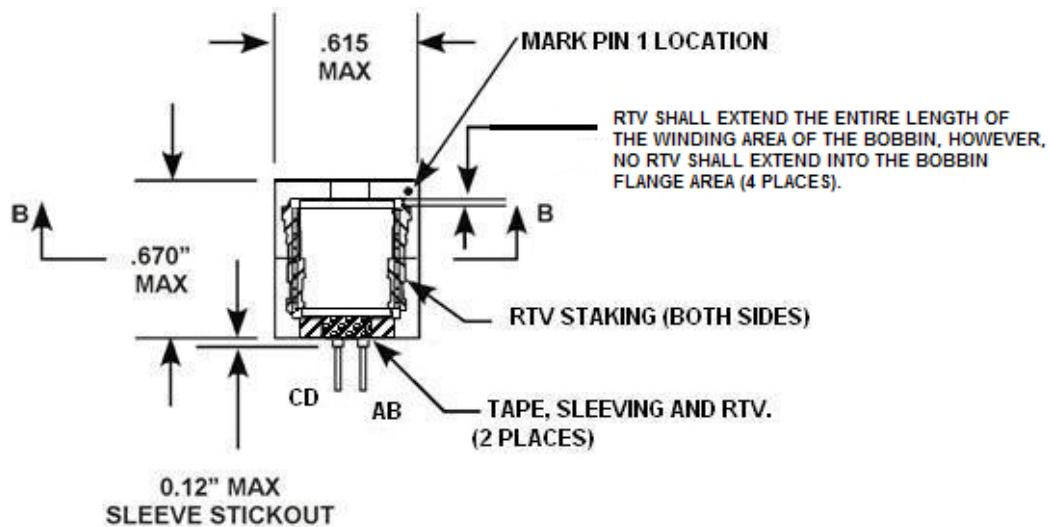
Wrap wound coil with Item 6 (Tape, Glass), one layer minimum, two layers maximum. Use wrap to orient leads A-D as needed to match location shown in P1 Outline drawing.

P1 Assembly Instructions and Notes

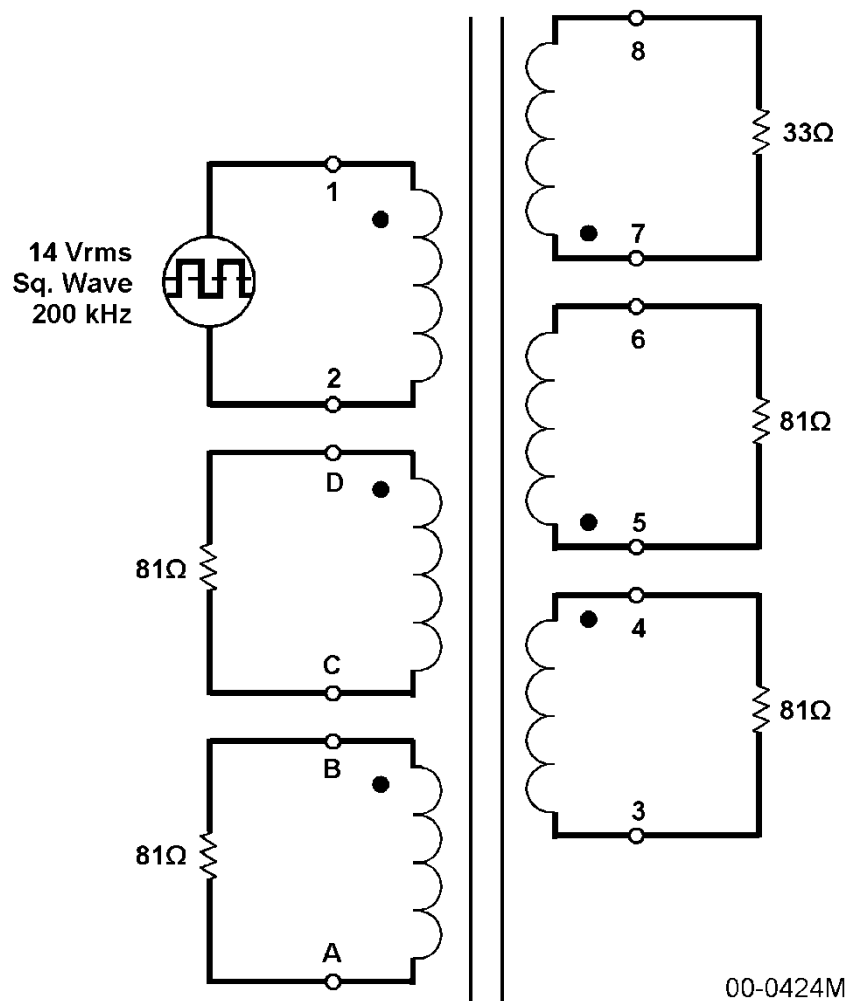
1. All materials shall be per the P1 Materials List.
2. Unless otherwise specified, all dimensions are in inches and include all materials used.
3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to $150^{\circ}\text{C} \pm 5^{\circ}\text{C}$ for 1 hour +10 minutes, -0 minutes.
4. Terminate windings Ns3 (5-6), Ns4 (4-3), Ns5 (7-8) and Np (1-2) to the appropriate pins using Item 10 (Solder). Ensure a gap of 0.020" is maintained between the solder joint and the bobbin's PCB standoffs (See Detail "A" of P1 Outline).
5. Vacuum impregnate wound bobbin assembly using Item 9 (Impregnation). Keep flying leads exiting the bobbin clear of impregnation.
6. Place coil into Item 4 (Core). Cement Item 4 (Core) together using Item 7 (Core Cement). **Do not bond center leg.** Once cement cures insure bobbin is not cemented to core. Ensure flying leads exit core per P1 Outline Figure.
7. Stake core and bobbin using Item 8 (RTV) at the top of the assembly as indicated in the P1 Outline Figure. Maintain a .01" to .05" gap in all corners. Do not allow the staking material to come in contact with inside corner edges.
8. Flying Leads Ns1 (D-C) and Ns2 (B-C) shall be $2" \pm 0.25"$ from core's edge. Strip and solder dip $0.2" \pm 0.025"$ of flying leads using Item 10 (Solder).
9. Apply Item 5 (Tape, Kapton) to the core where the flying leads exit.
10. Install Item 12 (Shrink Sleeving) on the portion of leads A-D that exit over the core. The sleeving is to extend beyond the core area but not exceed 0.12" from the outer core edge. Stake sleeved wire to core using Item 8 (RTV).
11. Mark each device per 20043779 using Item 11 (Marking Ink). Tag each lead within 1/2 inch of core body. Using Item 11 (Marking Ink), identify pin 1 on top of core with a white dot.



P1 Coil Former and Schematic



P1 Outline

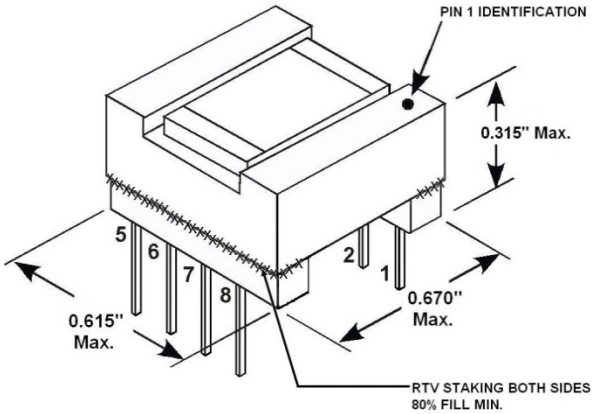


P1 Burn-In/Life Test Circuit

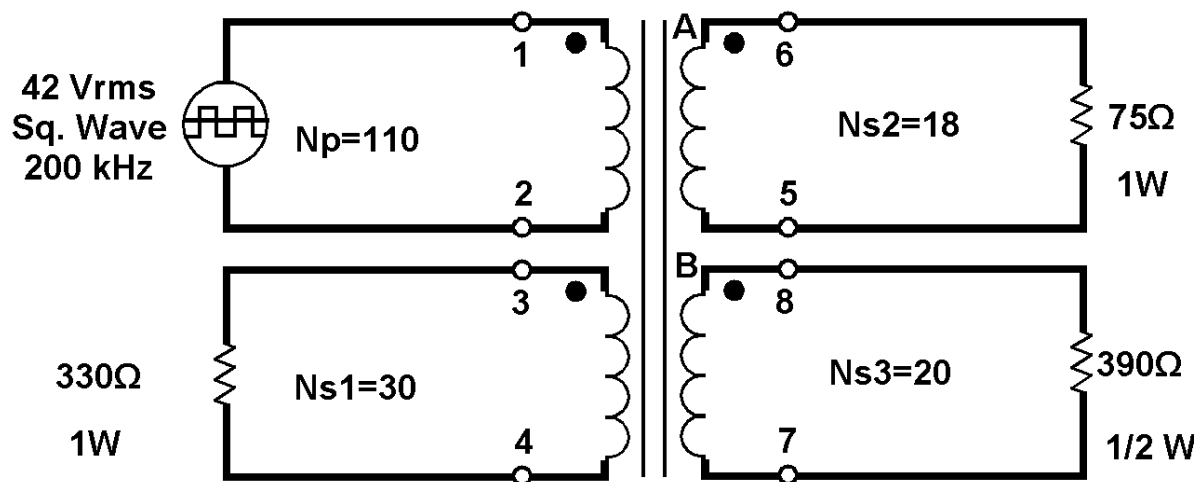
P2 Materials List			
Item	Material	Description	Source
1	Wire (2)	#35 AWG MW035CHX000035* or MW016CHX000035*	Elektrisola, Essex, MWS, or REA
2	Wire (2)	#32 AWG MW035CHX000032* or MW016CHX000032*	Elektrisola, Essex, MWS, or REA
3	Bobbin (3)	CSH-EFD15-1S-8P or CSH-EFD15-1S-8P-*Z	Ferroxcube
4	Core	EFD15-F3-A100-S or EFD15/8/5/3F3-A100-S	Ferroxcube
5	Tape, Kapton	P224 or K102per ASTM D5213	Per approved manufacturer's Baseline
6	Core Cement (1), (4)	Stycast 3050 w/CAT 11, Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved manufacturer's Baseline
7	RTV	RTV142 or CV-1142 or DC6-1104	Per approved manufacturer's Baseline
8	Impregnation	Scotchcast #281	Per approved manufacturer's Baseline
9	Solder	SN60 or SN63, per IPC J-STD-006	Per approved manufacturer's Baseline
10	Ink, Marking	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved manufacturer's Baseline
NOTES: 1. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW. 2. Acceptable magnet wire lubricant designations are "N" – No lubricant and "P" Paraffin Wax. 3. The "*" is a letter for the country of origin. Prior to winding, bobbin pins need to be solder dipped twice in their entirety using item 9. 4. Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.			

P2 Electrical Characteristics				
Winding Designation (start-finish)	Np1 (1-2)	Ns1 (3-4)	Ns2 (6-5)	Ns3 (8-7)
Number of Turns	110	30	18	20
Maximum Operating Current (Arms)	.038	.064	.152	.034
Working Voltage (Peak)	100 V	100 V	100 V	100 V
Dielectric Withstanding Voltage 2/	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) 3/	10,000 M Ω	10,000 M Ω	10,000 M Ω	10,000 M Ω
DC Resistance (Max.) 4/	4.0 Ω	500 m Ω	300 m Ω	350 m Ω
Inductance ($\pm 10\%$) 5/	1.21 mH	90 μ H	32 μ H	40 μ H
Leakage Inductance) 5/	<35 μ H	Shorted	Shorted	Shorted
Rated Voltage 1/	42V	25V	25V	25V
NOTES: 1. Induced voltage is required. 2. 300 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10 ⁻⁶ Torr. Qualifies unit to 100V peak working voltage. 3. 500 Vdc. 4. Corrected to +20°C. 5. Measured at 1 Vac, 10 kHz, 0 Adc bias.				

P2 Winding Instructions and Assembly Notes
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 9 (Solder).
Observe polarity during the winding process.
Using Item 2 (32 AWG Wire), wind Ns1 (3-4), Ns2 (6-5) and Ns3 (8-7) in parallel for 18 turns ($\approx$ 2 layers, 9 turns per layer). Minimize crossovers.
Break out winding Ns2 (6-5) (total of 18 turns) and route leads to appropriate lead locations. Continue winding Ns1 (3-4) and Ns3 (8-7) in parallel for 2 more turns.
Break out winding Ns3 (8-7) (total of 20 turns) and route leads to appropriate lead locations. Continue winding Ns1 (3-4) for 10 more turns.
Break out winding Ns1 (3-4) (total of 30 turns) and route leads to appropriate lead locations.
Wrap secondary windings with a single layer of Item 5 (Tape, Kapton).
Using Item 1 (35 AWG Wire), continuous wind Np (1-2) for 110 turns ($\approx$ 2 layers, 55 turns per layer). Minimize crossovers.
Break out winding Np (1-2) and route leads to appropriate lead locations.
Wrap wound coil with Item 5 (Tape, Kapton), 1 $\frac{1}{2}$ layers minimum, 2 layers maximum.
NOTES: 1. All materials shall be per the P2 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C $\pm$ 5°C for 1 hour +10 minutes, -0 minutes. 4. Terminate windings Np (1-2), Ns1 (3-4), Ns2 (6-5) and Ns3 (8-7) to the appropriate pins using item 9 (Solder). 5. Vacuum impregnate wound bobbin assembly using Item 8 (Impregnation). 6. Place coil into Item 4 (Core). Cement Item 4 (Core) together using Item 6 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 20043779 using Item 10 (Marking Ink). Using Item 10 (Marking Ink), identify pin 1 on top of core with a white dot.



Outline Dimensions and Schematic for P2



00-0459M

P2 Burn-In/Life Test Circuit

P3 Materials List			
Item	Material	Description	Suggested Source
1	Ultem 2300 (1)	Bobbin, Made to Bobbin Detail Figure	Per approved Manufacturer's Baseline
2	Wire (4)	#36 AWG MW035CHX000036* or MW016CHX000036*	Elektrisola, Essex, MWS, or REA
3	Core	EFD25-3F4	Phillips/Ferroxcube
4	Core Cement (2), (5)	Stycast 3050/Cat. 11, Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved Manufacturer's Baseline
5	Tape, Kapton	P224 or K102 per ASTM D5213	Per approved Manufacturer's Baseline
6	Impregnation	Scotchcast 280, clear	Per approved Manufacturer's Baseline
7	RTV	RTV142 or CV-1142 or DC6-1104	Per approved Manufacturer's Baseline
8	Solder	SN60 or SN63 per IPC J-STD-006	Per approved Manufacturer's Baseline
9	Terminal Pins	#26 AWG Nickel wire per MIL-STD-1276 (LW0.015N-34-N or LW0.015N-32-N)	Per approved Manufacturer's Baseline
10	Terminal Cement (3)	2280875-6 (Epon 828/Versamid-140/Cab-o-sil) or or Epon 828/Epikure 3140/Cab-o-sil	Per approved Manufacturer's Baseline
11	Ink	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline

NOTES:

1. Heat treat per requirements specified with bobbin detail.
2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40.
3. Mix ratio for Epon 828/V140/Cab-o-sil and Epon 828/Epikure 3140/Cab-o-sil is 60:40:4.
4. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax.
5. Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.

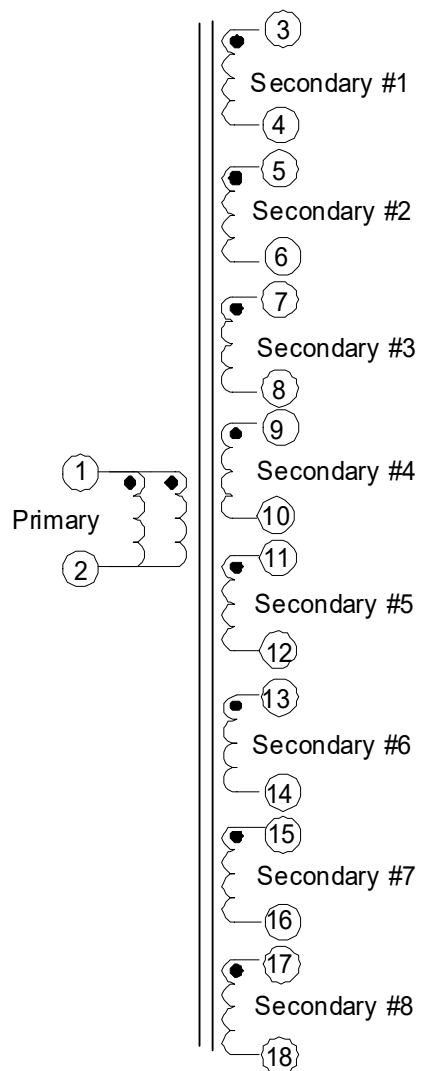
P3 Electrical Characteristics										
Winding #	Primary		Sec 1	Sec 2	Sec 3	Sec 4	Sec 5	Sec 6	Sec 7	Sec 8
Terminals (start-finish)	1-2		3-4	5-6	7-8	9-10	11-12	13-14	15-16	17-18
Winding Layer #	1	2	3	4	5	6	7	8	9	10
Number of turns	95	95	95	95	95	95	95	95	95	95
Wire Gauge	36	36	36	36	36	36	36	36	36	36
Maximum operating current (A_{RMS})	<1mA		<1mA	<1mA	<1mA	<1mA	<1mA	<1mA	<1mA	<1mA
Working Voltage (Peak)	100V		100V	100V	100V	100V	100V	100V	100V	100V
Dielectric withstanding voltage <u>1/</u>	No Breakdown									
Insulation Resistance <u>2/</u>	10, 000M Ω									
DC resistance (Ω , $\pm 10\%$) <u>3/</u> , <u>7/</u>	2.62		5.50	5.60	5.80	6.00	6.15	6.35	6.50	6.65
Inductance (mH, +66% / -0%) <u>4/</u> , <u>7/</u>	6.8		6.8	6.8	6.8	6.8	6.8	6.8	6.8	6.8
Leakage Inductance (Max.) (μ H) <u>4/</u> , <u>5/</u>	N/A		11.4	19.2	27.6	34.5	39.6	47.3	55	62.7
Rated Voltage <u>6/</u>	40V		40V	40V	40V	40V	40V	40V	40V	40V

NOTES:

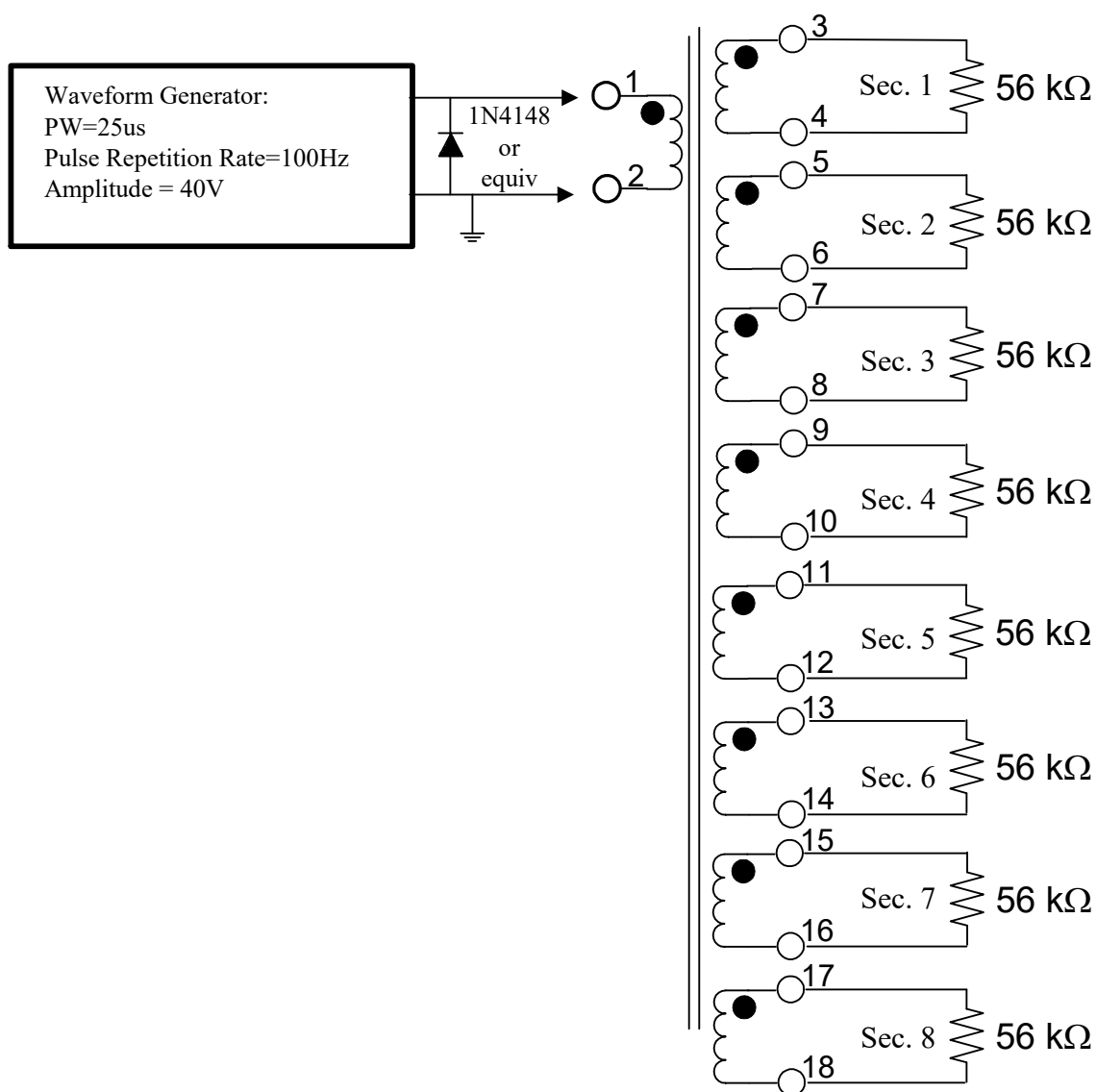
- 300 Vrms, 60 Hz at any pressure from one atmosphere to less than 5×10^{-6} Torr., Qualifies unit to 100V peak working voltage.
- 500 Vdc.
- Corrected to 20°C.
- Measure at 10 kHz, 0 Adc bias.
- Measure at 10 kHz, 0 Adc bias. Measure at winding 1-2, with only the winding of interest shorted**For example, for leakage between Pri and Secondary 1, short pin 3 to pin 4 and record (in column under winding "Sec 1") the inductance measured between pins 1 and 2.
- Induced voltage is required.
- Delta DCR and Delta L shall be $\pm 3\%$.

FABRICATION SEQUENCE FOR P3		
Layer	Turns	P3 Winding Instructions
N/A	-	Prior to installing Item 9 (Terminals), solder dip the entire terminal twice using Item 8 (Solder).
N/A	-	Install Item 9 (Terminals) into Item 1 (Bobbin) using Item 10 (Terminal Cement). Ensure bobbin terminal cut out is completely filled with cement.
N/A	-	Excluding layers 1 and 2, no "winding" shall touch another winding at any point. Stagger Kapton tape insulation start and finish points to avoid winding build up.
N/A	-	Observe polarity during the winding process.
1	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 1. Minimize crossovers. Route leads to appropriate lead locations.
1.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
2	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 2. Minimize crossovers. Route leads to appropriate lead locations.
2.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
3	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 1 (3-4) layer 3. Minimize crossovers. Route leads to appropriate lead locations.
3.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
4	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 2 (5-6) layer 4. Minimize crossovers. Route leads to appropriate lead locations.
4.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
5	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 3 (7-8) layer 5. Minimize crossovers. Route leads to appropriate lead locations.
5.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
6	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 4 (9-10) layer 6. Minimize crossovers. Route leads to appropriate lead locations.
6.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
7	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 5 (11-12) layer 7. Minimize crossovers. Route leads to appropriate lead locations.
7.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
8	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 6 (13-14) layer 8. Minimize crossovers. Route leads to appropriate lead locations.
8.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
9	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 7 (15-16) layer 9. Minimize crossovers. Route leads to appropriate lead locations.

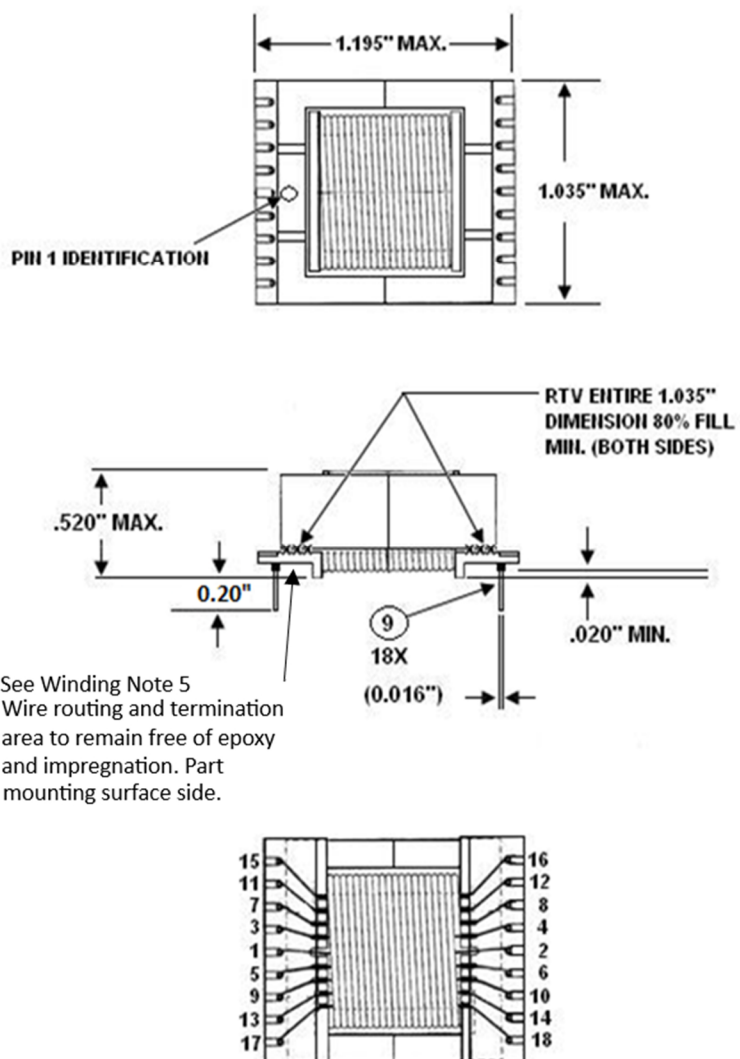
FABRICATION SEQUENCE FOR P3		
Layer	Turns	P3 Winding Instructions
9.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
10	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 8 (17-18) layer 10. Minimize crossovers. Route leads to appropriate lead locations.
10..5	1½ - 2	Final wrap winding with 1 ½ layers minimum, 2 layers maximum of Item 5 (Tape, Kapton).
P3 Assembly Instructions and Notes 1. All materials shall be per the P3 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 4. Wire routing to contain strain relief once clear of bobbin wire routing passage (Do not go straight from bobbin lead passage to pin). Pin termination to contain one insulated turn first wrap closest to bobbin and two uninsulated turns above. Terminate windings to the appropriate pins using item 8 (Solder). Ensure a gap of 0.020" is maintained between the solder joint and the bobbin's PCB standoffs. 5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation). Keep the area from the wire routing passage to the bobbin pins clear of impregnation (See Outline Figure). 6. Place coil into Item 3 (Core). Cement Item 3 (Core) together using Item 4 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 20043779 using Item 11 (Marking Ink). Using Item 11 (Marking Ink), identify pin 1 on top of core with a white dot. 9. ULTEM BOBBIN IS SOLUTION SENSITIVE. DO NOT CLEAN BOBBIN WITH ACETONE.		



P3 Schematic

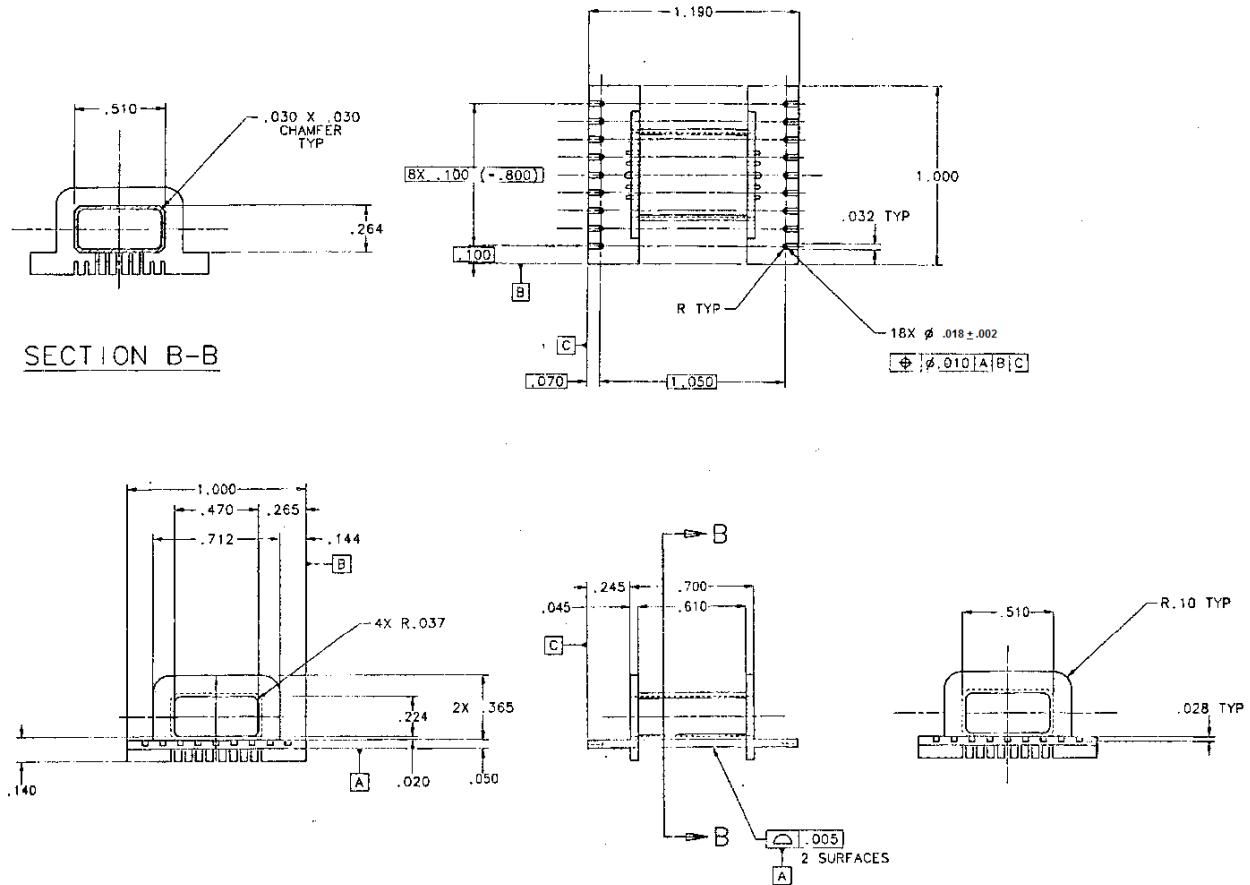


P3 Burn-in/Life Test Circuit



TOLERANCE (UNLESS OTHERWISE SPECIFIED)
 3 DECIMAL PLACES = $\pm .005"$
 2 DECIMAL PLACES = $\pm .010"$

P3 Outline



Bobbin Annealing Requirements

After final machining,

Bobbin shall be annealed to 300°F at a maximum heating rate of 20°F per hour.

Hold oven temperature at 300°F for 1 ½ hours.

Increase oven temperature to 400°F at a maximum heating rate of 20°F per hour.

Hold oven temperature at 400°F for 2 ½ hours.

Oven cool to 150°F over 25 hours with a maximum cool rate of 10°F per hour.

Turn off oven and allow to cool to room temperature with oven door closed before removing parts.

Tolerance

(Unless Otherwise Specified)

2 Decimal Places = $\pm .010$ "

P3 Bobbin Detail

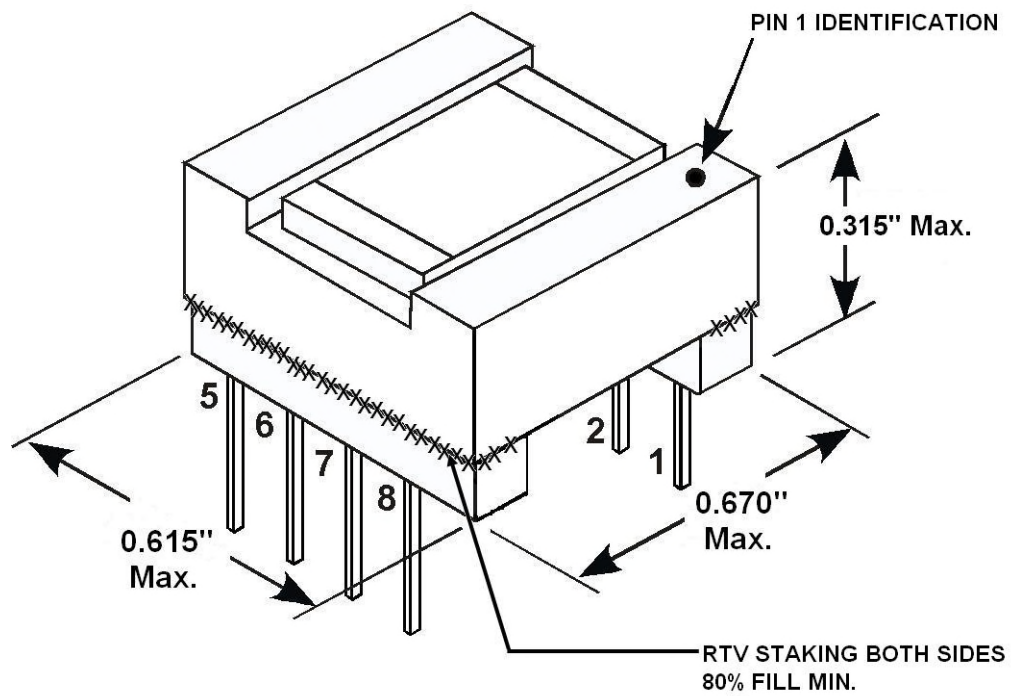
P4 Materials List			
Item	Material	Description	Source
1	Core, Gapped	EFD15-3F3-A160	Ferroxcube
2	Bobbin (4)	CSH-EFD15-1S-8P or CSH-EFD15-1S-8P-*Z	Ferroxcube
3	Wire (3)	#28 AWG MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Tape, Glass (1)	79 Glass per A-A-59770	Per approved manufacturer's Baseline
5	Core Cement (2) , (5)	Stycast 3050/Cat. 11, Epon 828/V140 or Epon 828/Epikure 3140	Per approved manufacturer's Baseline
6	Impregnation	Scotchcast 280	Per approved manufacturer's Baseline
7	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved manufacturer's Baseline
8	Solder	SN60 or SN63 per IPC J-STD-006	Per approved manufacturer's Baseline
9	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline

NOTE:

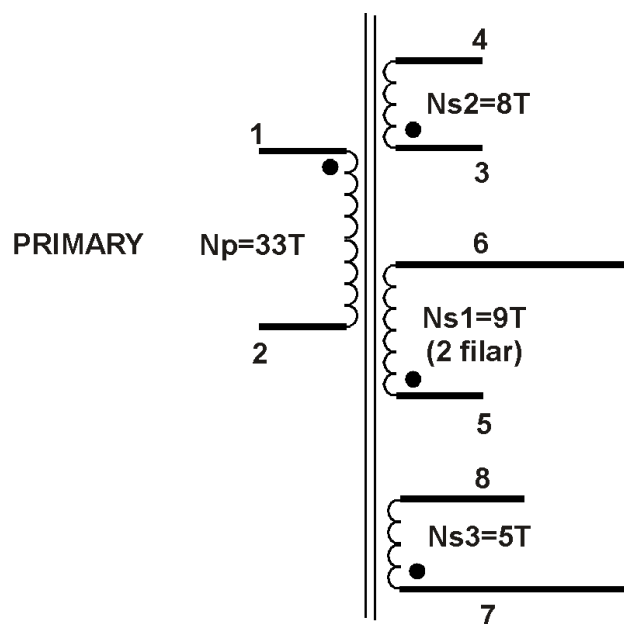
1. P213 removed. No longer available.
2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW.
3. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax.
4. The "*" is a letter for the country of origin. Prior to winding, bobbin pins need to be solder dipped twice in their entirety using item 8.
5. Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.

P4 Electrical Characteristics				
Characteristics	Limits	Limits	Limits	Limits
Winding Designation (Start-Finish)	Np (1-2)	Ns1 (5-6)	Ns2 (3-4)	Ns3 (7-8)
Number of Turns	33	9	8	5
Maximum Operating Current (Arms)	0.20	0.54	0.17	0.08
Working Voltage (Peak)	175 V	175 V	175 V	175V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	200 mΩ	30 mΩ	60 mΩ	40 mΩ
Inductance ($\pm 10\%$) <u>4/</u>	174 μH	12.9 μH	10.2 μH	4.0 μH
Leakage Inductance (Max) <u>5/</u>	1.5 μH	Shorted	Shorted	Shorted
	Shorted	150 nH	Shorted	Shorted
	Shorted	Shorted	150 nH	Shorted
	Shorted	Shorted	Shorted	150 nH
Cpri-sec <u>6/</u>	--	80.0 pF max.	80.0 pF max.	80.0 pF max.
Turns Ratio <u>7/</u>	Reference	0.273 $\pm 1.5\%$	0.242 $\pm 1.5\%$	0.152 $\pm 1.5\%$
Rated Voltage <u>8/</u>	80 V	15 V	15 V	8V
NOTES: 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10⁻⁶ Torr., Qualifies unit to 175V peak working voltage. 500 Vdc. Corrected to +20°C. - Measured at 1 Vac, 300kHz 0 Adc bias. - Measured at 300kHz. "Shorted" means all other windings shorted start-to-finish and shorted together. - At 300kHz. Capacitance is measured from primary winding to each secondary winding (with pins of each pair of windings shorted) all other windings open-circuited. - Turns ratio data may be taken in process prior to cementing cores. - Induced Voltage Required.				

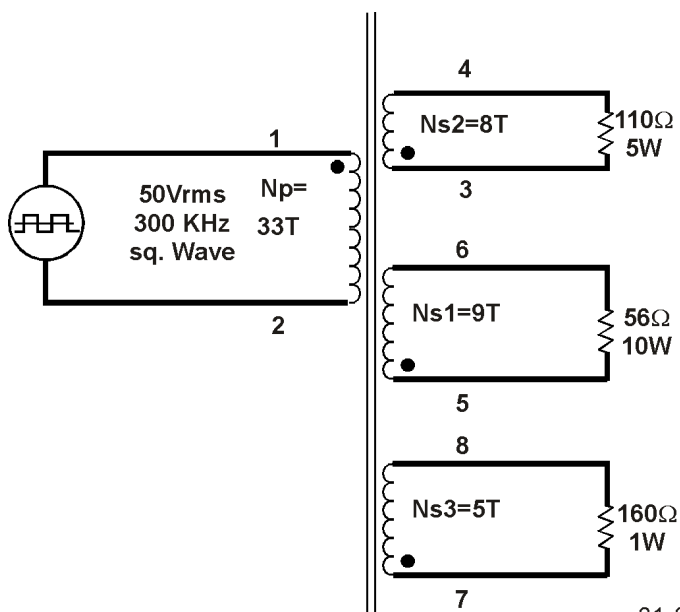
P4 Winding Instructions and Assembly Notes
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 8 (Solder).
Observe polarity during the winding process.
Using Item 3 (28 AWG Wire), wind Np (1-2) for 33 turns (2 layers). No crossovers. Route leads to appropriate lead locations.
Using Item 3 (28 AWG Wire), wind Ns1 (5-6) bi-filar for 9 turns (1 layer). No crossovers. Route leads to appropriate lead locations.
Using Item 3 (28 AWG Wire), wind Ns2 (3-4) for 8 turns (1 layer). No crossovers. Route leads to appropriate lead locations.
Using Item 3 (28 AWG Wire), wind Ns3 (7-8) for 5 turns (1 layer). No crossovers. Route leads to appropriate lead locations.
Wrap wound coil with Item 4 (Tape, Glass), 1 layer minimum, 1½ layers maximum.
NOTES: 1. All materials shall be per the P4 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 4. Terminate windings Np (1-2), Ns1 (5-6), Ns2 (3-4) and Ns3 (7-8) to the appropriate pins using item 8 (Solder). 5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation). 6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 5 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 20043779 using Item 9 (Marking Ink). Using Item 9 (Marking Ink), identify pin 1 on top of core with a white dot.



P4 Outline



01-0193pc



01-0196pc

P4 Schematic and Burn-in/Life Test Circuit

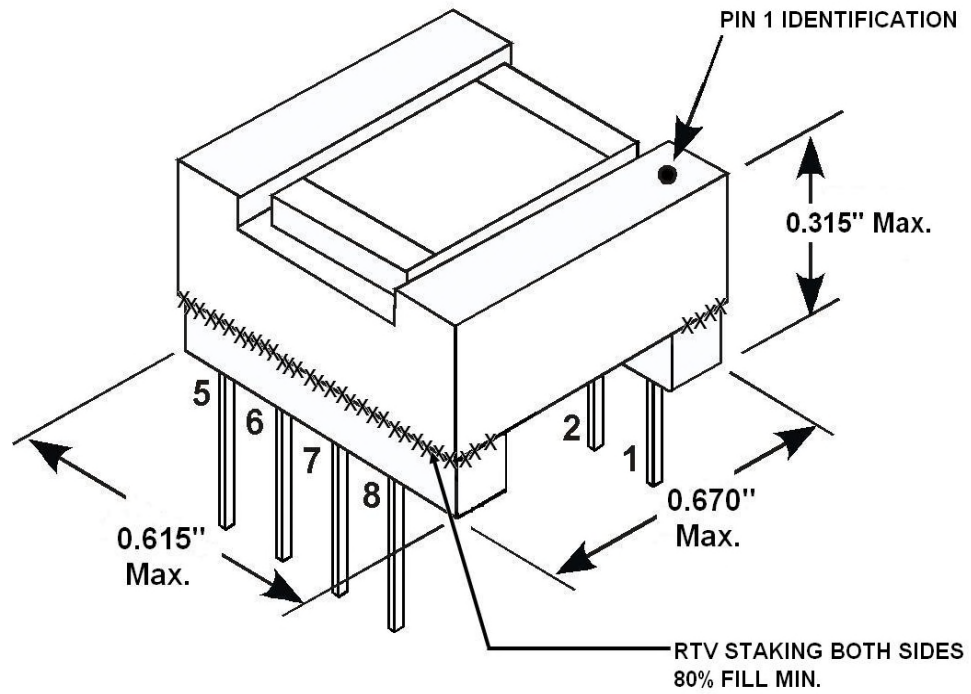
P5 Materials List			
Item	Material	Description	Source
1	Core, Gapped	EFD15-3F3-A160	Ferroxcube
2	Bobbin (4)	CSH-EFD15-1S-8P or CSH-EFD15-1S-8P-*Z	Ferroxcube
3	Wire (3)	#28 AWG MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Tape, Glass (1)	79 Glass per A-A-59770	Per approved manufacturer's Baseline
5	Core Cement (2), (5)	Stycast 3050/Cat. 11, Epon 828/V140 or Epon 828/Epikure 3140	Per approved manufacturer's Baseline
6	Impregnation	Scotchcast 280	Per approved manufacturer's Baseline
7	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved manufacturer's Baseline
8	Solder	SN60 or SN63 perIPC J-STD-006	Per approved manufacturer's Baseline
9	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline

NOTE:

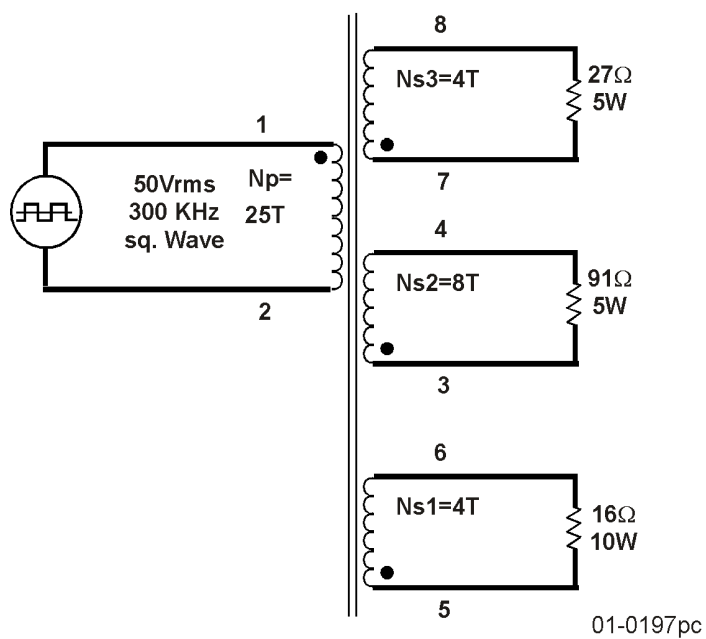
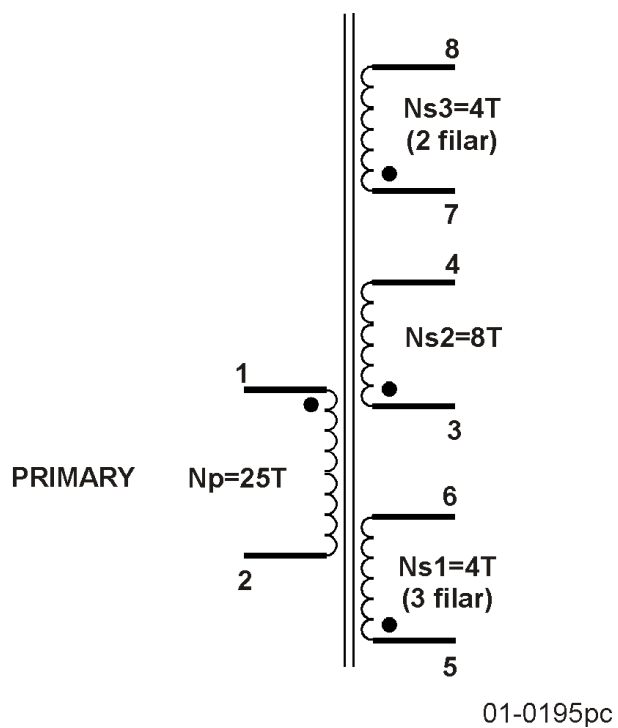
1. P213 removed. No longer available.
2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW.
3. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax.
4. The "*" is a letter for the country of origin. Prior to winding, bobbin pins need to be solder dipped twice in their entirety using item 8.
5. Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.

P5 Electrical Characteristics				
Characteristics	Limits	Limits	Limits	Limits
Winding Designation (Start-Finish)	Np (1-2)	Ns1 (5-6)	Ns2 (3-4)	Ns3 (7-8)
Number of Turns	25	4	8	4
Maximum Operating Current (Arms)	0.30	0.90	0.34	0.50
Working Voltage (IPeak)	175 V	175 V	175 V	175V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	180 mΩ	10 mΩ	50 mΩ	30 mΩ
Inductance ($\pm 10\%$) <u>4/</u>	100 μH	2.6 μH	10.2 μH	2.6 μH
Leakage Inductance (Max) <u>5/</u>	1.5 μH	Shorted	Shorted	Shorted
	Shorted	200 nH	Shorted	Shorted
	Shorted	Shorted	250 nH	Shorted
	Shorted	Shorted	Shorted	200 nH
Cpri-sec (Max.) <u>6/</u>	--	150.0 pF	180.0 pF	150.0 pF
Turns Ratio <u>7/</u>	Reference	0.160 $\pm$ 2.0%	0.320 $\pm$ 2.0%	0.160 $\pm$ 2.0%
Rated Voltage <u>8/</u>	80 V	9 V	17 V	9 V
NOTES: 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10⁻⁶ Torr., Qualifies unit to 175V peak working voltage. 500 Vdc. - Corrected to +20°C. - Measured at 1 Vac, 300kHz 0 Adc bias. - Measured at 300kHz. "Shorted" means all other windings shorted start-to-finish and shorted together. - At 300kHz. Capacitance is measured from primary winding to each secondary winding (with pins of each pair of windings shorted) all other windings open-circuited. - Turns ratio data may be taken in process prior to cementing cores. - Induced Voltage Required.				

P5 Winding Instructions and Assembly Notes	
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 8 (Solder).	
Observe polarity during the winding process.	
Using Item 3 (28 AWG Wire), wind Np (1-2) for 25 turns (1 layer). No crossovers. Route leads to appropriate lead locations.	
Using Item 3 (28 AWG Wire), wind Ns2 (3-4) for 8 turns (1 layer). No crossovers. Route leads to appropriate lead locations.	
Using Item 3 (28 AWG Wire), wind Ns1 (5-6) tri-filar for 4 turns (1 layer). No crossovers. Route leads to appropriate lead locations.	
Using Item 3 (28 AWG Wire), wind Ns3 (7-8) bi-filar for 4 turns (1 layer). No crossovers. Route leads to appropriate lead locations.	
Wrap wound coil with Item 4 (Tape, Glass), 1 layer minimum, 1½ layers maximum.	
NOTES: 1. All materials shall be per the P5 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 4. Terminate windings Np (1-2), Ns1 (5-6), Ns2 (3-4) and Ns3 (7-8) to the appropriate pins using item 8 (Solder). 5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation). 6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 5 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 20043779 using Item 9 (Marking Ink). Using Item 9 (Marking Ink), identify pin 1 on top of core with a white dot.	



P5 Outline



P5 Schematic and Burn-in/Life Test Circuit

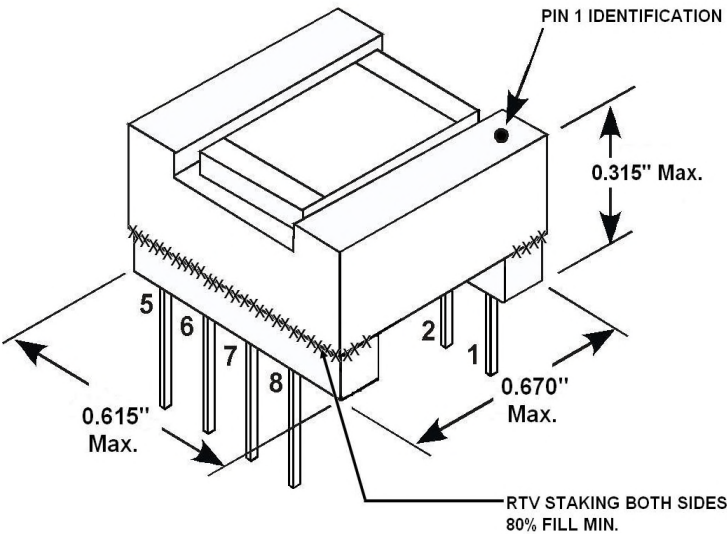
P5 Materials List			
Item	Material	Description	Source
1	Core, Gapped	EFD15-3F3-A160	Ferroxcube
2	Bobbin (4)	CSH-EFD15-1S-8P or CSH-EFD15-1S-8P-*Z	Ferroxcube
3	Wire (3)	#28 AWG MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Tape, Glass (1)	79 Glass per A-A-59770	Per approved manufacturer's Baseline
5	Core Cement (2), (5)	Stycast 3050/Cat. 11, Epon 828/V140 or Epon 828/Epikure 3140	Per approved manufacturer's Baseline
6	Impregnation	Scotchcast 280	Per approved manufacturer's Baseline
7	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved manufacturer's Baseline
8	Solder	SN60 or SN63 perIPC J-STD-006	Per approved manufacturer's Baseline
9	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline

NOTE:

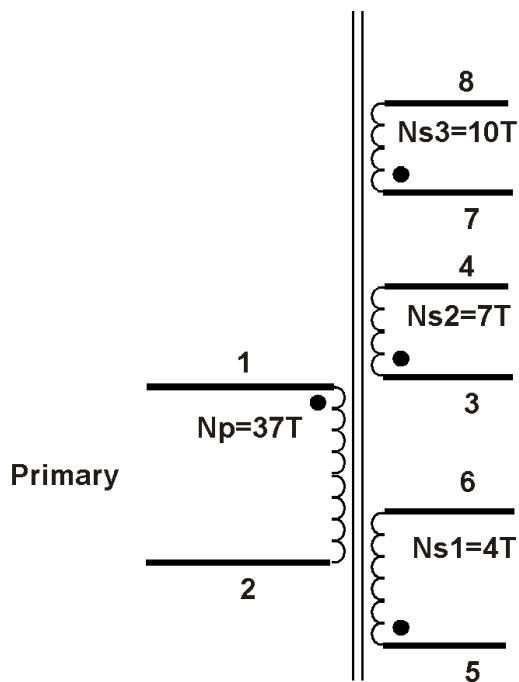
1. P213 removed. No longer available.
2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW.
3. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax.
4. The "*" is a letter for the country of origin. Prior to winding, bobbin pins need to be solder dipped twice in their entirety using item 8.
5. Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.

P6 Electrical Characteristics				
Characteristics	Limits	Limits	Limits	Limits
Winding Designation (Start-Finish)	Np (1-2)	Ns1 (5-6)	Ns2 (3-4)	Ns3 (7-8)
Number of Turns	37	4	7	10
Maximum Operating Current (Arms)	0.150	0.80	0.30	0.10
Working Voltage (I _{Peak})	175 V	175 V	175 V	175V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	270 mΩ	15 mΩ	60 mΩ	80 mΩ
Inductance ($\pm 10\%$) <u>4/</u>	220 μH	2.6 μH	7.9 μH	16.1 μH
Leakage Inductance (Max) <u>5/</u>	3.0 μH	Shorted	Shorted	Shorted
	Shorted	100 nH	Shorted	Shorted
	Shorted	Shorted	150 nH	Shorted
	Shorted	Shorted	Shorted	275 nH
Cpri-sec (Max.) <u>6/</u>	--	80.0 pF	80.0 pF	80.0 pF
Turns Ratio <u>7/</u>	Reference	0.108 $\pm 2.0\%$	0.189 $\pm 2.0\%$	0.270 $\pm 2.0\%$
Rated Voltage <u>8/</u>	80 V	7 V	10 V	15 V
NOTES: 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10⁻⁶ Torr., Qualifies unit to 175V peak working voltage. 500 Vdc. - Corrected to +20°C. - Measured at 1 Vac, 300kHz 0 Adc bias. - Measured at 300kHz. "Shorted" means all other windings shorted start-to-finish and shorted together. - At 300kHz. Capacitance is measured from primary winding to each secondary winding (with pins of each pair of windings shorted) all other windings open-circuited. - Turns ratio data may be taken in process prior to cementing cores. - Induced Voltage Required.				

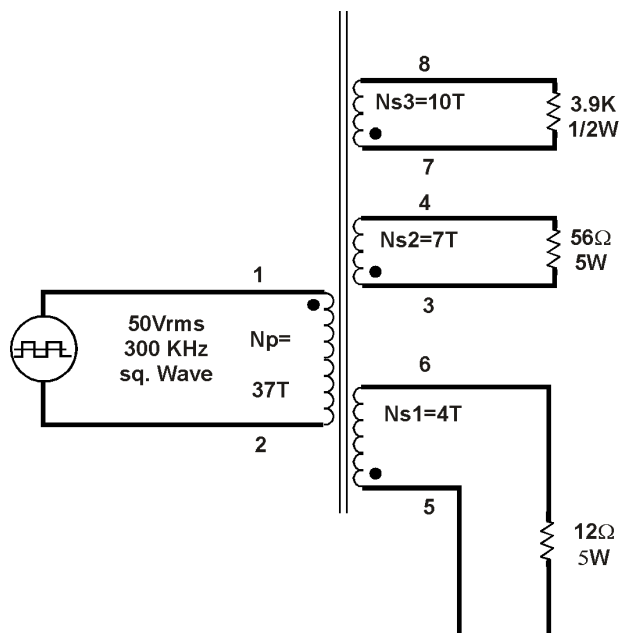
P6 Winding Instructions and Assembly Notes
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 8 (Solder).
Observe polarity during the winding process.
Using Item 3 (28 AWG Wire), wind Np (1-2) for 37 turns (2 layers). No crossovers. Route leads to appropriate lead locations.
Using Item 3 (28 AWG Wire), wind Ns3 (7-8) for 10 turns (1 layer). No crossovers. Route leads to appropriate lead locations.
Using Item 3 (28 AWG Wire), wind Ns2 (3-4) for 7 turns (1 layer). No crossovers. Route leads to appropriate lead locations.
Using Item 3 (28 AWG Wire), wind Ns1 (5-6) tri-filar for 4 turns (1 layer). No crossovers. Route leads to appropriate lead locations.
Wrap wound coil with Item 4 (Tape, Glass), 1 layer minimum, 1½ layers maximum.
NOTES: 1. All materials shall be per the P6 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 4. Terminate windings Np (1-2), Ns1 (5-6), Ns2 (3-4) and Ns3 (7-8) to the appropriate pins using item 8 (Solder). 5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation). 6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 5 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 20043779 using Item 9 (Marking Ink). Using Item 9 (Marking Ink), identify pin 1 on top of core with a white dot.



P6 Outline



01-0199pc



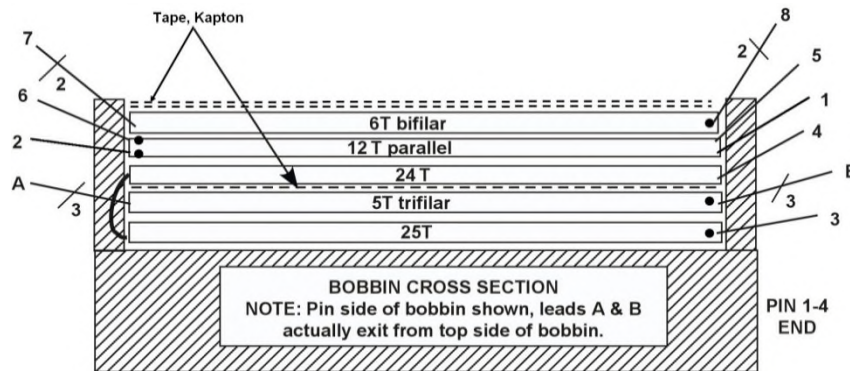
01-0200pc

P6 Schematic and Burn-in/Life Test Circuit

P7 Materials List			
Item	Material	Description	Source
1	Core, Gapped (1)	EFD15-3F3-A90-S, EFD15/8/5/3F3/AL=90	Ferroxcube
2	Bobbin (5)	CSH-EFD15-1S-8P or CSH-EFD15-1S-8P-*Z	Ferroxcube
3	Wire (4)	#29 AWG MW035CHX000029* or MW016CHX000029*	Elektrisola, Essex, MWS, or REA
4	Wire (4)	#25 AWG MW035CHX000025* or MW016CHX000025*	Essex, MWS, or REA
5	Tape, Kapton	P224 or K102per ASTM D5213	Per approved manufacturer's Baseline
6	Core Cement (2), (6)	Stycast 3050/Cat. 11, Epon 828/V140 or Epon 828/Epikure 3140	Per approved manufacturer's Baseline
7	Impregnation	Scotchcast 280 or Stycast 3050/Cat. 11	Per approved manufacturer's Baseline
8	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved manufacturer's Baseline
9	Solder	SN60 or SN63 per IPC J-STD-006	Per approved manufacturer's Baseline
10	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline
11	Shrink Sleeving	FIT-500	Per approved manufacturer's Baseline
12	Epoxy (3)	Ablebond 77-2LTC or Scotchweld 2216B/A	Per approved manufacturer's Baseline
NOTE: - Suppliers may procure a core with a higher AL and gap center leg to meet inductance. - Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW. - No longer used. Left for history purposes only. Use Item 8 (RTV) to stake shrink sleeving in its place. - Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax. - The "*" is a letter for the country of origin. Prior to winding, bobbin pins need to be solder dipped twice in their entirety using item 8. - Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.			

P7 Electrical Characteristics					
Characteristics					
Winding Designation (Start-Finish)	Np (3-4)	Ns1 (B-A)	Ns2 (2-1)	Ns3 (6-5)	Ns4 (8-7)
Number of Turns	49	5	12	12	6
Maximum Operating Current (Arms)	0.160	1.28	0.060	0.020	0.050
Working Voltage (Peak)	100 V	100 V	100 V	100 V	100V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	400 mΩ	10 mΩ	120 mΩ	120 mΩ	75 mΩ
Inductance (± 10%) <u>4/</u>	216 μH	2.3 μH	13 μH	13 μH	3.2 μH
Leakage Inductance (Max) <u>5/</u>	3 nH	Shorted	Shorted	Shorted	Shorted
Rated Voltage <u>6/</u>	72 V	8 V	14 V	14 V	8 V
NOTES: 1. 300 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10 ⁻⁶ Torr., Qualifies unit to 100V peak working voltage. 2. 500 Vdc. Apply voltage for 60 seconds minimum duration across each winding and ground with all other windings connected to ground. 3. Corrected to +20°C. Self leads include a lead length of 2.0 ±0.25". 4. Measured at 250kHz 0 Adc bias. 5. Measured at 250kHz. "Shorted" means all other windings shorted start-to-finish and shorted together. 6. Induced Voltage Required.					

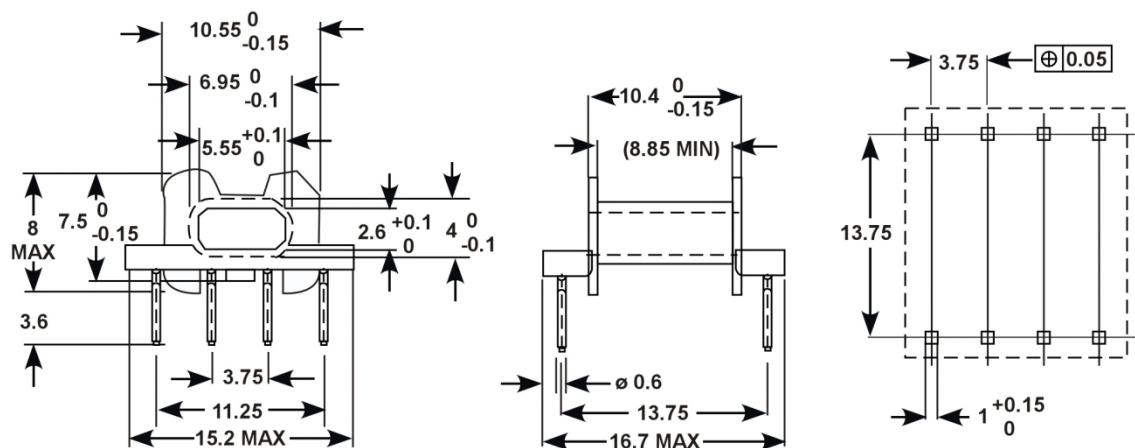
P7 Winding Table, Instructions and Figure					
Winding Designation	Np (3-4)	Ns1 (B-A)	Ns2 (2-1)	Ns3 (6-5)	Ns4 (8-7)
No. of Turns	49	5	12	12	6
Wire Size AWG	29, Single	25, Trifilar	29, Single	29, Single	29, Bifilar



Refer to winding figure for clarification.
Spread all windings across winding width.
All lead routing shall be performed after final winding layer.
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 9 (Solder).
Observe polarity during the winding process.
Begin winding Np (3-4) using Item 3 (29 AWG Wire) onto Item 2 (Bobbin). Start on Pin 1-4 side of the bobbin and wind 25 turns counter-clockwise. Pull out leaving enough wire to complete the remaining turns of Np winding.
Using Item 4 (25 AWG Wire) tri-filar and starting on Pin 1-4 side of the bobbin, wind 5 turns clockwise of winding Ns1 (B-A). Leads B-A are flying leads which exit through the top bobbin notches (opposite pin side), B on the pin 1-4 side, A on the pin 5-8 side.
Install one layer of Item 5 (Tape, Kapton) to provide a flat surface for additional layers.
Finish winding Np (3-4) with 24 turns counter-clockwise (49 turns total), finishing on the pin 1-4 side of the bobbin.
Using Item 3 (29 AWG Wire), wind 12 turns of Ns2 (2-1) and Ns3 (6-5) in parallel clockwise over the previous winding, starting on the pin 5-8 side of the bobbin.
Using Item 3 (29 AWG Wire) bi-filar and starting on Pin 1-4 side of the bobbin, wind 6 turns clockwise of winding Ns4 (8-7).
Wrap wound coil with one layer of Item 5 (Tape, Kapton).
Route leads to appropriate pin and fly lead locations. Use corners to crossover leads on the incorrect side to minimize build up on bottom of wound bobbin.
Wrap wound coil with Item 5 (Tape, Kapton).one layer minimum, two layers maximum. Use wrapping to orient leads B-A as needed to match the location shown in Outline Dimensions for P7.

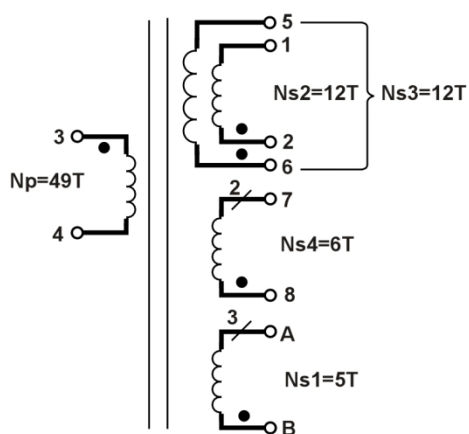
P7 Assembly Instructions and Notes

1. All materials shall be per the P7 Materials List.
2. Unless otherwise specified, all dimensions are in inches and include all materials used.
3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C $\pm$ 5°C for 1 hour +10 minutes, -0 minutes
4. Terminate windings Ns2 (2-1), Ns3 (6-5), Ns4 (8-7) and Np (3-4) to the appropriate pins using item 9 (Solder). A maximum termination height from pin end (bobbin side) is 0.09" max. (See Detail "A" of P7 Outline).
5. Vacuum impregnate wound bobbin assembly using Item 7 (Impregnation). Keep bobbin pins and flying leads exiting the bobbin clear of impregnation.
6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 6 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. Ensure flying leads exit core per P7 Outline figure.
7. Stake core and bobbin using Item 8 (RTV) at the top of the assembly as indicated in the P7 Outline Figure. Do not force material into gap. (See Section B-B for maximum depth). Maintain a .01" to .05" gap in all corners. Do not allow the staking material to come in contact with inside corner edges.
8. Flying Leads Ns1 (B-A) shall be 2.00" $\pm$ 0.25" from core's edge. Strip and solder dip 0.200" $\pm$ 0.025" of flying leads using Item 9 (Solder).
9. Apply Item 5 (Tape, Kapton) to the core where the flying leads exit.
10. Install Item 11 (Shrink Sleeve) on the portion of leads A-B that exit over the core. The sleeving is to extend beyond the core area but not exceed 0.12" from the outer core edge. Stake sleeved wire to core using Item 8 (RTV).
11. Mark each device per 20043779 using Item 10 (Marking Ink). Tag each lead within 1/2 inch of core body. Using Item 10 (Marking Ink), identify pin 1 on top of core with a white dot.



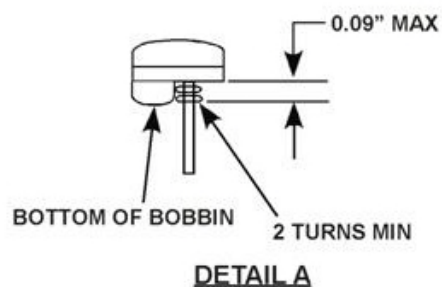
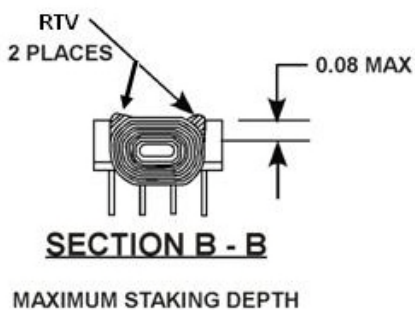
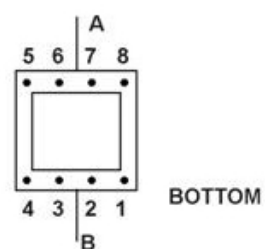
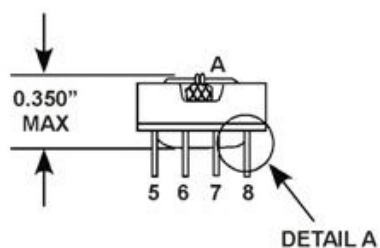
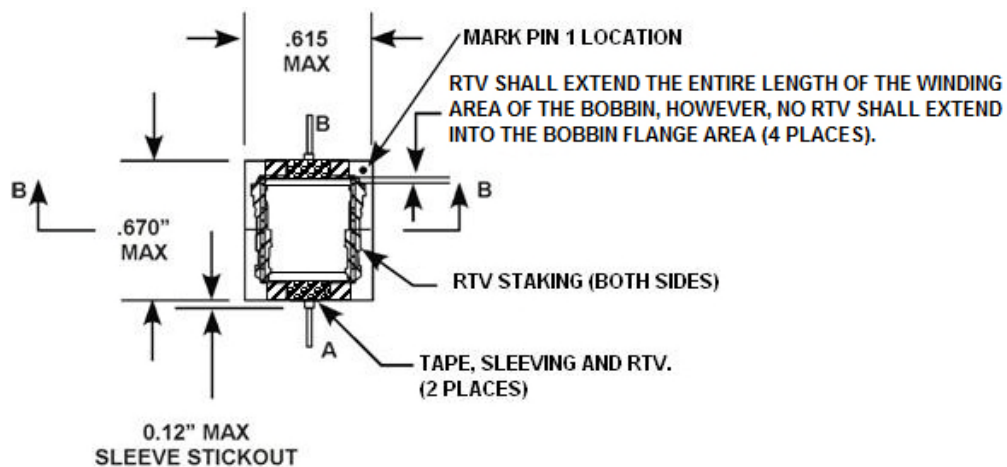
DIMENSIONS IN MM

EFD 15 COIL FORMER; 8-PINS

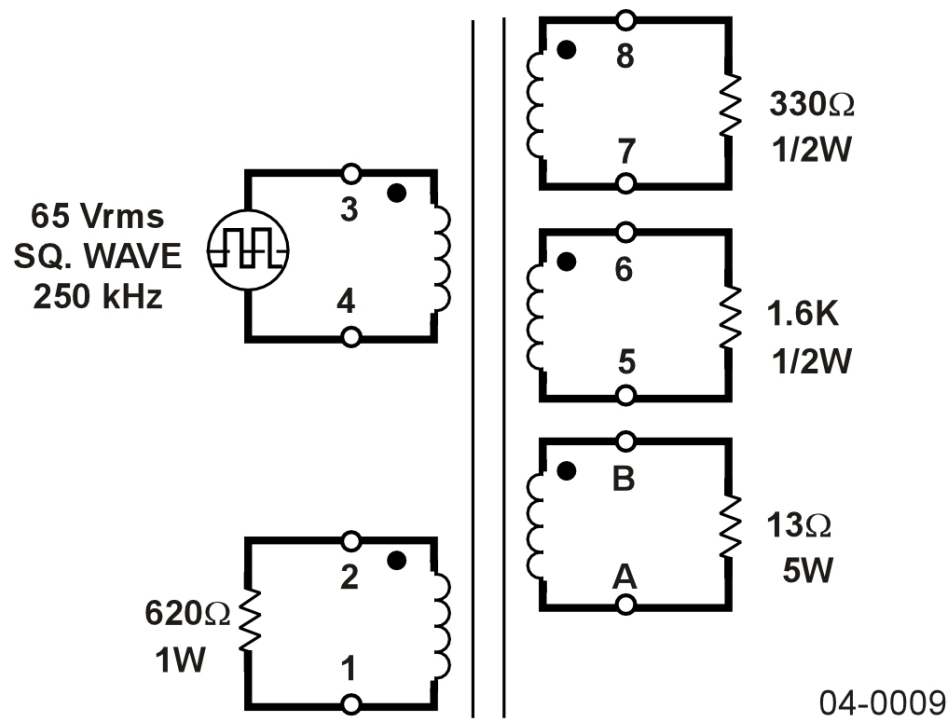


04-0007

P7 Coil Former and Schematic



P7 Outline



P7 Burn-In/Life Test Circuit

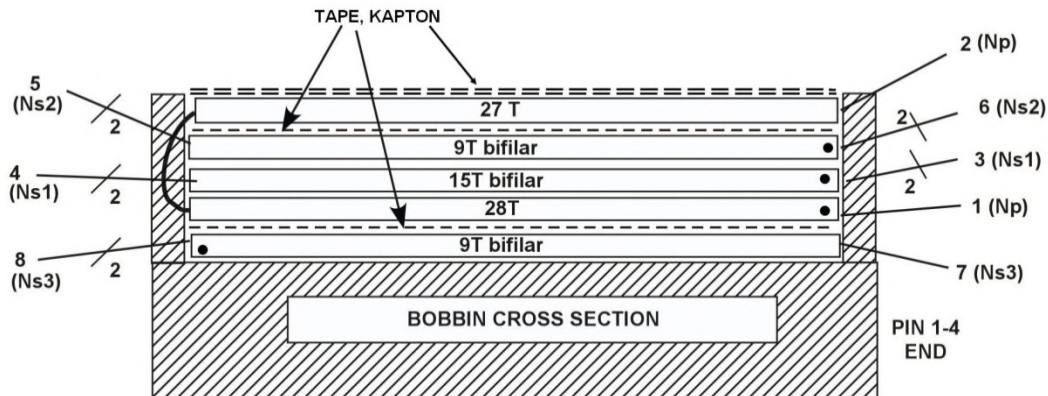
P8 Materials List			
Item	Material	Description	Source
1	Core, Gapped (1)	EFD15-3F3-A100-S, EFD15/8/5/3F3/AL=100	Ferroxcube
2	Bobbin (4)	CSH-EFD15-1S-8P or CSH-EFD15-1S-8P-*Z	Ferroxcube
3	Wire (3)	#30 AWG MW035CHX000030* or MW016CHX000030*	Elektrisola, Essex, MWS, or REA
4	Wire (3)	#26 AWG MW035CHX000026* or MW016CHX000026+*	Elektrisola, Essex, MWS, or REA
5	Tape, Kapton	P224 or K102per ASTM D5213	Per approved manufacturer's Baseline
6	Core Cement (2), (5)	Stycast 3050/Cat. 11, Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved manufacturer's Baseline
7	Impregnation	Scotchcast 280 or Stycast 3050/Cat. 11	Per approved manufacturer's Baseline
8	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved manufacturer's Baseline
9	Solder	SN60 or SN63 per IPC J-STD-006	Per approved manufacturer's Baseline
10	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline

NOTE:

- Suppliers may procure a core with a higher AL and gap center leg to meet inductance.
- Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW.
- Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax.
- The "*" is a letter for the country of origin. Prior to winding, bobbin pins need to be solder dipped twice in their entirety using item 9.
- Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.

P8 Electrical Characteristics				
Characteristics				
Winding Designation (Start-Finish)	Np (1-2)	Ns1 (3-4)	Ns2 (6-5)	Ns3 (8-7)
Number of Turns	55	15	9	9
Maximum Operating Current (Arms)	0.140	0.070	0.070	0.040
Working Voltage (Peak)	175 V	175 V	175 V	175V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	550 mΩ	80 mΩ	25 mΩ	25 mΩ
Inductance (± 10%) <u>4/</u>	300 μH	22.3 μH	8.0 μH	8.0 μH
Leakage Inductance (Max) <u>5/</u>	3 nH	Shorted	Shorted	Shorted
Rated Voltage <u>6/</u>	70 V	15 V	9 V	9 V
NOTES: 1. 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10 ⁻⁶ Torr., Qualifies unit to 175V peak working voltage. 2. 500 Vdc. Corrected to +20°C. 3. Measured at 200kHz 0 Adc bias. 4. Measured at 200kHz. "Shorted" means all other windings shorted start-to-finish and shorted together. 5. Induced Voltage Required.				

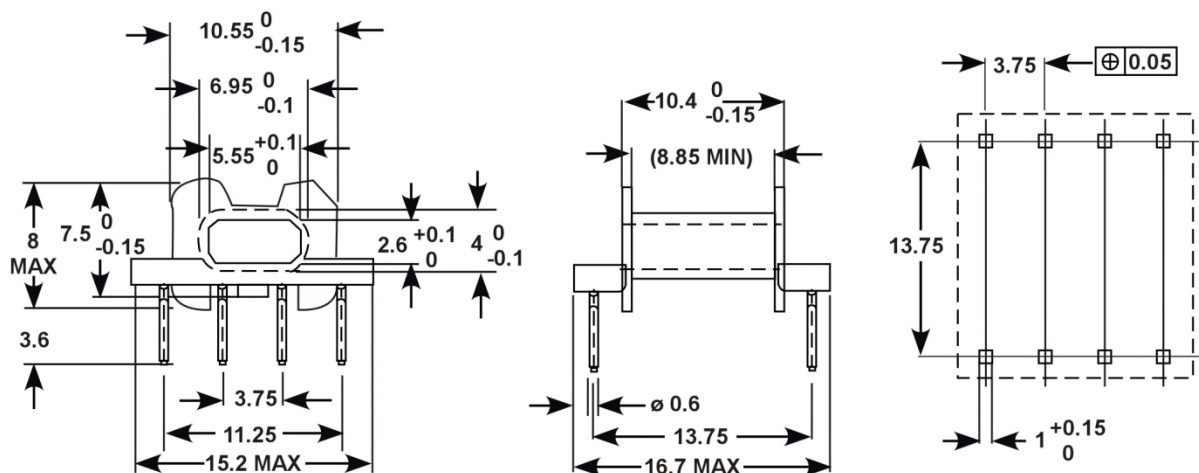
P8 Winding Table, Instructions and Figure				
Winding Designation	Np (1-2)	Ns1 (3-4)	Ns2 (6-5)	Ns3 (8-7)
No. of Turns	55	15	9	9
Wire Size, AWG	30, Single	30, Bifilar	26, Bifilar	26, Bifilar



Refer to Winding Figure for Clarification.
Spread all windings across winding width.
All lead routing shall be performed after final winding layer.
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 9 (Solder).
Observe polarity during the winding process.
Clockwise and counter-clockwise directions are as viewed from the pin 1-4 end of the bobbin.
Using item 4 (26AWG Wire), wind 9 turns of Ns3 (8-7) bi-filar, clockwise onto Item 2 (Bobbin). Start on the pin 1-4 side of the bobbin.
Install one layer of Item 5 (Tape, Kapton) to provide a flat surface for additional layers.
Begin winding Np (1-2) using Item 3 (30 AWG Wire) onto Item 2 (Bobbin). Start on Pin 1-4 side of the bobbin and wind 28 turns counter-clockwise. Pull out leaving enough wire to complete the remaining turns of Np winding.
Using Item 3 (30 AWG Wire) bi-filar and starting on Pin 1-4 side of the bobbin, wind 15 turns counter-clockwise of winding Ns1 (3-4).
Using Item 4 (26 AWG Wire) bi-filar and starting on Pin 5-8 side of the bobbin, wind 9 turns clockwise of winding Ns2 (6-5).
Install one layer of Item 5 (Tape, Kapton) to provide a flat surface for additional layers.
Finish winding Np (1-2) with 27 turns counter-clockwise (55 turns total), finishing on the pin 1-4 side of the bobbin.
Wrap wound coil with one layer of Item 5 (Tape, Kapton).
Route leads to appropriate pin locations. Use corners to crossover leads on the incorrect side to minimize build up on bottom of wound bobbin.
Wrap wound coil with Item 5 (Tape, Kapton).one layer minimum, two layers maximum.

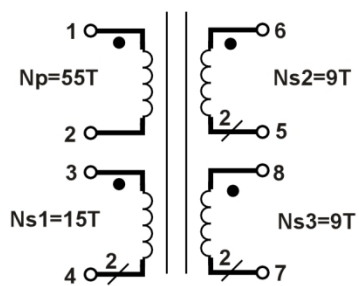
P8 Assembly Instructions and Notes:

1. All materials shall be per the P8 Materials List.
2. Unless otherwise specified, all dimensions are in inches and include all materials used.
3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C $\pm$ 5°C for 1 hour +10 minutes, -0 minutes
4. Terminate windings Np (1-2), Ns1 (3-4), Ns2 (6-5) and Ns3 (8-7) to the appropriate pins using item 9 (Solder). A maximum termination height from pin end (bobbin side) is 0.09" max. (See Detail "A" of P8 Outline).
5. Vacuum impregnate wound bobbin assembly using Item 7 (Impregnation). Keep bobbin pins clear of impregnation.
6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 6 (Core Cement). **Do not bond center leg.** Once cement cures insure bobbin is not cemented to core.
7. Stake core and bobbin using Item 8 (RTV) at the top of the assembly as indicated in the P8 Outline Figure. Do not force material into gap. (See Section B-B for maximum depth). Maintain a .01" to .05" gap in all corners. Do not allow the staking material to come in contact with inside corner edges.
8. Mark each device per 20043779 using Item 10 (Marking Ink). Using Item 10 (Marking Ink), identify pin 1 on top of core with a white dot.



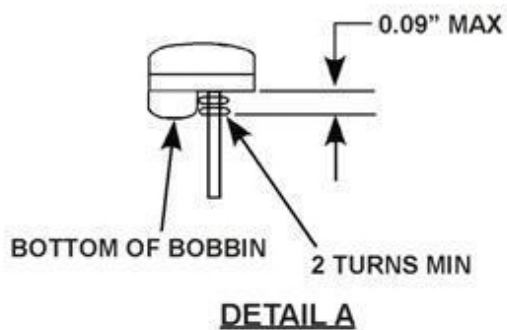
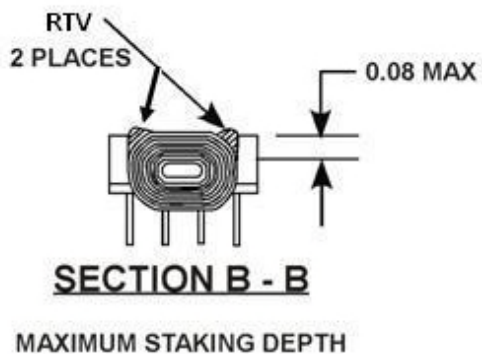
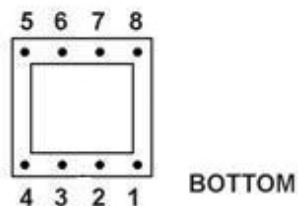
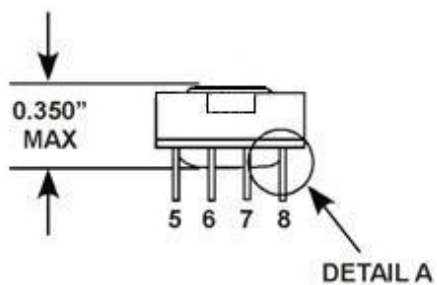
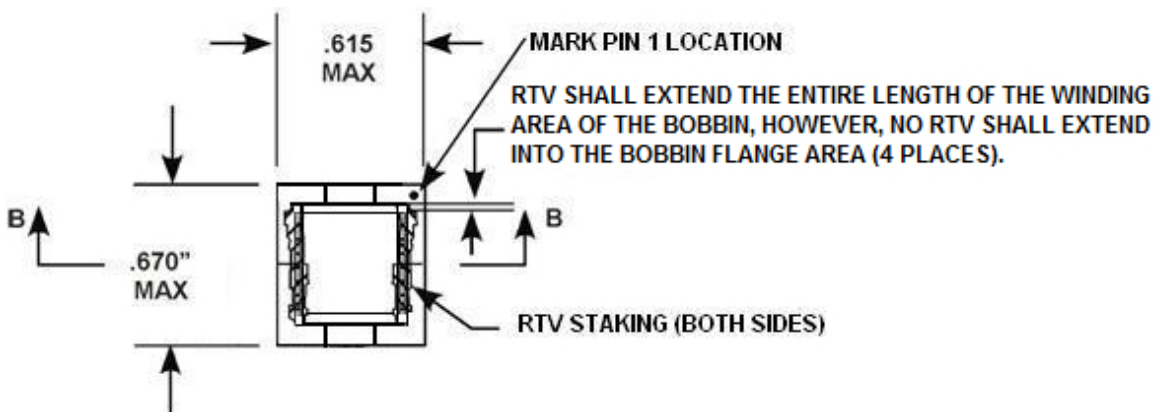
DIMENSIONS IN MM

EFD 15 COIL FORMER; 8-PINS

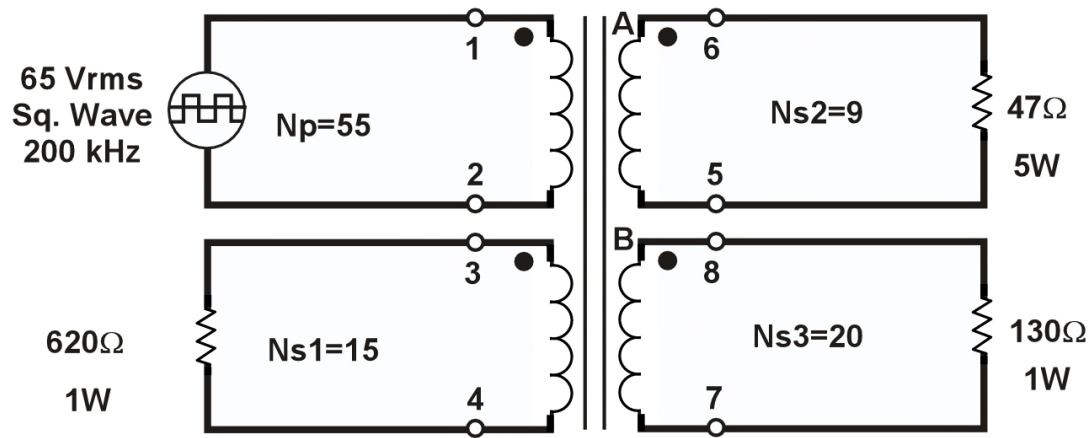


04-0010

P8 Coil Former and Schematic



P8 Outline

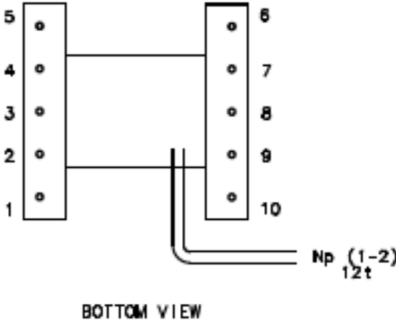
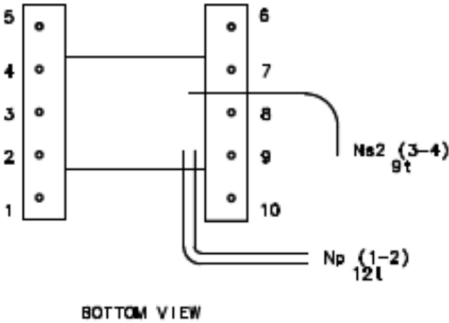


04-0012

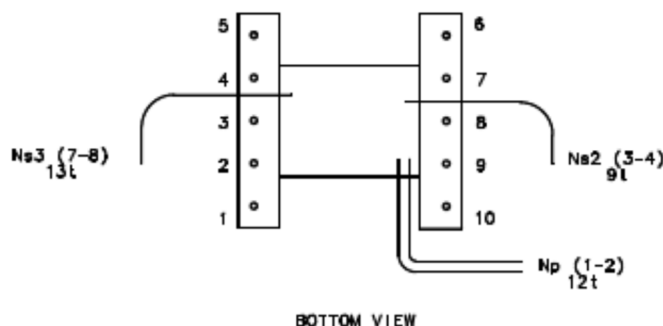
P8 Burn-In/Life Test Circuit

P9 Materials List			
Item	Material	Description	Source
1	Core, Gapped	EFD 20/10/7-3F3-A160	Ferroxcube
2	Bobbin	CPH-EFD20-1S-10PD-Z	Ferroxcube
3	Wire (1)	#28 AWG, NEMA MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Tape, Glass	#79 Glass per A-A-59770	Per approved Manufacturer's Baseline
5	Core Cement (2)	Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved Manufacturer's Baseline
6	Impregnation	Scotchcast 280	Per approved Manufacturer's Baseline
7	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved Manufacturer's Baseline
8	Solder	SN60 or SN63 per J-STD-006	Per approved Manufacturer's Baseline
9	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline
10	Tape, Kapton	Permacel P224 or K102	Per approved Manufacturer's Baseline
NOTE: 1. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax. 2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW.			

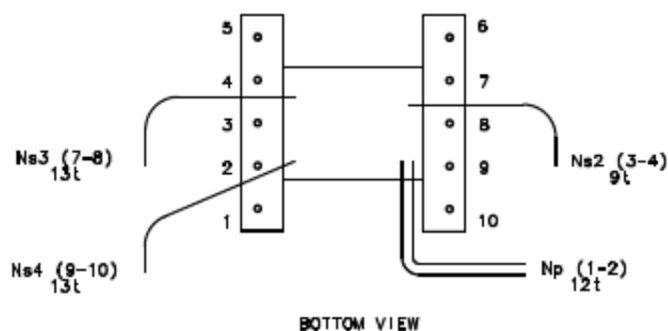
P9 Electrical Characteristics					
Characteristics	Limits	Limits	Limits	Limits	Limits
Winding Designation (Start-Finish)	Np (1-2)	Ns1 (5-6)	Ns2 (3-4)	Ns3 (7-8)	Ns4 (9-10)
Number of Turns	23	5	9	13	13
Maximum Operating Current (Arms)	0.266	0.798	0.127	0.276	0.064
Working Voltage (Peak)	175V	175V	175V	175V	175V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	126mΩ	14mΩ	99mΩ	142mΩ	142mΩ
Inductance ($\pm 10\%$) <u>4/</u>	84.64μH	4.00μH	12.96μH	27.04μH	27.04μH
Leakage Inductance (Max) <u>5/</u>	425nH Shorted Shorted Shorted Shorted	Shorted 135nH Shorted Shorted Shorted	Shorted Shorted 45nH Shorted Shorted	Shorted Shorted Shorted 325nH Shorted	Shorted Shorted Shorted Shorted 255nH
Cpri-sec (Max) <u>6/</u>	--	75.0pF	75.0pF	75.0pF	75.0pF
Turns Ratio <u>7/</u>	Reference	0.217 $\pm 1.5\%$	0.391 $\pm 1.5\%$	0.565 $\pm 1.5\%$	0.565 $\pm 1.5\%$
Rated Voltage <u>8/</u>	80V	18V	32V	46V	46V
NOTES: 1. 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10 ⁻⁶ Torr., Qualifies unit to 175V peak working voltage. 2. 500 Vdc. 3. Corrected to +20°C. 4. Measured at 1Vac, 300kHz, 0Adc bias. 5. Measured at 1Vac, 300kHz, 0Adc bias. "Shorted" means all other windings shorted start-to-finish and shorted together. 6. Measured at 1Vac, 300kHz, 0Adc bias. Capacitance is measured from primary winding to each secondary winding (with pins of each pair of windings shorted) all other windings open-circuited. 7. Turns ratio data may be taken in process prior to cementing cores. 8. Induced Voltage Test required.					

P9 Winding Instructions and Assembly Notes
Prior to beginning winding, solder dip all bobbin pins in their entirety twice, using Item 8 (Solder).
Observe polarity during the winding process. . Use bobbin slots to route wires. Ensure no wire extends below the bobbin standoff.
Position bobbin so view is of bottom-side with pin 1 in bottom left-hand corner and set winding direction “top-going”. Per the outline drawing, identity Pin 1 on bobbin with a white dot using Item 9 (Marking Ink).
Using Item 3 (28 AWG Wire) and starting at pin 1of bobbin, wind Np (1-2) bifilar for 12 turns (1 layer) evenly and with no crossovers. Break out wire(s) from bobbin as shown in figure. Ensure there is sufficient length remaining to complete the winding at a later step in the “Winding Instructions”.  BOTTOM VIEW
Using Item 10 (Tape, Kapton), wind 1 layer making sure to overlap Np windings by ~0.150”.
Using Item 3 (28 AWG Wire) and starting at pin 5 of bobbin, wind Ns1a (5a-6a) bifilar for 5 turns (1 layer) evenly and with no crossovers. Break out finish leads to pin 6 of bobbin.
Using Item 3 (28 AWG Wire) and starting at pin 3 of bobbin, wind Ns2 (3-4) for 9 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #4 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the “Winding Instruction”.  BOTTOM VIEW

Using Item 3 (28 AWG Wire) and starting at pin 7 of bobbin, wind Ns3 (7-8) for 13 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #8 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instruction".



Using Item 3 (28 AWG Wire) and starting at pin 9 of bobbin, wind Ns4 (9-10) for 13 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #10 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instruction".



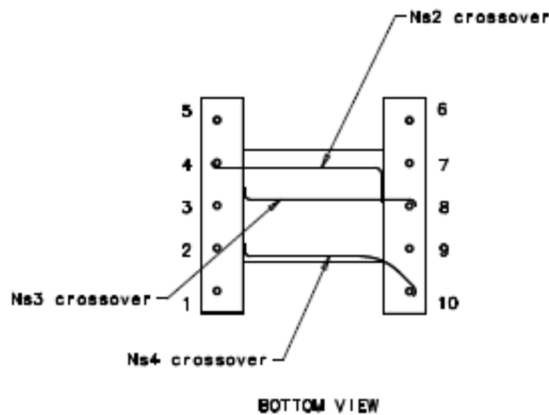
Using Item 3 (28 AWG Wire) and starting at pin 5 of bobbin, wind Ns1b (5b-6b) bifilar for 5 turns (1 layer) evenly and with no crossovers. Terminate finish leads to pin 6 of bobbin.

Using Item 10 (Tape, Kapton), wind 1 layer making sure to overlap Ns1 windings by ~0.150".

Using Item 3 (28 AWG Wire), wind the balance of Np (1-2) bifilar for 11 turns (1 layer) evenly and with no crossovers. Break out finish lead to pin 2 of bobbin.

Using Item 10 (Tape, Kapton), wind 1 layer(s) making sure to overlap Np windings by ~0.150".

NOTE: During the following winding instructions, when crossing over wires, ensure that none of the crossovers overlap one another. See figure.



Pick up wire labeled #4 and crossover Ns2 (3-4) as depicted in the figure above and terminate to pin 4 of bobbin.

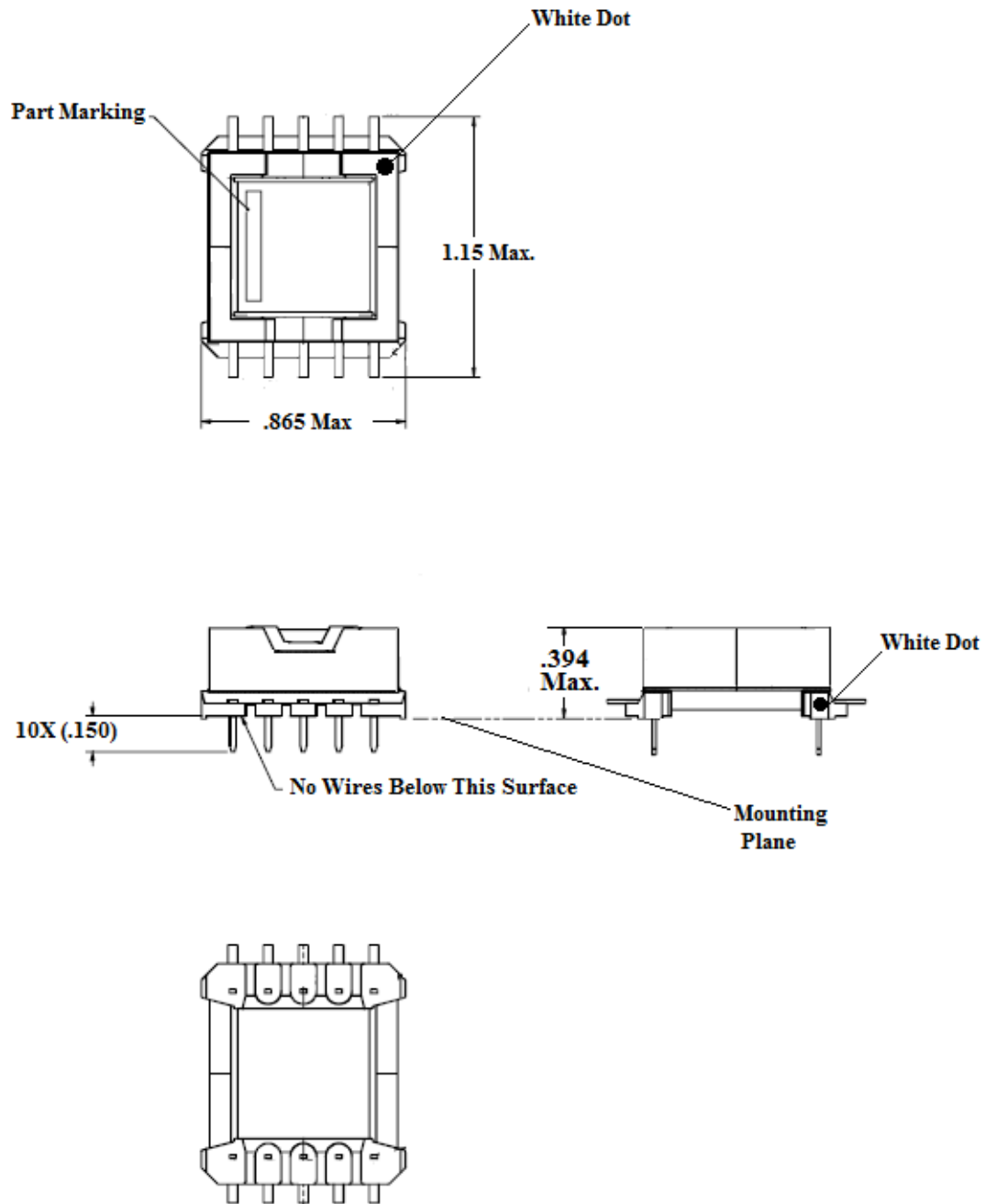
Pick up wire labeled #10 and crossover Ns4 (9-10) as depicted in the figure above and terminate to pin 10 of bobbin.

Pick up wire labeled #8 and crossover Ns3 (7-8) as depicted in the figure above and terminate to pin 8 of bobbin.

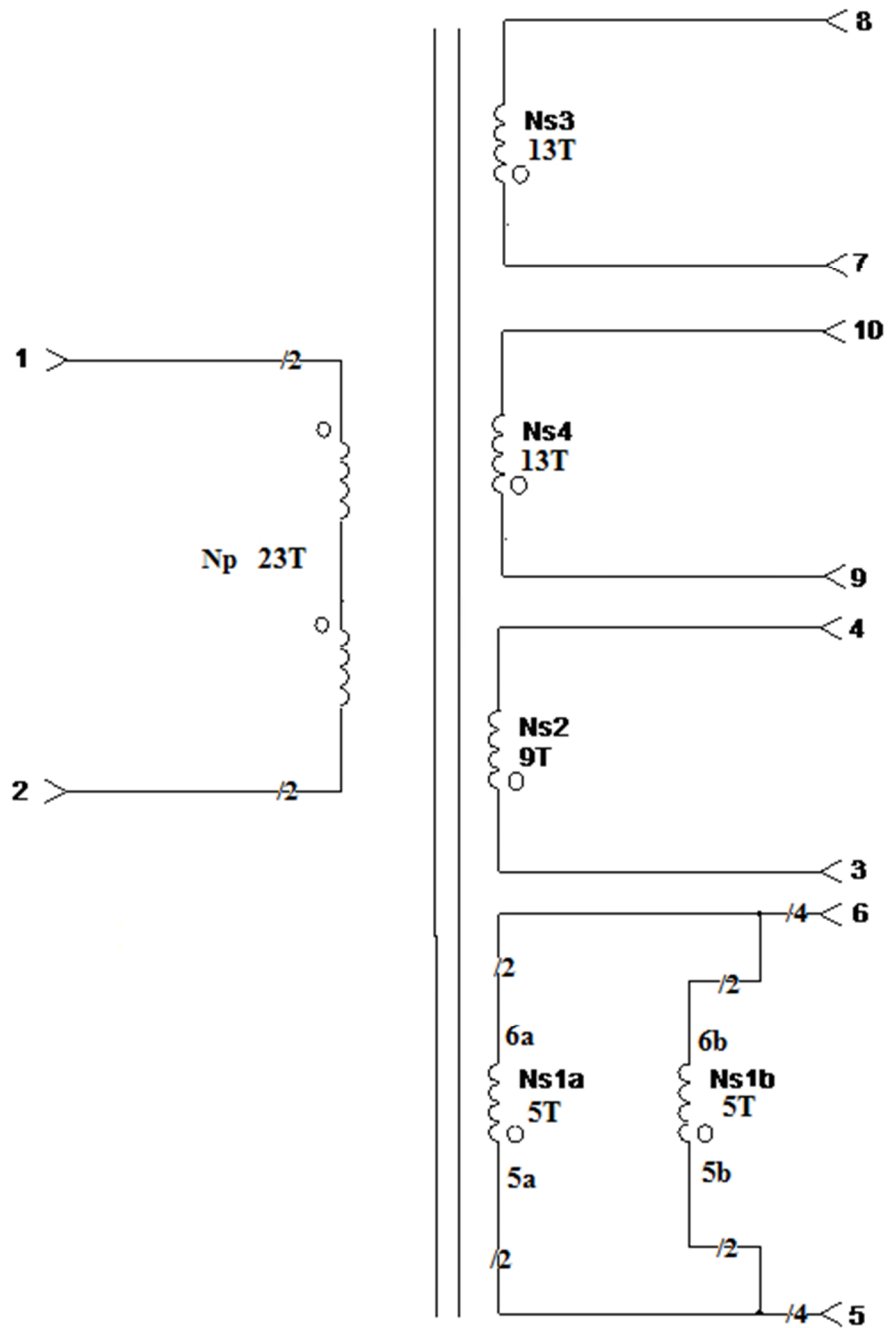
Wrap wound coil with Item 4 (Tape, Glass), 1 layer minimum, 1½ layers maximum.

NOTES:

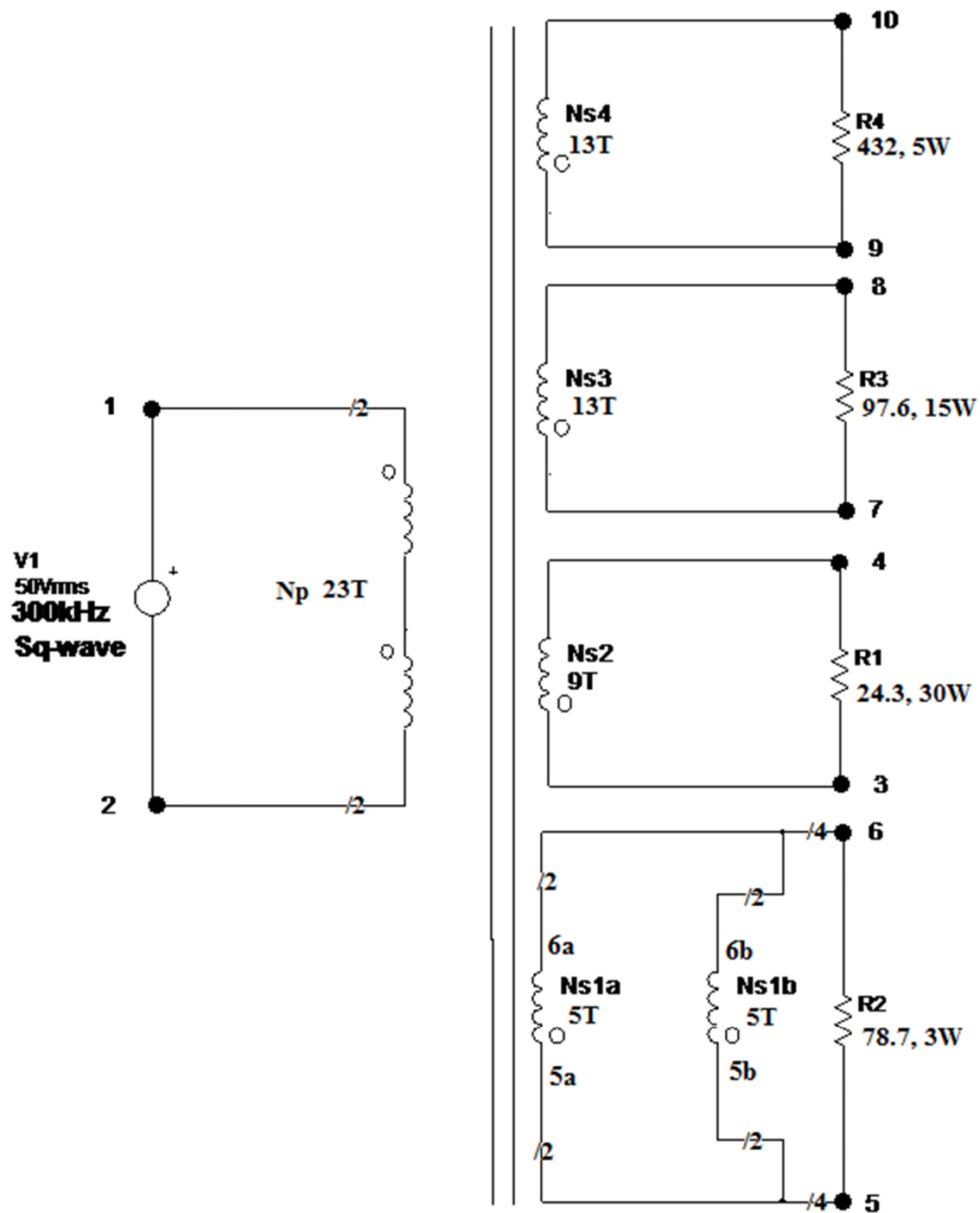
1. All materials shall be per the P9 Materials List.
2. All dimensions are in inches and include all materials used.
3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes.
4. Terminate windings Np (1-2), Ns1 (5-6), Ns2 (3-4), Ns3 (7-8) and Ns4 (9-10) to the appropriate pins using Item 8 (Solder).
5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation).
6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 5 (Core Cement). **Do not bond center leg.** Once cement cures, ensure bobbin is not cemented to core.
7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required.
8. Mark each device per 20043779 using Item 9 (Marking Ink). Using Item 9 (Marking Ink), identify pin 1 on top of core with a white dot.



P9 Outline



P9 Schematic



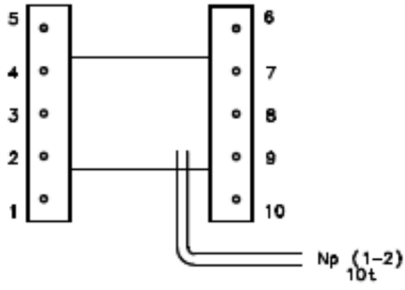
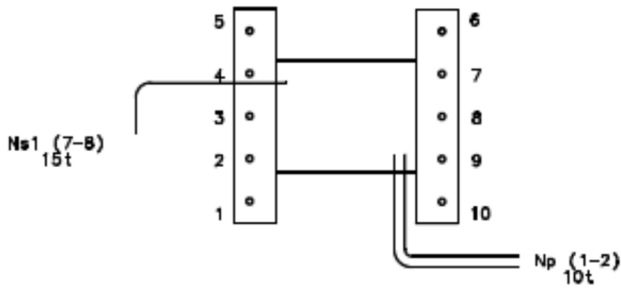
P9 Burn-in/Life Test Circuit

P10 Materials List			
Item	Material	Description	Source
1	Core, Gapped	EFD 20/10/7-3F3-A160	Ferroxcube
2	Bobbin	CPH-EFD20-1S-10PD-Z	Ferroxcube
3	Wire (1)	#28 AWG, NEMA MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Tape, Glass	79 Glass per A-A-59770	Per approved Manufacturer's Baseline
5	Core Cement (2)	Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved Manufacturer's Baseline
6	Impregnation	Scotchcast 280	Per approved Manufacturer's Baseline
7	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved Manufacturer's Baseline
8	Solder	SN60 or SN63 per IPC J-STD-006	Per approved Manufacturer's Baseline
9	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline
10	Tape, Kapton	Permacel P224 or K102	Per approved Manufacturer's Baseline
NOTE: 1. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax. 2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW.			

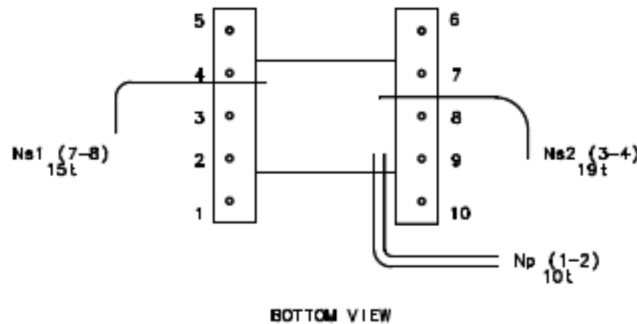
P10 Electrical Characteristics				
Characteristics	Limits	Limits	Limits	Limits
Winding Designation (Start-Finish)	Np (1-2)	Ns1 (7-8)	Ns2 (3-4)	Ns3 (9-10)
Number of Turns	20	15	18	14
Maximum Operating Current (Arms)	0.336	0.331	0.252	0.064
Working Voltage (I _{Peak})	175 V	175 V	175 V	175V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	110mΩ	41mΩ	197mΩ	153mΩ
Inductance ($\pm 10\%$) <u>4/</u>	64.00μH	36.00μH	51.84 μH	31.36μH
Leakage Inductance (Max) <u>5/</u>	325nH Shorted Shorted Shorted	Shorted 170nH Shorted Shorted	Shorted Shorted 280nH Shorted	Shorted Shorted Shorted 220nH
C _{pri-sec} (Max.) <u>6/</u>	--	105.0 pF	82.0 pF	71.0 pF
Turns Ratio <u>7/</u>	Reference	0.750 $\pm 1.5\%$	0.900 $\pm 1.5\%$	0.700 $\pm 1.5\%$
Rated Voltage <u>8/</u>	80V	60V	72V	56V

NOTES:

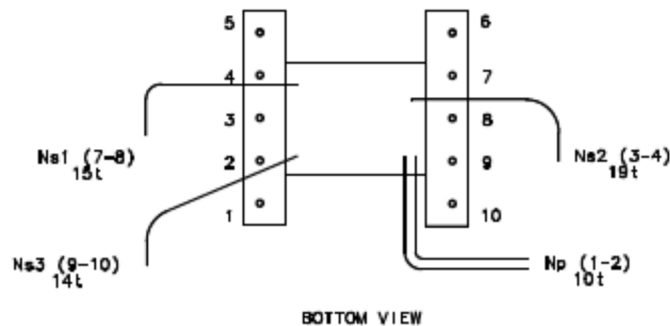
1. 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10⁻⁶ Torr., Qualifies unit to 175V peak working voltage.
2. 500 Vdc.
3. Corrected to +20°C.
4. Measured at 1Vac, 300kHz, 0Adc bias.
5. Measured at 1Vac, 300kHz, 0Adc bias. "Shorted" means all other windings shorted start-to-finish and shorted together.
6. Measured at 1Vac, 300kHz, 0Adc bias. Capacitance is measured from primary winding to each secondary winding (with pins of each pair of windings shorted) all other windings open-circuited.
7. Turns ratio data may be taken in process prior to cementing cores.
8. Induced Voltage Test required.

P10 Winding Instructions and Assembly Notes
Prior to beginning winding, solder dip all bobbin pins in their entirety twice using Item 8 (Solder).
Observe polarity during the winding process. . Use bobbin slots to route wires. Ensure no wire extends below the bobbin standoff.
Position bobbin so view is of bottom-side with pin 1 in bottom left-hand corner and set winding direction "top-going". Per the outline drawing, identity Pin 1 on bobbin with a white dot using Item 9 (Marking Ink).
Using Item 3 (28 AWG Wire) and starting at pin 1 of bobbin, wind Np (1-2) bifilar for 10 turns (1 layer) evenly and with no crossovers. Break out wire(s) from bobbin as shown in figure. Ensure there is sufficient length remaining to complete the winding at a later step in the "Winding Instruction".  BOTTOM VIEW
Using Item 10 (Tape, Kapton), wind 1 layer making sure to overlap Np windings by ~0.150".
Using Item 3 (28 AWG Wire) and starting at pin 7 of bobbin, wind Ns1a (7a-8a) for 15 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #8 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instruction".  BOTTOM VIEW

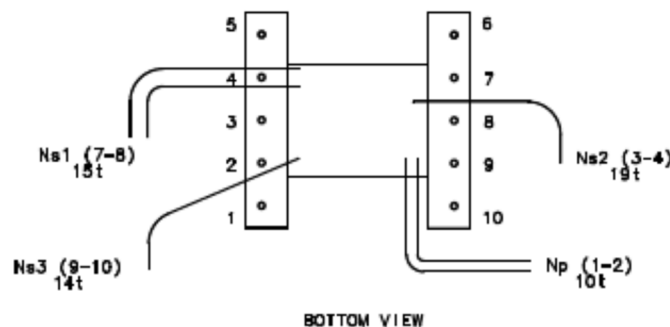
Using Item 3 (28 AWG Wire) and starting at pin 3 of bobbin, wind Ns2 (3-4) for 18 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #4 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instruction".



Using Item 3 (28 AWG Wire) and starting at pin 9 of bobbin, wind Ns3 (9-10) for 14 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #10 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instruction".



Using Item 3 (28 AWG Wire) and starting at pin 7 of bobbin, wind Ns1b (7b-8b) for 15 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #8 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instruction".

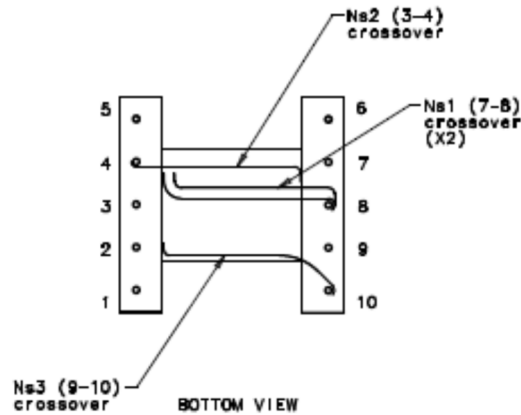


Using Item 10 (Tape, Kapton), wind 1 layer(s) making sure to overlap Ns1b windings by ~0.150".

Using Item 3 (28 AWG Wire), wind the balance of Np (1-2) bifilar for 10 turns (1 layer) evenly and with no crossovers. Break out finish lead to pin 2 of bobbin.

Using Item 10 (Tape, Kapton), wind 1 layer(s) making sure to overlap Np windings by ~0.150".

NOTE: During the following winding instructions, when crossing over wires, ensure that none of the crossovers overlap one another. See figure.



Pick up wires labeled #8 and crossover both Ns1 (7-8) wires as depicted in the figure above and terminate to pin 8 of bobbin.

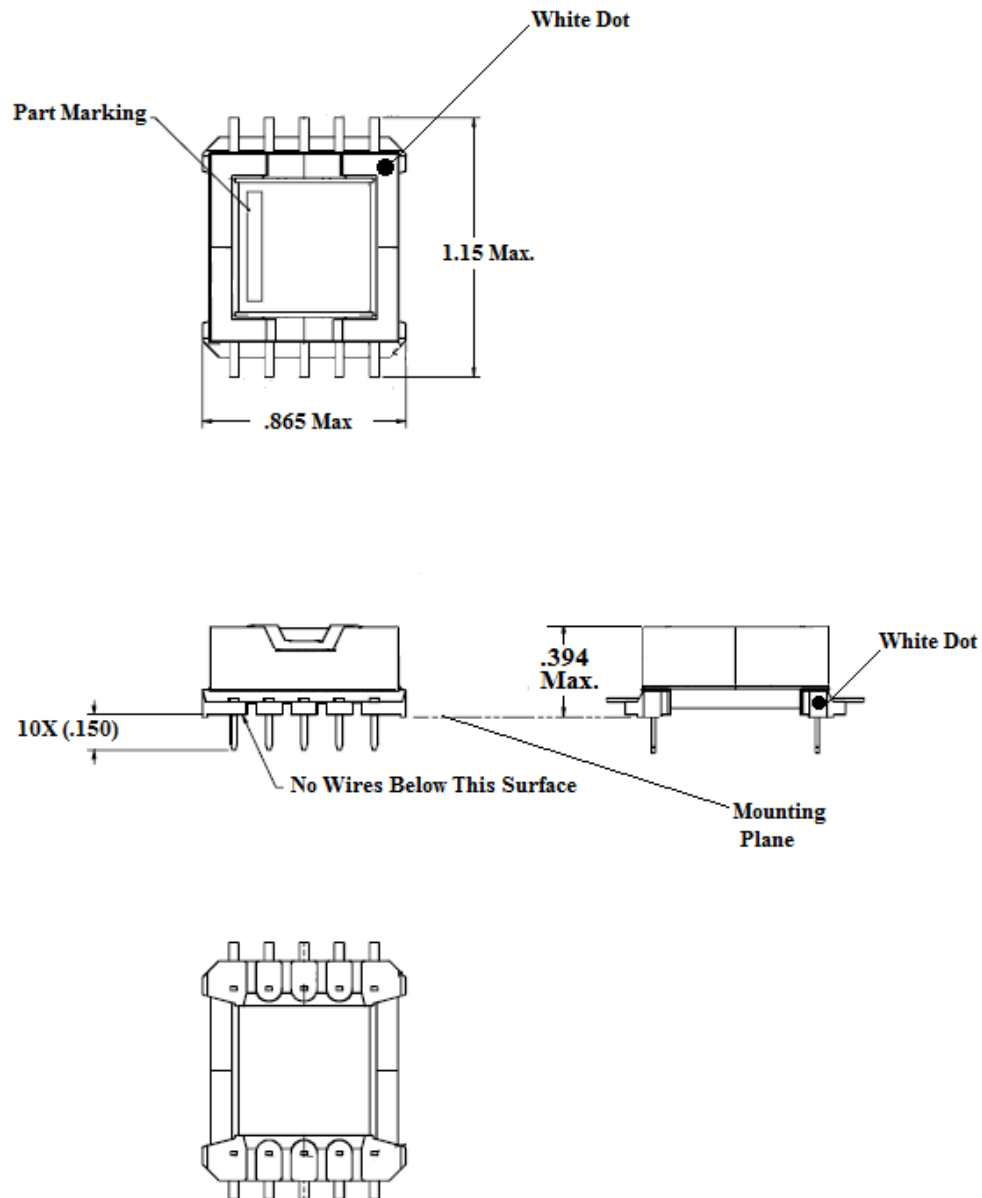
Pick up wire labeled #10 and crossover Ns3 (9-10) as depicted in the figure above and terminate to pin 10 of bobbin.

Pick up wire labeled #4 and crossover Ns2 (3-4) as depicted in the figure above and terminate to pin 4 of bobbin.

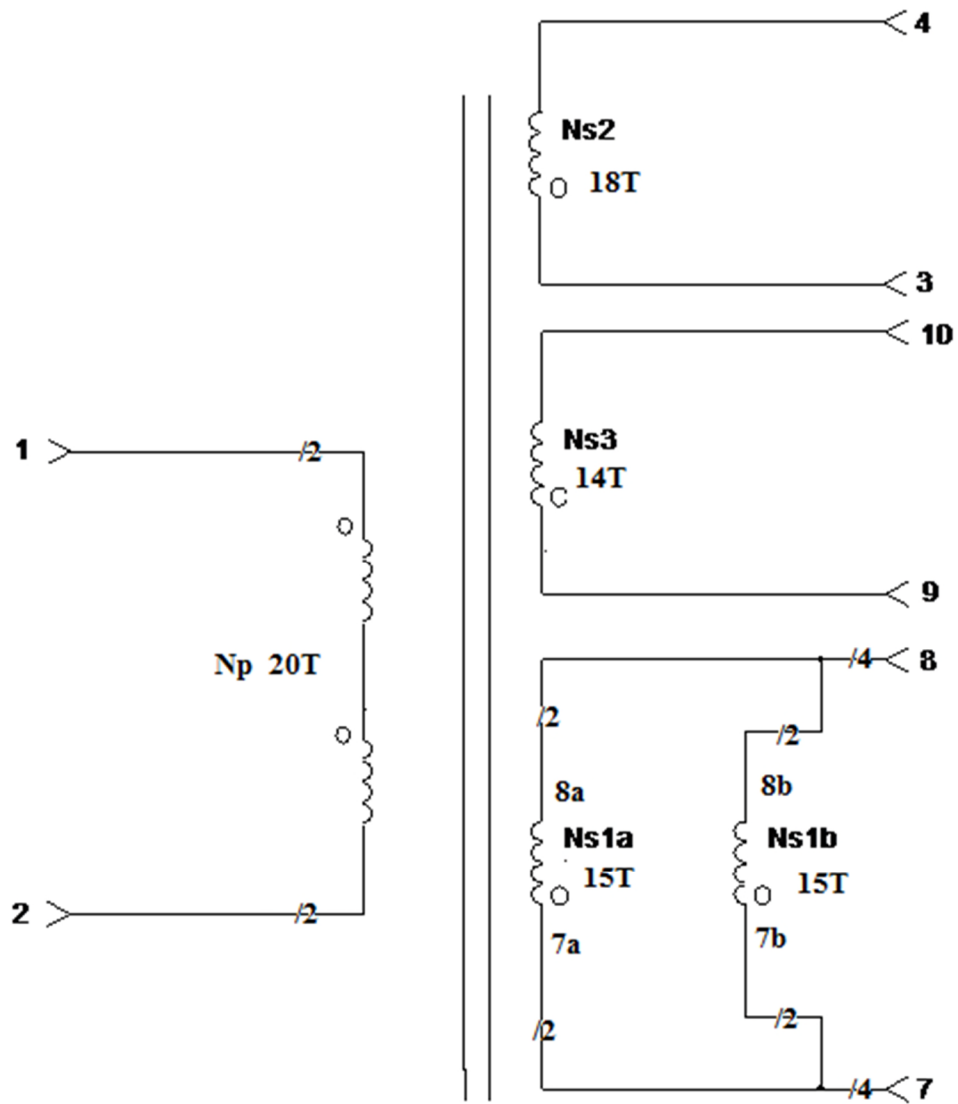
Wrap wound coil with Item 4 (Tape, Glass), 1 layer minimum, 1½ layers maximum.

NOTES:

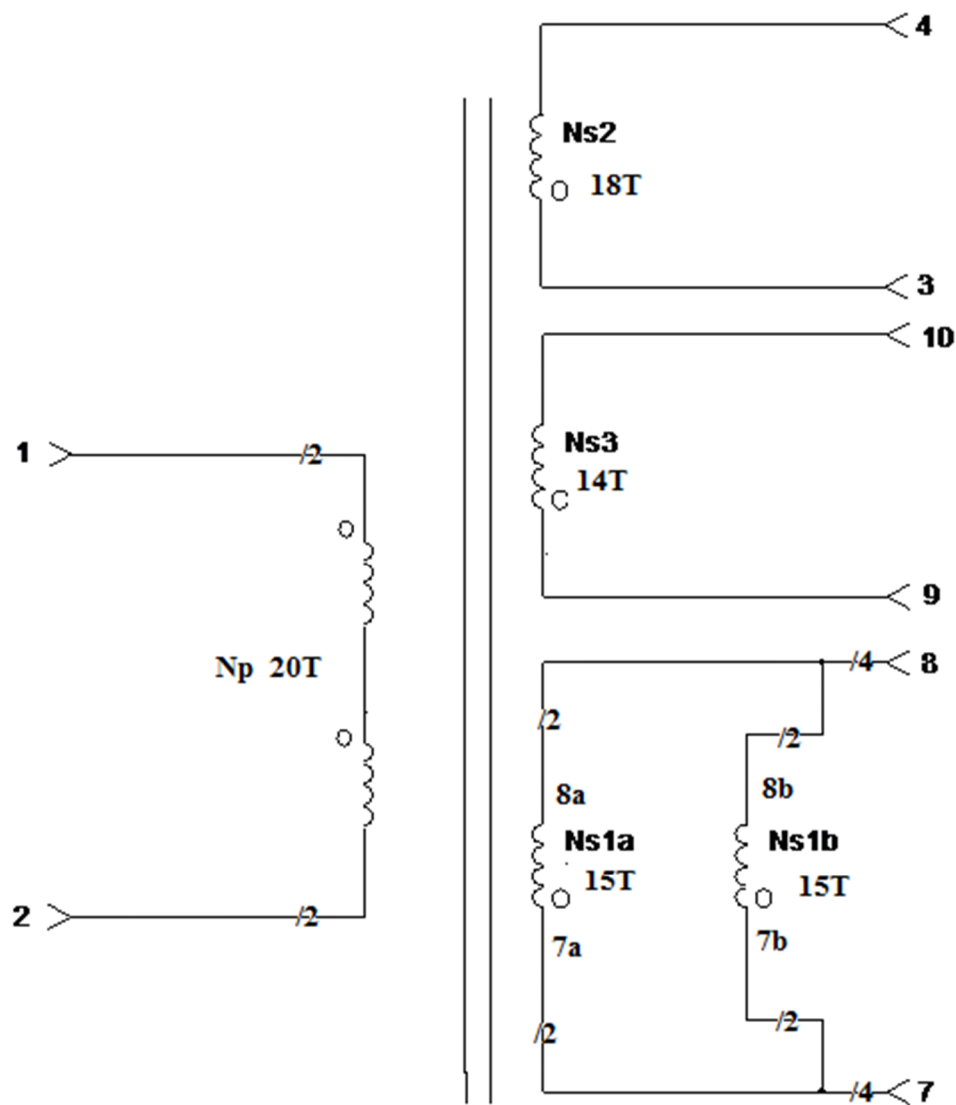
1. All materials shall be per the P10 Materials List.
2. All dimensions are in inches and include all materials used.
3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes.
4. Terminate windings Np (1-2), Ns1 (7-8), Ns2 (3-4) and Ns3 (9-10) to the appropriate pins using Item 8 (Solder).
5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation).
6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 5 (Core Cement). **Do not bond center leg.** Once cement cures insure bobbin is not cemented to core.
7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required.
8. Mark each device per 20043779 using Item 9 (Marking Ink). Using Item 9 (Marking Ink), identify pin 1 on top of core with a white dot.



P10 Outline



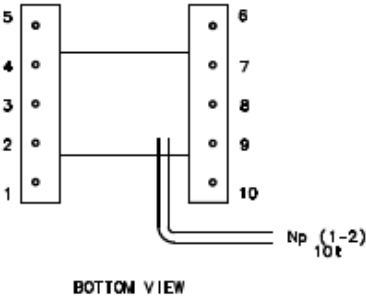
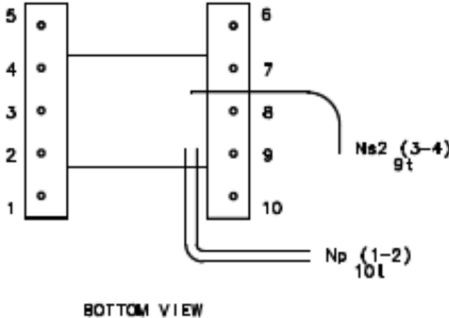
P10 Schematic



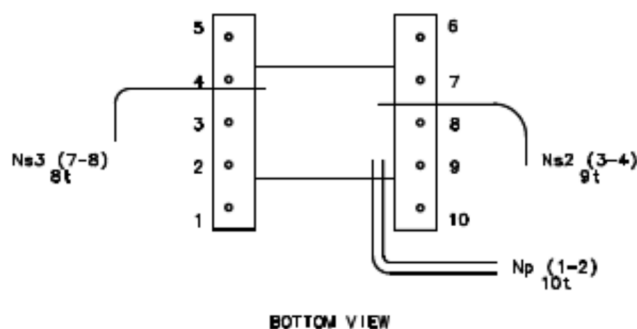
P10 Burn-in/Life Test Circuit

P11 Materials List			
Item	Material	Description	Source
1	Core, Gapped	EFD 20/10/7-3F3-A160	Ferroxcube
2	Bobbin	CPH-EFD20-1S-10PD-Z	Ferroxcube
3	Wire (1)	#28 AWG, NEMA MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Tape, Glass	79 Glass per A-A-59770	Per approved Manufacturer's Baseline
5	Core Cement (2)	Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved Manufacturer's Baseline
6	Impregnation	Scotchcast 280	Per approved Manufacturer's Baseline
7	Staking Compound	RTV 142 or CV-1142-1 or DC6-1104	Per approved Manufacturer's Baseline
8	Solder	SN60 or SN63 per IPCJ-STD-006	Per approved Manufacturer's Baseline
9	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline
10	Tape, Kapton	Permacel P224 or K102	Per approved Manufacturer's Baseline
NOTE: 1. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax. 2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40 PBW.			

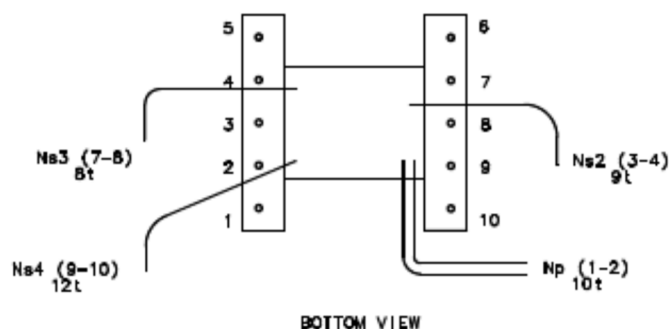
P11 Electrical Characteristics					
Characteristics	Limits	Limits	Limits	Limits	Limits
Winding Designation (Start-Finish)	Np (1-2)	Ns1 (5-6)	Ns2 (3-4)	Ns3 (7-8)	Ns4 (9-10)
Number of Turns	20	7	9	8	12
Maximum Operating Current (Arms)	0.348	1.167	0.127	0.276	0.067
Working Voltage (Peak)	175V	175V	175V	175V	175V
Dielectric Withstanding Voltage <u>1/</u>	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (Min.) <u>2/</u>	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ	10,000 MΩ
DC Resistance (Max.) <u>3/</u>	110mΩ	19mΩ	99mΩ	88mΩ	131mΩ
Inductance ($\pm 10\%$) <u>4/</u>	64.00μH	7.84μH	12.96μH	10.24μH	23.04μH
Leakage Inductance (Max) <u>5/</u>	350nH Shorted Shorted Shorted Shorted	Shorted 50nH Shorted Shorted Shorted	Shorted Shorted 125nH Shorted Shorted	Shorted Shorted Shorted 150nH Shorted	Shorted Shorted Shorted Shorted 185nH
Cpri-sec (Max) <u>6/</u>	--	90.0pF	50.0pF	50.0pF	55.0pF
Turns Ratio <u>7/</u>	Reference	0.350 $\pm 1.5\%$	0.450 $\pm 1.5\%$	0.400 $\pm 1.5\%$	0.600 $\pm 1.5\%$
Rated Voltage <u>8/</u>	80V	28V	36V	32V	48V
NOTES: 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5 x 10⁻⁶ Torr., Qualifies unit to 175V peak working voltage. 500 Vdc. - Corrected to +20°C. - Measured at 1Vac, 300kHz, 0Adc bias. - Measured at 1Vac, 300kHz, 0Adc bias. "Shorted" means all other windings shorted start-to-finish and shorted together. - Measured at 1Vac, 300kHz, 0Adc bias. Capacitance is measured from primary winding to each secondary winding (with pins of each pair of windings shorted) all other windings open-circuited. - Turns ratio data may be taken in process prior to cementing cores. - Induced Voltage Test required.					

P11 Winding Instructions and Assembly Notes
Prior to beginning winding, solder dip all bobbin pins in their entirety twice, using Item 8 (Solder).
Observe polarity during the winding process. Use bobbin slots to route wires. Ensure no wire extends below the bobbin standoff.
Position bobbin so view is of bottom-side with pin 1 in the bottom left-hand corner and set winding direction “top-going”. Per the outline drawing, identity Pin 1 on bobbin with a white dot using Item 9 (Marking Ink).
Using Item 3 (28 AWG Wire) and starting at pin 1 of bobbin, wind Np (1-2) bifilar for 10 turns (1 layer) evenly and with no crossovers. Break out wire(s) from bobbin as shown in figure. Ensure there is sufficient length remaining to complete the winding at a later step in the “Winding Instructions”. 
Using Item 10 (Tape, Kapton), wind 1 layer making sure to overlap Np windings by ~0.150”.
Using Item 3 (28 AWG Wire) and starting at pin 5 of bobbin, wind Ns1a (5a-6a) bifilar for 7 turns (1 layer) evenly and with no crossovers. Break out finish leads to pin 6 of bobbin.
Using Item 3 (28 AWG Wire) and starting at pin 3 of bobbin, wind Ns2 (3-4) for 9 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #4 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the “Winding Instructions”. 

Using Item 3 (28 AWG Wire) and starting at pin 7 of bobbin, wind Ns3 (7-8) for 8 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #8 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instructions".



Using Item 3 (28 AWG Wire) and starting at pin 9 of bobbin, wind Ns4 (9-10) for 12 turns (1 layer) evenly and with no crossovers. Break out wire from bobbin and tag #10 as shown in figure. Ensure there is sufficient length remaining to allow the wire to crossover and terminate at a later step in the "Winding Instructions".



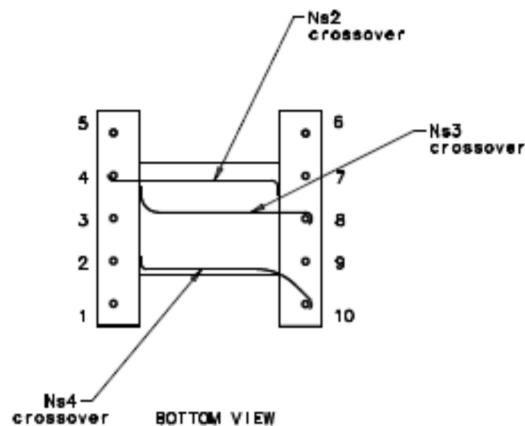
Using Item 3 (28 AWG Wire) and starting at pin 5 of bobbin, wind Ns1b (5b-6b) bifilar for 7 turns (1 layer) evenly and with no crossovers. Terminate finish leads to pin 6 of bobbin.

Using Item 10 (Tape, Kapton), wind 1 layer(s) making sure to overlap Ns1b windings by ~0.150".

Using Item 3 (28 AWG Wire), wind the balance of Np (1-2) bifilar for 10 turns (1 layer) evenly and with no crossovers. Break out finish lead to pin 2 of bobbin.

Using Item 10 (Tape, Kapton), wind 1 layer(s) making sure to overlap Np windings by ~0.150".

NOTE: During the following winding instructions, when crossing over wires, ensure that none of the crossovers overlap one another. See figure.



Pick up wire labeled #10 and crossover Ns4 (9-10) as depicted in the figure above and terminate to pin 10 of bobbin.

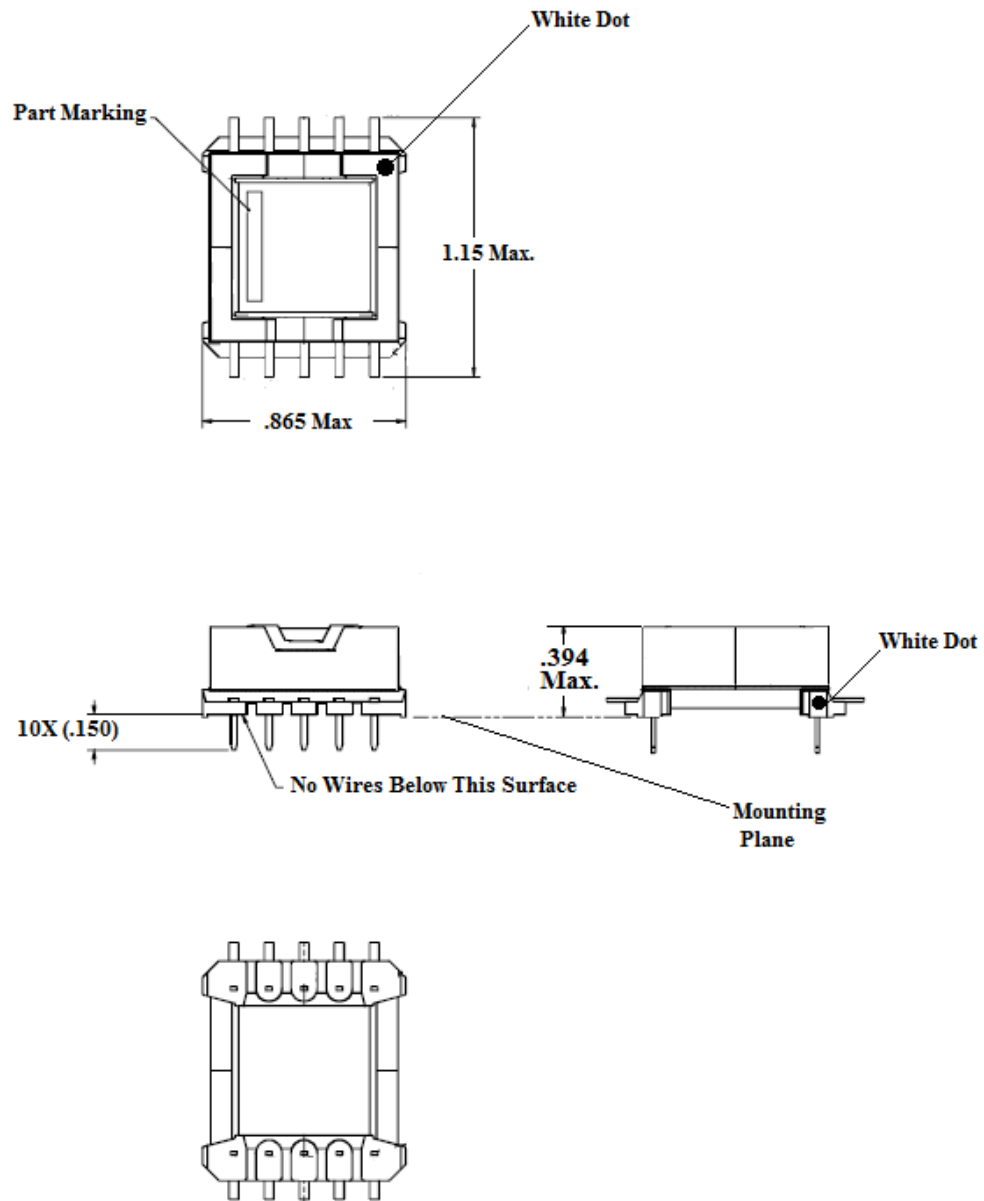
Pick up wire labeled #8 and crossover Ns3 (7-8) as depicted in the figure above and terminate to pin 8 of bobbin.

Pick up wire labeled #4 and crossover Ns2 (3-4) as depicted in the figure above and terminate to pin 4 of bobbin.

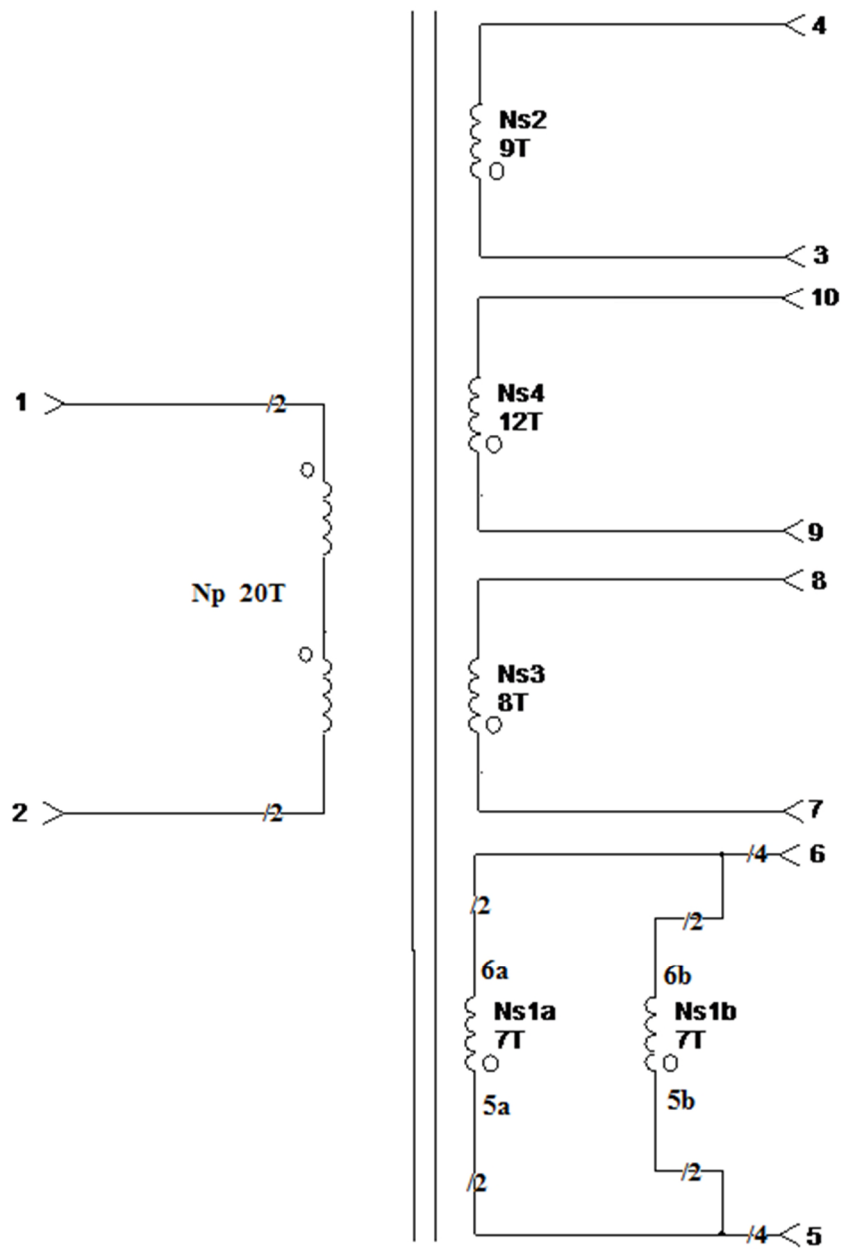
Wrap wound coil with Item 4 (Tape, Glass), 1 layer minimum, 1½ layers maximum.

NOTES:

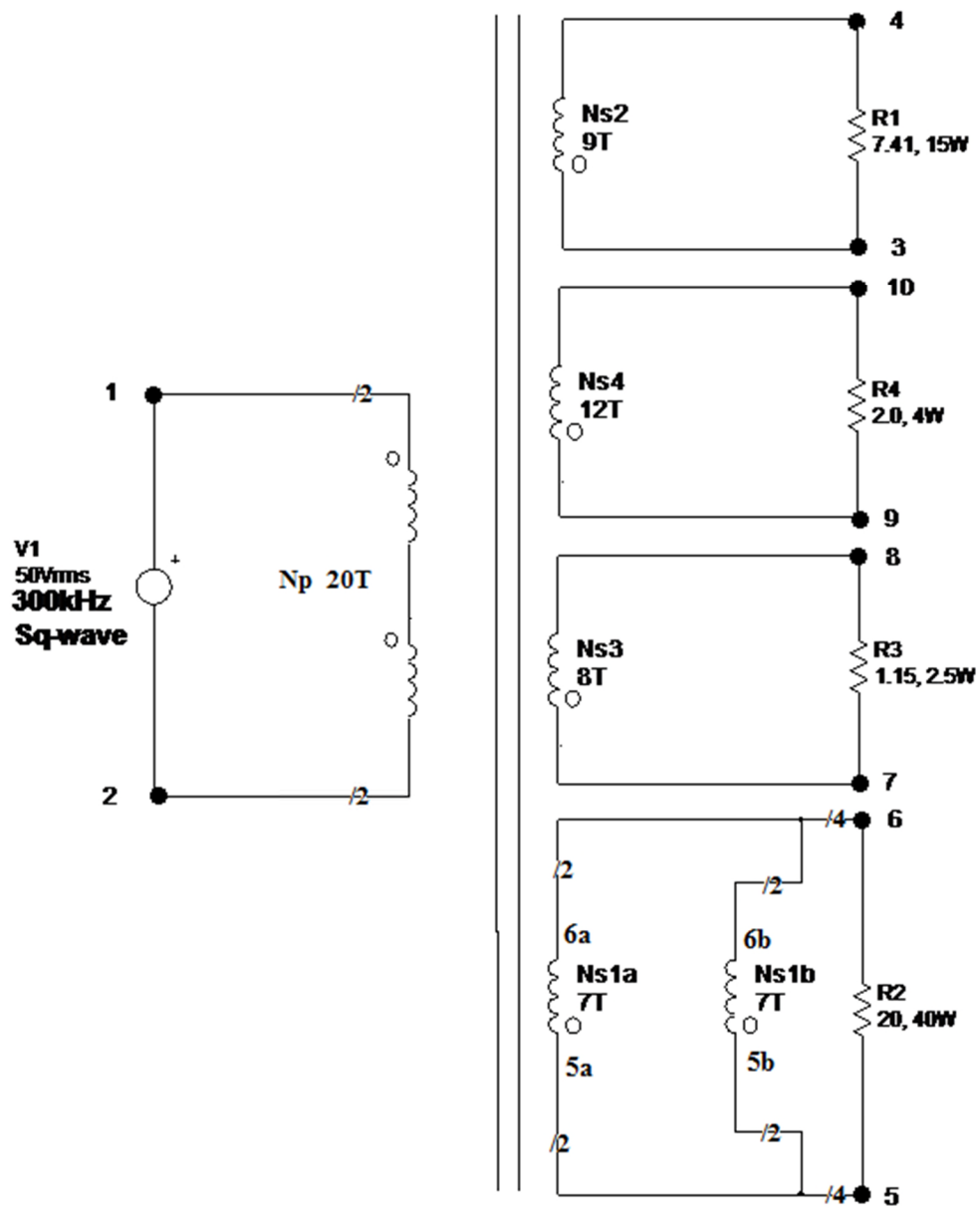
1. All materials shall be per the P11 Materials List.
2. All dimensions are in inches and include all materials used.
3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes.
4. Terminate windings Np (1-2), Ns1 (5-6), Ns2 (3-4), Ns3 (7-8) and Ns4 (9-10) to the appropriate pins using Item 8 (Solder).
5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation).
6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 5 (Core Cement). **Do not bond center leg.** Once cement cures, ensure bobbin is not cemented to core.
7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required.
8. Mark each device per 20043779 using Item 9 (Marking Ink). Using Item 9 (Marking Ink), identify pin 1 on top of core with a white dot.



P11 Outline



P11 Schematic



P11 Burn-in/Life Test Circuit

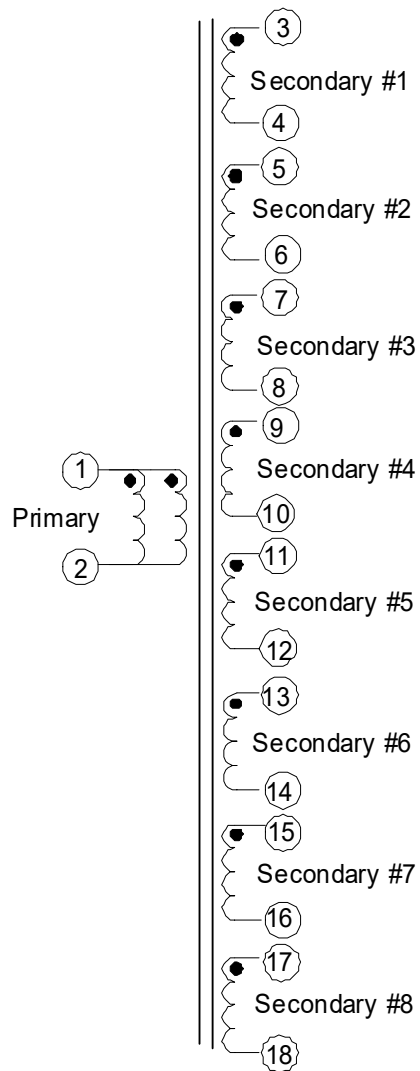
P12 Materials List			
Item	Material	Description	Suggested Source
1	Ultem 2300 (1)	Bobbin, Made to Bobbin Detail Figure	Per approved Manufacturer's Baseline
2	Wire (4)	#36 AWG MW035CHX000036* or MW016CHX000036*	Elektrisola, Essex, MWS, or REA
3	Core	EFD25-3F4	Phillips/Ferroxcube
4	Core Cement (2), (5)	Stycast 3050/Cat. 11, Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved Manufacturer's Baseline
5	Tape, Kapton	P224 or K102 per ASTM D5213	Per approved Manufacturer's Baseline
6	Impregnation	Scotchcast 280, clear	Per approved Manufacturer's Baseline
7	RTV	RTV142 or CV-1142 or DC6-1104	Per approved Manufacturer's Baseline
8	Solder	SN60 or SN63 per IPC J-STD-006	Per approved Manufacturer's Baseline
9	Terminal Pins	#26 AWG Nickel wire per MIL-STD-1276 (LW0.015N-34-N or LW0.015N-32-N)	Per approved Manufacturer's Baseline
10	Terminal Cement (3)	2280875-6 (Epon 828/Versamid-140/Cab-o-sil) or or Epon 828/Epikure 3140/Cab-o-sil	Per approved Manufacturer's Baseline
11	Ink	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline
NOTES: - Heat treat per requirements specified with bobbin detail. - Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40. - Mix ratio for Epon 828/V140/Cab-o-sil and Epon 828/Epikure 3140/Cab-o-sil is 60:40:4. - Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax. - Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.			

NOTES:

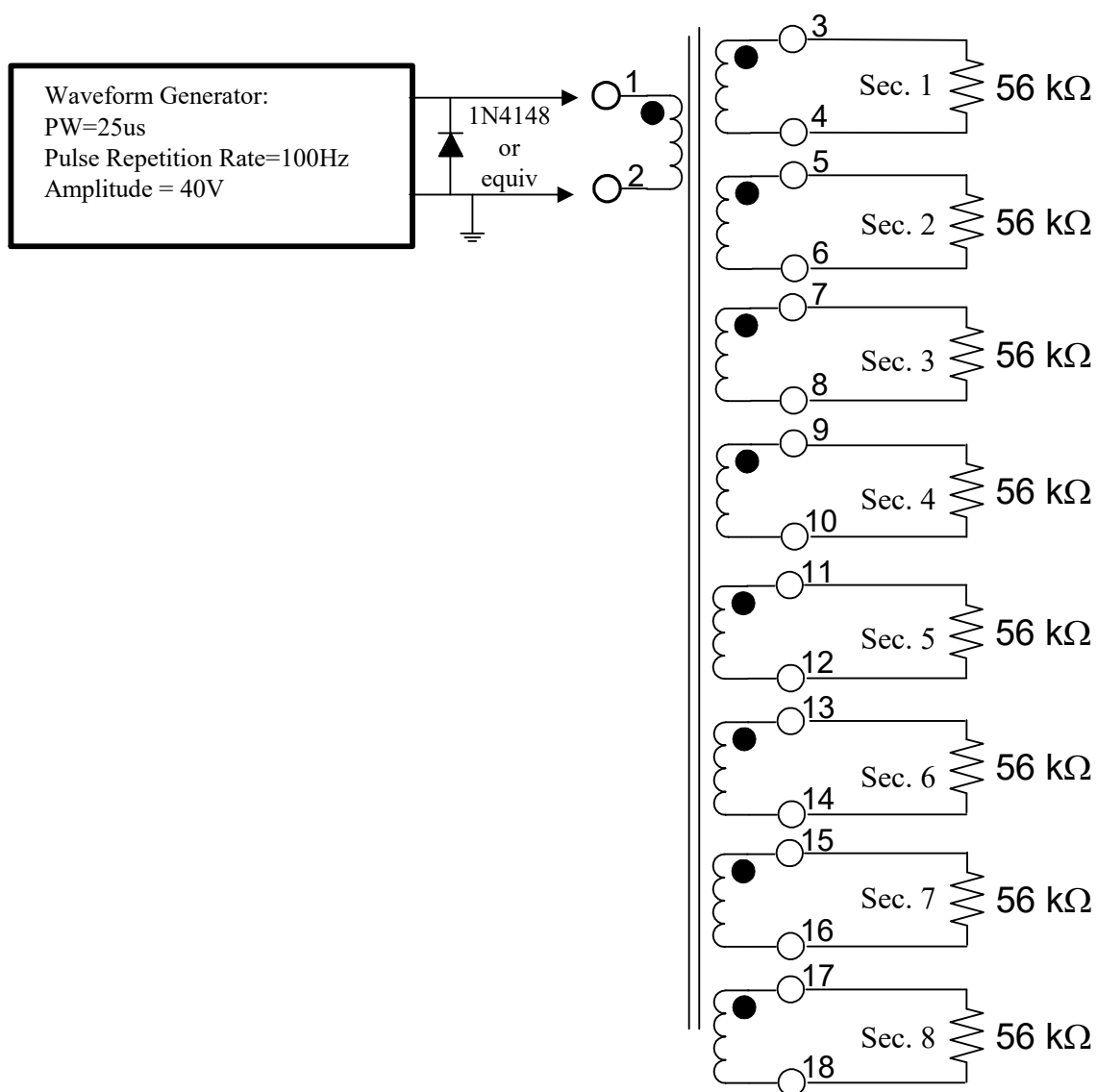
1. 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5×10^{-6} Torr., Qualifies unit to 175V peak working voltage.
2. 500 Vdc.
3. Corrected to 20°C.
4. Measure at 10 kHz, 0 Adc bias.
5. Measure at 10 kHz, 0 Adc bias. Measure at winding 1-2, with only the winding of interest shorted**For example, for leakage between Pri and Secondary 1, short pin 3 to pin 4 and record (in column under winding "Sec 1") the inductance measured between pins 1 and 2.
6. Induced voltage is required.
7. Delta DCR and Delta L shall be $\pm 3\%$.

FABRICATION SEQUENCE FOR P12		
Layer	Turns	P12 Winding Instructions
N/A	-	Prior to installing Item 9 (Terminals), solder dip the entire terminal twice using Item 8 (Solder).
N/A	-	Install Item 9 (Terminals) into Item 1 (Bobbin) using Item 10 (Terminal Cement). Ensure bobbin terminal cut out is completely filled with cement.
N/A	-	Excluding layers 1 and 2, no "winding" shall touch another winding at any point. Stagger Kapton tape insulation start and finish points to avoid winding build up.
N/A	-	Observe polarity during the winding process.
1	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 1. Minimize crossovers. Route leads to appropriate lead locations.
1.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
2	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 2. Minimize crossovers. Route leads to appropriate lead locations.
2.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
3	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 1 (3-4) layer 3. Minimize crossovers. Route leads to appropriate lead locations.
3.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
4	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 2 (5-6) layer 4. Minimize crossovers. Route leads to appropriate lead locations.
4.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
5	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 3 (7-8) layer 5. Minimize crossovers. Route leads to appropriate lead locations.
5.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
6	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 4 (9-10) layer 6. Minimize crossovers. Route leads to appropriate lead locations.
6.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
7	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 5 (11-12) layer 7. Minimize crossovers. Route leads to appropriate lead locations.
7.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
8	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 6 (13-14) layer 8. Minimize crossovers. Route leads to appropriate lead locations.
8.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
9	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 7 (15-16) layer 9. Minimize crossovers. Route leads to appropriate lead locations.

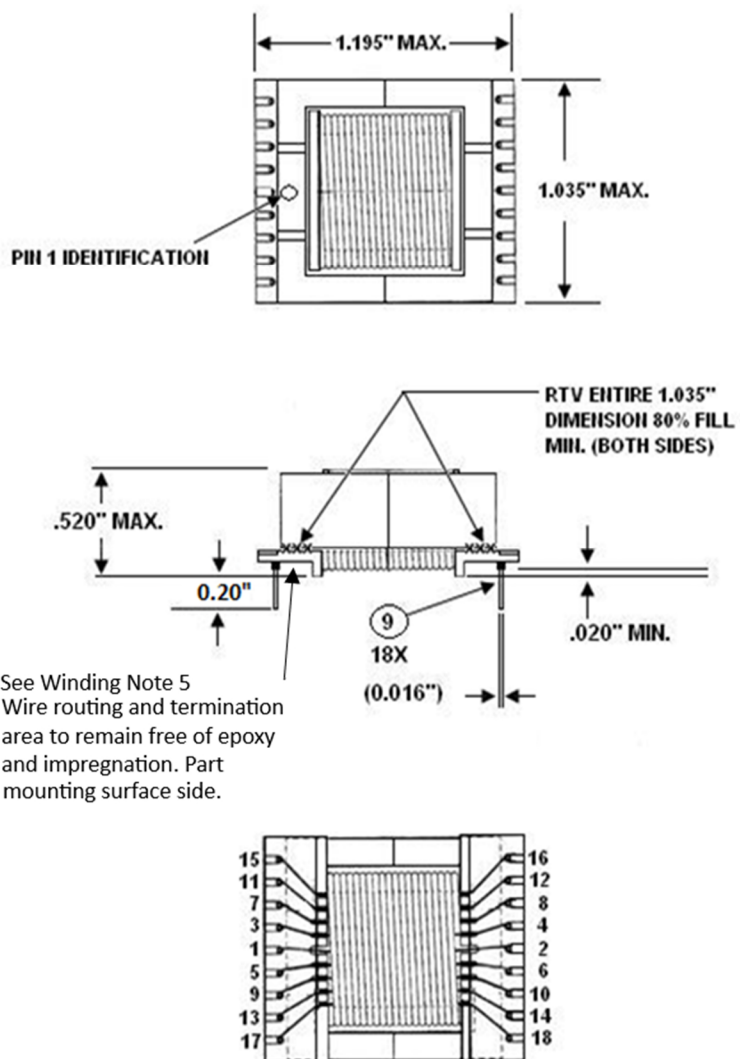
FABRICATION SEQUENCE FOR P12		
Layer	Turns	P12 Winding Instructions
9.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
10	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 8 (17-18) layer 10. Minimize crossovers. Route leads to appropriate lead locations.
10..5	1½ - 2	Final wrap winding with 1 ½ layers minimum, 2 layers maximum of Item 5 (Tape, Kapton).
P12 Assembly Instructions and Notes 1. All materials shall be per the P12 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 4. Wire routing to contain strain relief once clear of bobbin wire routing passage (Do not go straight from bobbin lead passage to pin). Pin termination to contain one insulated turn first wrap closest to bobbin and two uninsulated turns above. Terminate windings to the appropriate pins using item 8 (Solder). Ensure a gap of 0.020" is maintained between the solder joint and the bobbin's PCB standoffs. 5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation). Keep the area from the wire routing passage to the bobbin pins clear of impregnation (See Outline Figure). 6. Place coil into Item 3 (Core). Cement Item 3 (Core) together using Item 4 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 20043779 using Item 11 (Marking Ink). Using Item 11 (Marking Ink), identify pin 1 on top of core with a white dot. 9. ULTEM BOBBIN IS SOLUTION SENSITIVE. DO NOT CLEAN BOBBIN WITH ACETONE.		



P12 Schematic

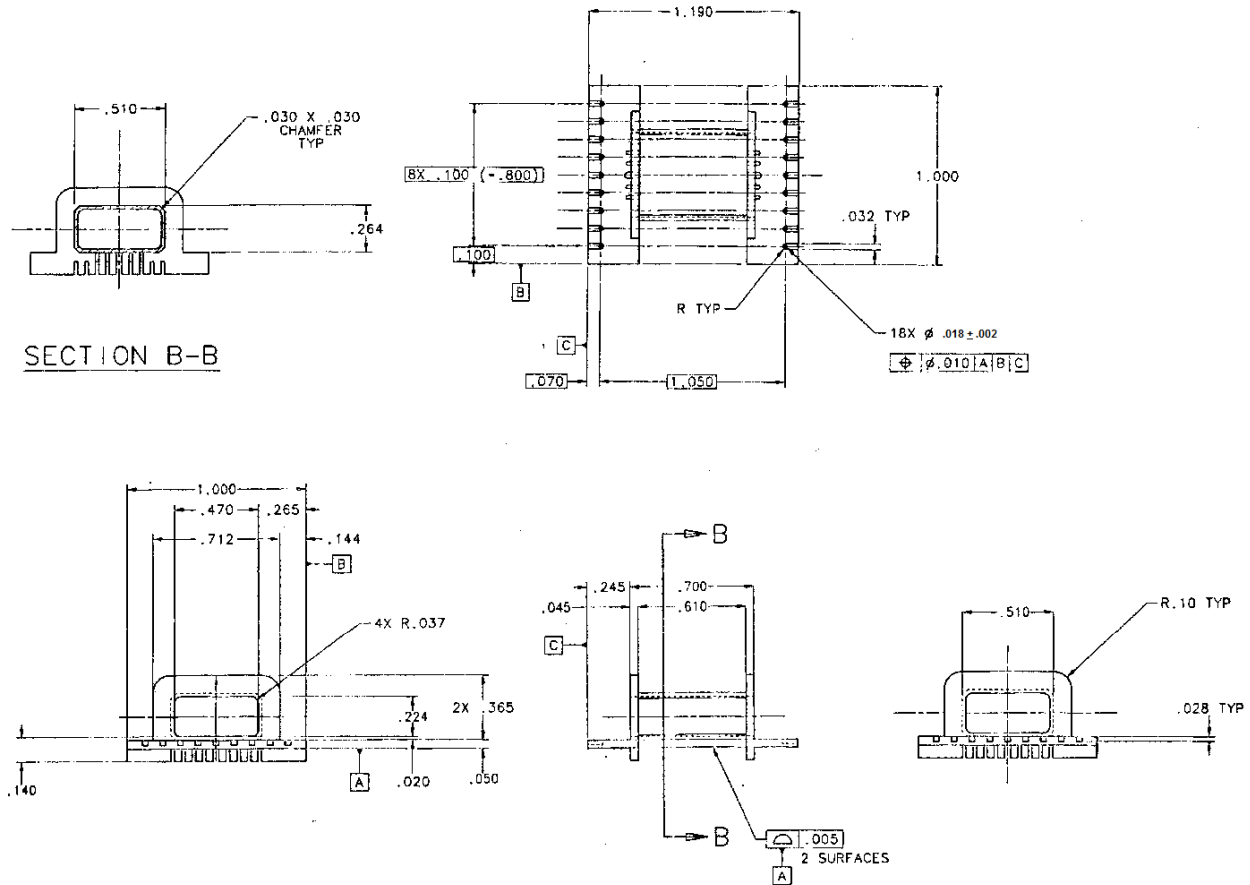


P12 Burn-in/Life Test Circuit



TOLERANCE (UNLESS OTHERWISE SPECIFIED)
 3 DECIMAL PLACES = $\pm .005"$
 2 DECIMAL PLACES = $\pm .010"$

P12 Outline



Bobbin Annealing Requirements

After final machining,

Bobbin shall be annealed to 300°F at a maximum heating rate of 20°F per hour.

Hold oven temperature at 300°F for 1 ½ hours.

Increase oven temperature to 400°F at a maximum heating rate of 20°F per hour.

Hold oven temperature at 400°F for 2 ½ hours.

Oven cool to 150°F over 25 hours with a maximum cool rate of 10°F per hour.

Turn off oven and allow to cool to room temperature with oven door closed before removing parts.

Tolerance

(Unless Otherwise Specified)

2 Decimal Places = ± 0.010 "

P12 Bobbin Detail

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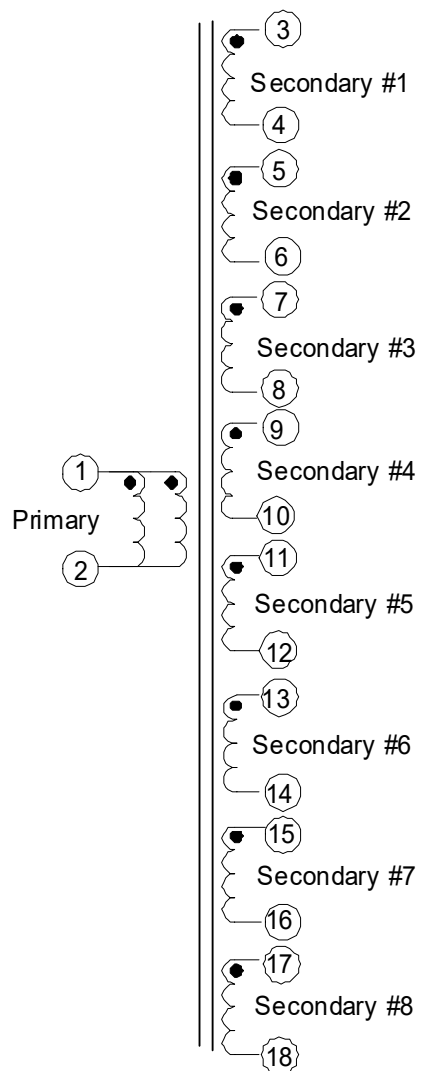
Break between Non-Torr and Torr Requirements Magnetics

P503 Materials List			
Item	Material	Description	Suggested Source
1	Ultem 2300 (1)	Bobbin, Made to Bobbin Detail Figure	Per approved Manufacturer's Baseline
2	Wire (2)	#36 AWG MW035CHX000036* or MW016CHX000036*	Elektrisola, Essex, MWS, or REA
3	Core	EFD25-3F4 or EFD25/13/9	Phillips/Ferroxcube
4	Core Cement (3)	Epon 828/V140, RE2039/HD0243 or Epon 828/Epikure 3140	Per approved Manufacturer's Baseline
5	Tape, Kapton	P224 or K102 per ASTM D5213	Per approved Manufacturer's Baseline
6	Impregnation	Scotchcast 280, clear	Per approved Manufacturer's Baseline
7	RTV	RTV142 or CV-1142 or DC6-1104	Per approved Manufacturer's Baseline
8	Solder	SN60 or SN63 per IPC J-STD-006	Per approved Manufacturer's Baseline
9	Terminal Pins	#26 AWG Nickel wire per MIL-STD-1276 (LW0.015N-34-N or LW0.015N-32-N)	Per approved Manufacturer's Baseline
10	Terminal Cement (4)	2280875-6 (Epon 828/Versamid-140/Cab-o-sil) or Epon 828/Epikure 3140/Cab-o-sil	Per approved Manufacturer's Baseline
11	Ink	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline
NOTES: 1. Heat treat per requirements specified with bobbin detail. 2. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax. 3. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40. 4. Mix ratio for Epon 828/V140/Cab-o-sil and Epon 828/Epikure 3140/Cab-o-sil is 60:40:4.			

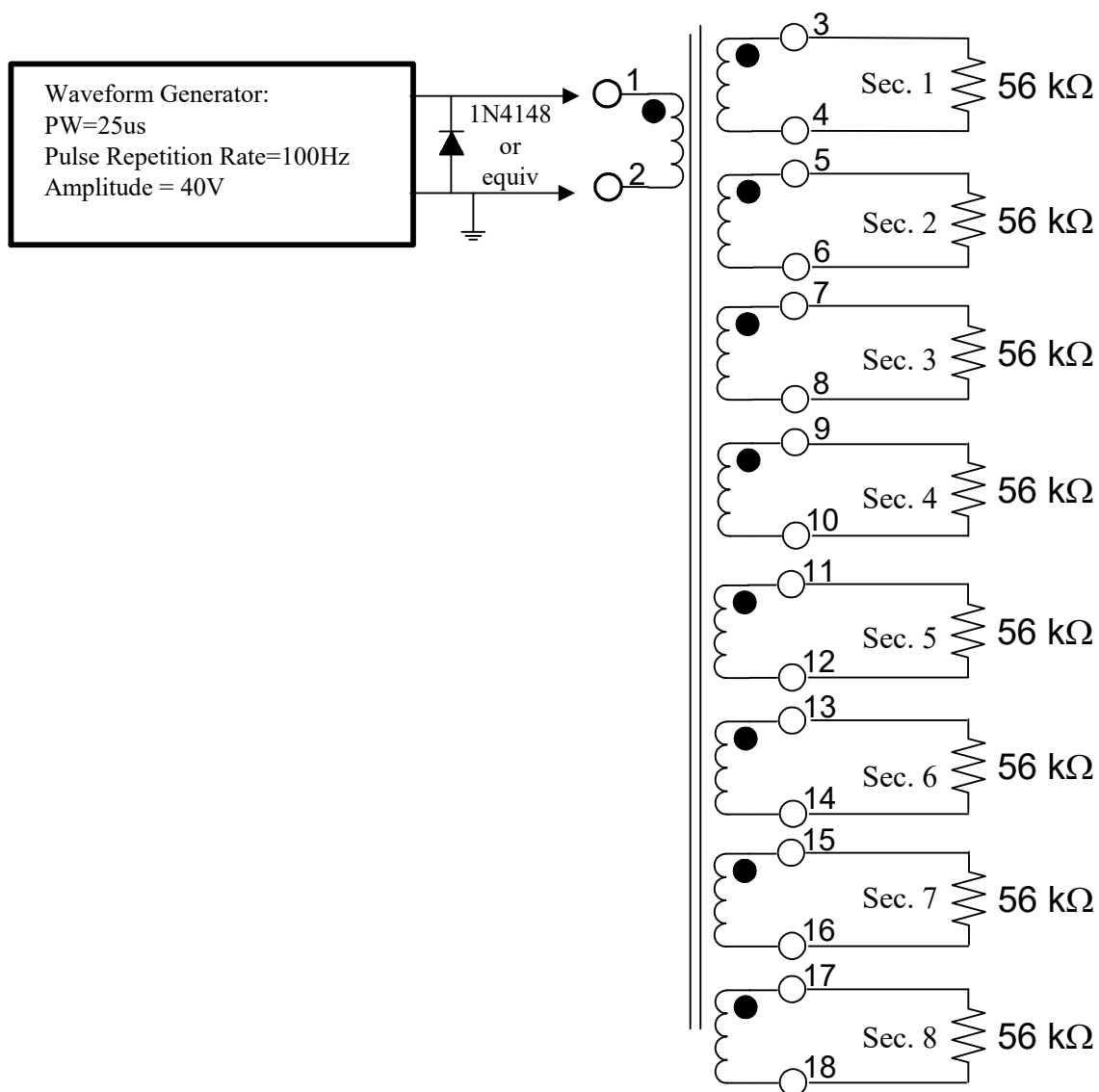
NOTES:

FABRICATION SEQUENCE FOR P503		
Layer	Turns	P503 Winding Instructions
N/A	-	Prior to installing Item 9 (Terminals), solder dip the entire terminal twice using Item 8 (Solder).
N/A	-	Install Item 9 (Terminals) into Item 1 (Bobbin) using Item 10 (Terminal Cement). Ensure bobbin terminal cut out is completely filled with cement.
N/A	-	Excluding layers 1 and 2, no "winding" shall touch another winding at any point. Stagger Kapton tape insulation start and finish points to avoid winding build up.
N/A	-	Observe polarity during the winding process.
1	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 1. Minimize crossovers. Route leads to appropriate lead locations.
1.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
2	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 2. Minimize crossovers. Route leads to appropriate lead locations.
2.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
3	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 1 (3-4) layer 3. Minimize crossovers. Route leads to appropriate lead locations.
3.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
4	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 2 (5-6) layer 4. Minimize crossovers. Route leads to appropriate lead locations.
4.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
5	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 3 (7-8) layer 5. Minimize crossovers. Route leads to appropriate lead locations.
5.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
6	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 4 (9-10) layer 6. Minimize crossovers. Route leads to appropriate lead locations.
6.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
7	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 5 (11-12) layer 7. Minimize crossovers. Route leads to appropriate lead locations.
7.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
8	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 6 (13-14) layer 8. Minimize crossovers. Route leads to appropriate lead locations.
8.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
9	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 7 (15-16) layer 9. Minimize crossovers. Route leads to appropriate lead locations.

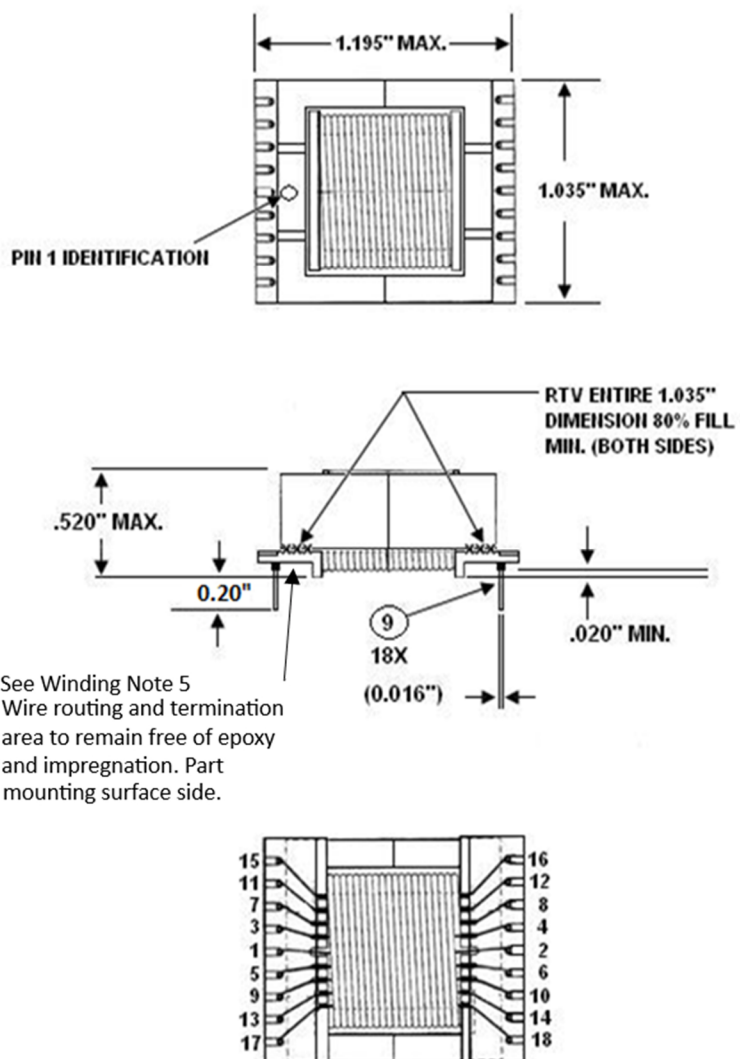
FABRICATION SEQUENCE FOR P503		
Layer	Turns	P503 Winding Instructions
9.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
10	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 8 (17-18) layer 10. Minimize crossovers. Route leads to appropriate lead locations.
10..5	1½ - 2	Final wrap winding with 1 ½ layers minimum, 2 layers maximum of Item 5 (Tape, Kapton).
Assembly Instructions and Notes 1. All materials shall be per the P503 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 4. Wire routing to contain strain relief once clear of bobbin wire routing passage (Do not go straight from bobbin lead passage to pin). Pin termination to contain one insulated turn first wrap closest to bobbin and two uninsulated turns above. Terminate windings to the appropriate pins using item 8 (Solder). Ensure a gap of 0.020" is maintained between the solder joint and the bobbin's PCB standoffs. 5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation). Keep the area from the wire routing passage to the bobbin pins clear of impregnation (See Outline Figure). 6. Place coil into Item 3 (Core). Cement Item 3 (Core) together using Item 4 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 3GPDE001 using Item 11 (Marking Ink). Using Item 11 (Marking Ink), identify pin 1 on top of core with a white dot. 9. ULTEM BOBBIN IS SOLUTION SENSITIVE. DO NOT CLEAN BOBBIN WITH ACETONE.		



P503 Schematic



P503 Burn-in/Life Test Circuit



TOLERANCE (UNLESS OTHERWISE SPECIFIED)
 3 DECIMAL PLACES = $\pm .005"$
 2 DECIMAL PLACES = $\pm .010"$

P503 Outline



Turn off oven and allow to cool to room temperature with oven door closed before removing parts.

2 Decimal Places = $\pm 0.010''$

P503 Bobbin Detail

P512 Materials List			
Item	Material	Description	Suggested Source
1	Ultem 2300 (1)	Bobbin, Made to Bobbin Detail Figure	Per approved Manufacturer's Baseline
2	Wire (4)	#36 AWG MW035CHX000036* or MW016CHX000036*	Elektrisola, Essex, MWS, or REA
3	Core	EFD25-3F4	Phillips/Ferroxcube
4	Core Cement (2), (5)	Stycast 3050/Cat. 11 or Epon 828/V140 or RE2039/HD0243 or Epon 828/Epikure 3140	Per approved Manufacturer's Baseline
5	Tape, Kapton	P224 or K102 per ASTM D5213	Per approved Manufacturer's Baseline
6	Impregnation	Scotchcast 280, clear	Per approved Manufacturer's Baseline
7	RTV	RTV142 or CV-1142 or DC6-1104	Per approved Manufacturer's Baseline
8	Solder	SN60 or SN63 per IPC J-STD-006	Per approved Manufacturer's Baseline
9	Terminal Pins	#26 AWG Nickel wire per MIL-STD-1276 (LW0.015N-34-N or LW0.015N-32-N)	Per approved Manufacturer's Baseline
10	Terminal Cement (3)	2280875-6 (Epon 828/Versamid-140/Cab-o-sil) or Epon 828/Epikure 3140/Cab-o-sil	Per approved Manufacturer's Baseline
11	Ink	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's Baseline

NOTES:

1. Heat treat per requirements specified with bobbin detail.
2. Mix ratio for Epon 828/V140 and Epon 828/Epikure 3140 is 60:40.
3. Mix ratio for Epon 828/V140/Cab-o-sil and Epon 828/Epikure 3140/Cab-o-sil is 60:40:4.
4. Acceptable magnet wire lubricant designations are "N" – No Lubricant and "P" Paraffin Wax.
5. Stycast 3050/Cat. 11 no longer an acceptable core cement. Left for historical purposes only.

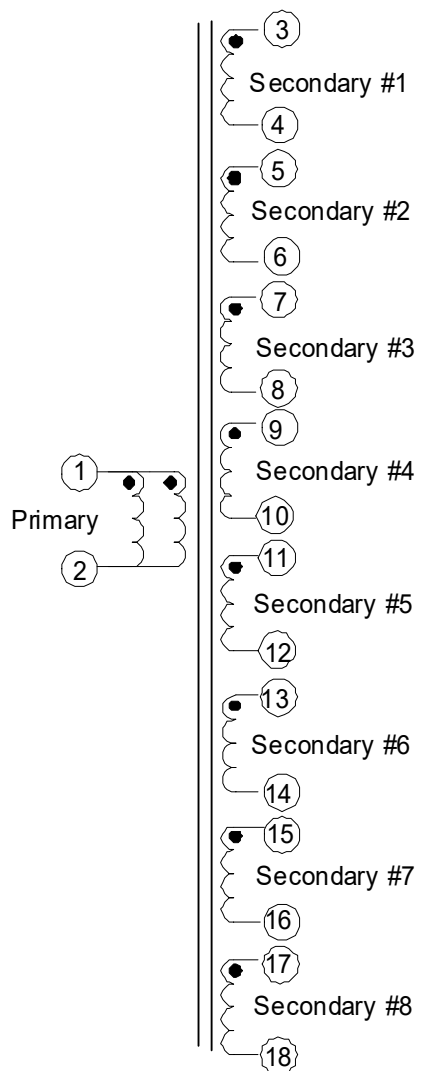
P512 Electrical Characteristics										
Winding #	Primary		Sec 1	Sec 2	Sec 3	Sec 4	Sec 5	Sec 6	Sec 7	Sec 8
Terminals (start-finish)	1-2		3-4	5-6	7-8	9-10	11-12	13-14	15-16	17-18
Winding Layer #	1	2	3	4	5	6	7	8	9	10
Number of turns	95	95	95	95	95	95	95	95	95	95
Wire Gauge	36	36	36	36	36	36	36	36	36	36
Maximum operating current (A _{RMS})	<1mA		<1mA	<1mA	<1mA	<1mA	<1mA	<1mA	<1mA	<1mA
Working Voltage (Peak)	175V		175V	175V	175V	175V	175V	175V	175V	175V
Dielectric withstanding voltage 1/	No Breakdown									
Insulation Resistance 2/	10, 000MΩ									
DC resistance (Ω, ±10%) 3/, <u>7</u> /	2.62		5.50	5.60	5.80	6.00	6.15	6.35	6.50	6.65
Inductance (mH, +66% / -0%) 4/, <u>7</u> /	6.8		6.8	6.8	6.8	6.8	6.8	6.8	6.8	6.8
Leakage Inductance (Max.) (μH) 4/, <u>5</u> /	N/A		11.4	19.2	27.6	34.5	39.6	47.3	55	62.7
Rated Voltage <u>6</u> /	40V		40V	40V	40V	40V	40V	40V	40V	40V

NOTES:

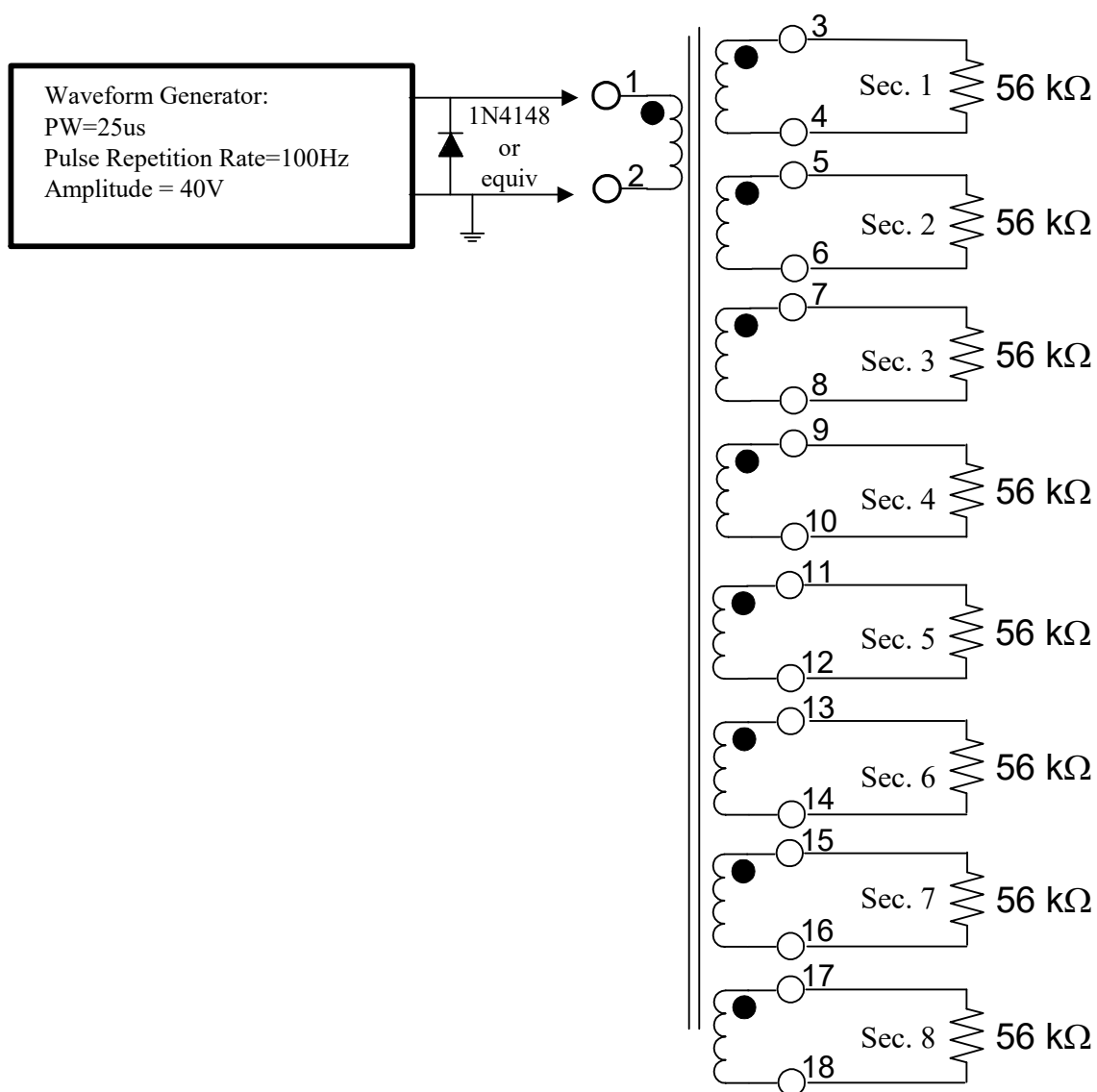
1. 500 Vrms, 60 Hz at any pressure from one atmosphere to less than 5×10^{-6} Torr., Qualifies unit to 175V peak working voltage.
2. 500 Vdc.
3. Corrected to 20°C.
4. Measure at 10 kHz, 0 Adc bias.
5. Measure at 10 kHz, 0 Adc bias. Measure at winding 1-2, with only the winding of interest shorted**For example, for leakage between Pri and Secondary 1, short pin 3 to pin 4 and record (in column under winding "Sec 1") the inductance measured between pins 1 and 2.
6. Induced voltage is required.
7. Delta DCR and Delta L shall be $\pm 3\%$.

FABRICATION SEQUENCE FOR P512		
Layer	Turns	P512 Winding Instructions
N/A	-	Prior to installing Item 9 (Terminals), solder dip the entire terminal twice using Item 8 (Solder).
N/A	-	Install Item 9 (Terminals) into Item 1 (Bobbin) using Item 10 (Terminal Cement). Ensure bobbin terminal cut out is completely filled with cement.
N/A	-	Excluding layers 1 and 2, no "winding" shall touch another winding at any point. Stagger Kapton tape insulation start and finish points to avoid winding build up.
N/A	-	Observe polarity during the winding process.
1	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 1. Minimize crossovers. Route leads to appropriate lead locations.
1.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
2	95	Using Item 2 (36 AWG Wire), wind 95 turns of primary (1-2) layer 2. Minimize crossovers. Route leads to appropriate lead locations.
2.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
3	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 1 (3-4) layer 3. Minimize crossovers. Route leads to appropriate lead locations.
3.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
4	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 2 (5-6) layer 4. Minimize crossovers. Route leads to appropriate lead locations.
4.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
5	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 3 (7-8) layer 5. Minimize crossovers. Route leads to appropriate lead locations.
5.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
6	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 4 (9-10) layer 6. Minimize crossovers. Route leads to appropriate lead locations.
6.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
7	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 5 (11-12) layer 7. Minimize crossovers. Route leads to appropriate lead locations.
7.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
8	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 6 (13-14) layer 8. Minimize crossovers. Route leads to appropriate lead locations.
8.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
9	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 7 (15-16) layer 9. Minimize crossovers. Route leads to appropriate lead locations.

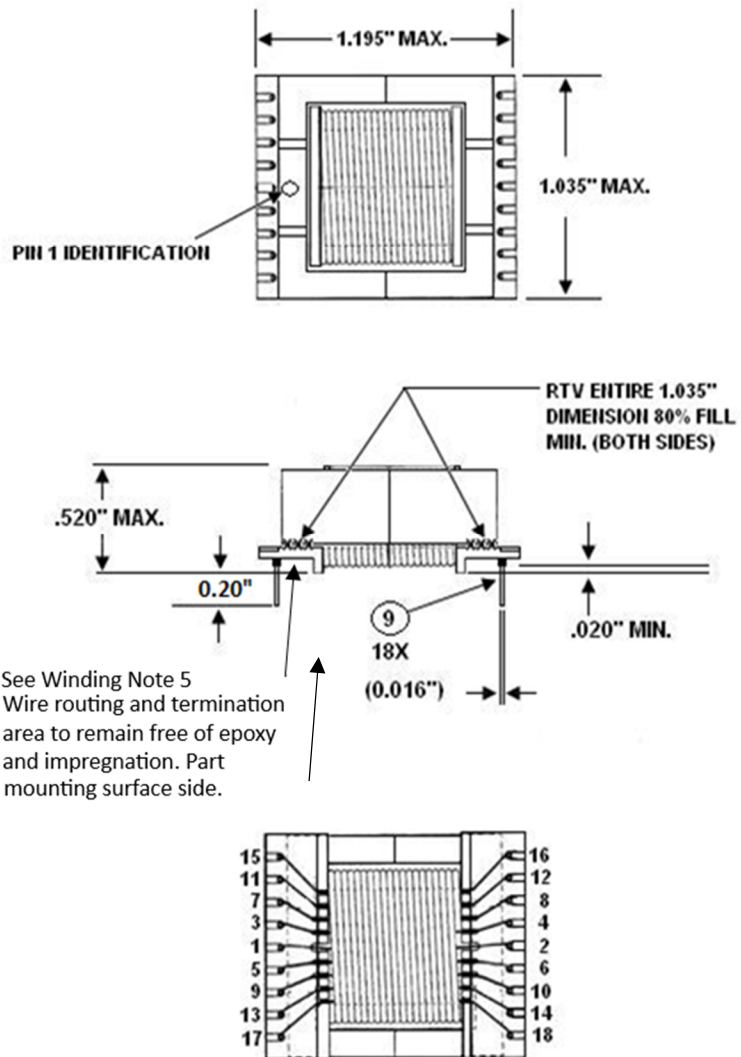
FABRICATION SEQUENCE FOR P512		
Layer	Turns	P512 Winding Instructions
9.5	1	Wrap winding with a single layer of Item 5 (Tape, Kapton). Finish shall overlap start between 0.25" to 0.50".
10	95	Using Item 2 (36 AWG Wire), wind 95 turns of Sec 8 (17-18) layer 10. Minimize crossovers. Route leads to appropriate lead locations.
10..5	1½ - 2	Final wrap winding with 1 ½ layers minimum, 2 layers maximum of Item 5 (Tape, Kapton).
Assembly Instructions and Notes 1. All materials shall be per the P512 Materials List. 2. All dimensions are in inches and include all materials used. 3. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetic in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 4. Wire routing to contain strain relief once clear of bobbin wire routing passage (Do not go straight from bobbin lead passage to pin). Pin termination to contain one insulated turn first wrap closest to bobbin and two uninsulated turns above. Terminate windings to the appropriate pins using item 8 (Solder). Ensure a gap of 0.020" is maintained between the solder joint and the bobbin's PCB standoffs. 5. Vacuum impregnate wound bobbin assembly using Item 6 (Impregnation). Keep the area from the wire routing passage to the bobbin pins clear of impregnation (See Outline Figure). 6. Place coil into Item 3 (Core). Cement Item 3 (Core) together using Item 4 (Core Cement). Do not bond center leg. Once cement cures insure bobbin is not cemented to core. 7. Stake the bottom bobbin to core interface using Item 7 (RTV). 80% minimum fill of interface required. 8. Mark each device per 3GPDE001 using Item 11 (Marking Ink). Using Item 11 (Marking Ink), identify pin 1 on top of core with a white dot. 9. ULTEM BOBBIN IS SOLUTION SENSITIVE. DO NOT CLEAN BOBBIN WITH ACETONE.		



P512 Schematic

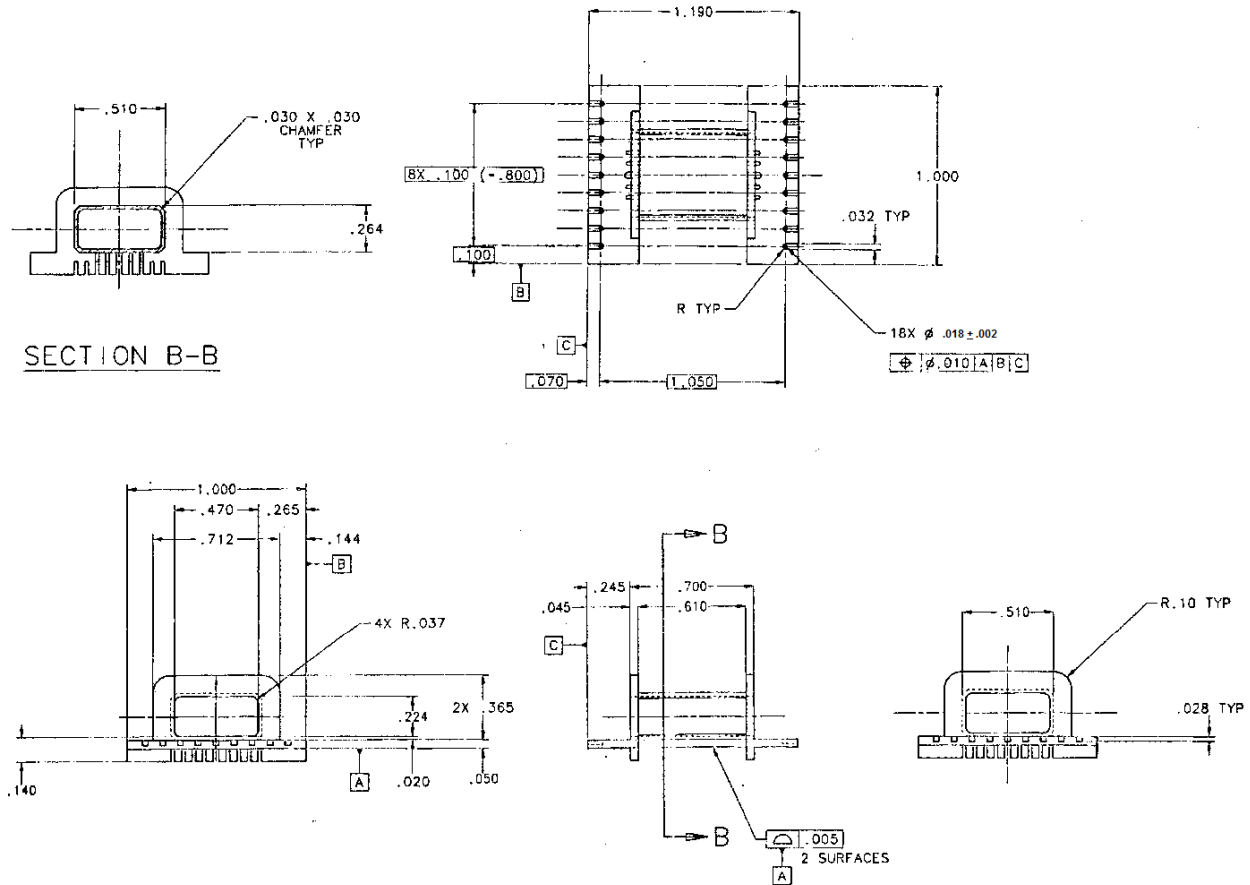


P512 Burn-in/Life Test Circuit



TOLERANCE (UNLESS OTHERWISE SPECIFIED)
 3 DECIMAL PLACES = $\pm .005"$
 2 DECIMAL PLACES = $\pm .010"$

P512 Outline



Bobbin Annealing Requirements

After final machining,

Bobbin shall be annealed to 300°F at a maximum heating rate of 20°F per hour.

Hold oven temperature at 300°F for 1 ½ hours.

Increase oven temperature to 400°F at a maximum heating rate of 20°F per hour.

Hold oven temperature at 400°F for 2 ½ hours.

Oven cool to 150°F over 25 hours with a maximum cool rate of 10°F per hour.

Turn off oven and allow to cool to room temperature with oven door closed before removing parts.

Tolerance

(Unless Otherwise Specified)

2 Decimal Places = $\pm$.010"

P512 Bobbin Detail

P515 Materials List

Item	Material	Description	Source
1	Core (1)	EFD25/13/9-3F3	Ferroxcube
2	Bobbin (2)	CSH-EFD25-1S-10P-IZ	Ferroxcube
3	Wire (2)	#28 AWG MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Wire (3)	#29 AWG MW035CHX000029* or MW016CHX000029*	Elektrisola, Essex, MWS, or REA
5	Wire (3)	#30 AWG MW035CHX000030* or MW016CHX000030*	Elektrisola, Essex, MWS, or REA
6	Tape, Kapton	P224 or K102 per ASTM D5213	Per approved Manufacturer's baseline
7	Tape, Glass	79 Glass per A-A-59770	Per approved Manufacturer's baseline
8	Impregnation	Scotchcast 280	Per approved Manufacturer's baseline
9	Core Glue (3)	RE2039/HD0243 or Epon 828/V140 or Epon 828/Epikure 3140	Per approved Manufacturer's baseline
10	Staking (Bobbin/Core)	RTV142 or CV-1142 or DC6-1104	Per approved Manufacturer's baseline
11	Solder	SN60 or SN63 per IPC J-STD-006	Per approved Manufacturer's baseline
12	Ink/Catalyst	6050 or Markem 7224 or Markem 7906 or M-0-NC or M-9-N/Cat 77 or B3 per AA-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's baseline
Notes - Grind center legs of cores to achieve N1 inductance ($Al \approx 110nH$). - Prior to winding, terminal pins shall be solder dipped twice in their entirety. - Approved magnet wire lubricant - * = "P" or "N". - Epon 828/V140 and Epon 828/Epikure 3140 Mix Ratio: 60:40 PBW.			

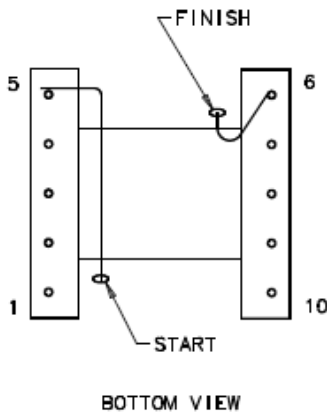
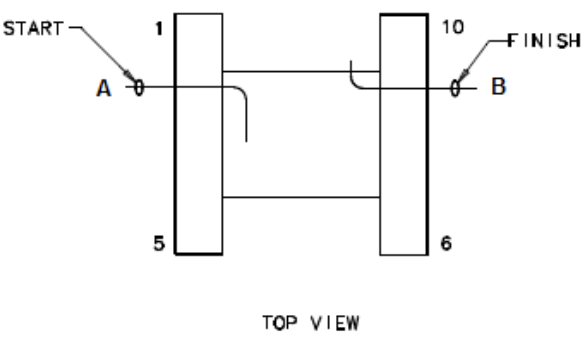
P515 Electrical Characteristics

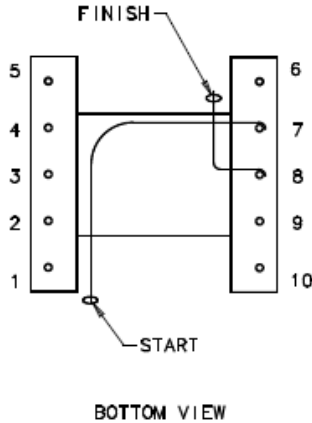
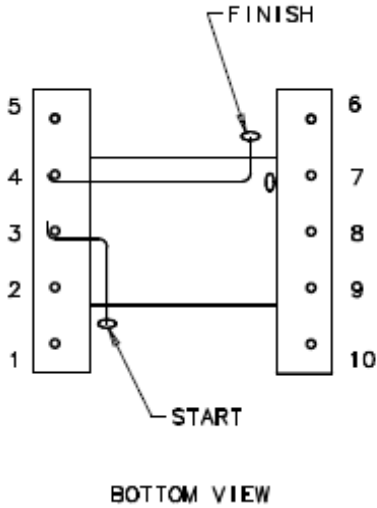
Characteristic	Limits	Limits	Limits	Limits	Limits	Limits
Winding Designation (start-finish)	N1 (5-6)	N2 (A-B)	N3 (7-8)	N4 (3-4)	N5 (9-10)	N6 (1-2)
Turns	18	7	17	17	17	29
Wire (AWG) X Strands	28 X 2	28 X 5	29 X 2	28 X 2	30 X 2	30 X 1
Maximum Operating Current	1.4 Arms	1.80 Arms	0.30 Arms	0.60 Arms	0.10 Arms	0.15 Arms
Working Voltage (RMS)	175V	175V	175V	175V	175V	175V
Dielectric Withstanding Voltage (1)	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (2)	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.
DC Resistance Max. (3)	82 m Ω	18 m Ω	110 m Ω	90 m Ω	150 m Ω	500 m Ω
Inductance ($\pm 4\%$) (4)	35.6 μ H (8)	5.4 μ H	32.0 μ H	32.0 μ H	32.0 μ H	93 μ H
Leakage Inductance (Max.) (5)	500 nH	Shorted	Shorted	Shorted	Shorted	Shorted
Turns Ratio (6)	Reference	0.389 $\pm 2.0\%$	0.944 $\pm 2.0\%$	0.944 $\pm 2.0\%$	0.944 $\pm 2.0\%$	1.61 $\pm 2.0\%$
Inductance Delta Limits Percent (7)	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$
Resistance Delta Limits Percent (7)	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%

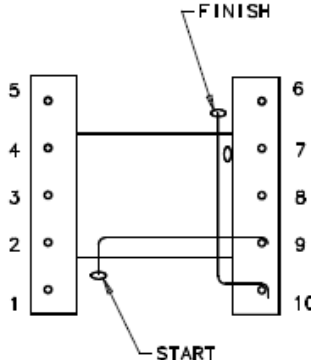
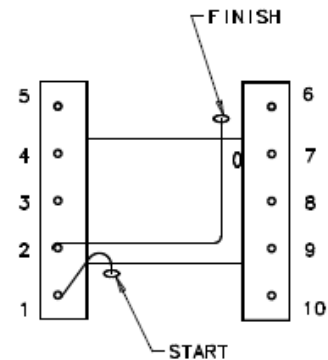
Notes:

1. 500Vrms, 60 Hz, at 28 to 32 inches of Hg qualifies unit to 175V peak working voltage.
2. 500Vdc shall be applied between each winding and the core with all other windings grounded to the core
3. Corrected to +20°C.
4. Measured at 1.0 Vrms, 100 KHz, 0 Adc and all other windings open.
5. Measured at 100 KHz. "Shorted" means all other windings shorted start to finish and shorted together.
6. Turns ratio data may be taken during the assembly process, prior to cementing cores.
7. Delta limits apply to burn-in and life test.
8. Grind center legs of cores symmetrical to achieve N1 inductance.

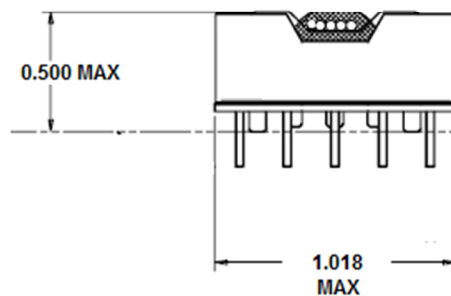
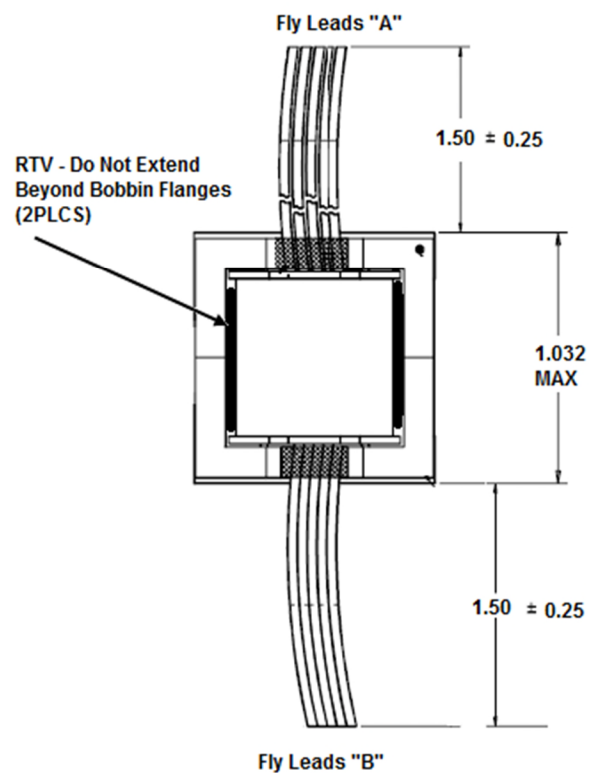
P515 Winding Instructions and Notes

P515	
Layers	Winding Instructions
-	OVERALL HEIGHT MAGNETIC IS VERY CRITICAL. WIND AS TIGHTLY AS POSSIBLE AND MAINTAIN WIRE AND LEAD TERMINATIONS BELOW BOBBIN STANDOFFS.
-	Observe polarity per schematic diagram.
-	Prior to winding, solder dip bobbin pins in their entirety twice using Item 11 (Solder).
-	Winding Item 2 (Bobbin) "Top Going" with pins 1 to 5 against the winding machine. Unless otherwise stated all windings shall be wound left to right.
1	Wind 18 turns of Winding "N1" (5-6) bi-filar, using Item 3 (Wire, Magnet 28AWG) spread evenly across Item 2 (Bobbin) in a single layer starting at pin 5 and finishing at pin 6, no crossovers allowed. See view below:  BOTTOM VIEW Note: - Wire from the winding window to the pins shall be contained in the bobbin slots provided for such wires. Wire wraps on the pins and termination solder shall be confined to the height of the bobbin standoffs so that when the finished assembly is mounted on the printed wire board, the assembly sits on the standoffs and does not exceed a total maximum height of 0.500".
2	Starting on the top of the bobbin pin 1 side, wind 7 turns of Windings "N2" (A-B) five filar, using Item 3 (Wire, Magnet 28 AWG) spread evenly across Item 2 (Bobbin) in a single layer no crossovers allowed and finishing on pin 10 side. Flying leads shall exit the bobbin in the slots provided for fly leads.  TOP VIEW

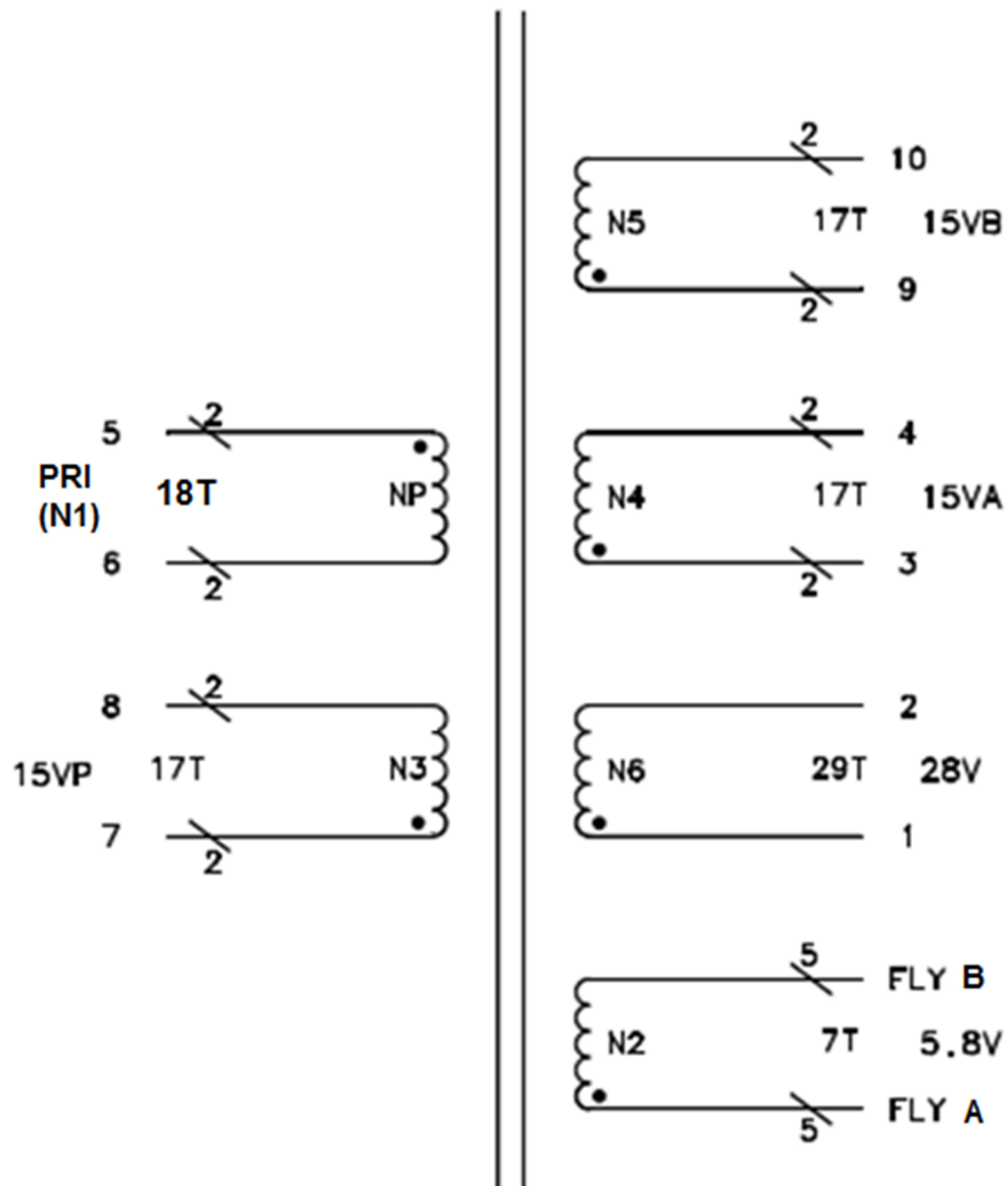
P515	
Layers	Winding Instructions
3	Begin winding "N3" by starting at pin 7 and crossing over to the left side of the bobbin as pictured below. Wind 17 turns of Winding "N3" (7-8) bi-filar, using Item 4 (Wire, Magnet 29AWG) spread evenly across Item 2 (Bobbin) in a single layer finishing at pin 8, no other crossovers allowed  Note: - Wire from the winding window to the pins shall be contained in the bobbin slots provided for such wires. Wire wraps on the pins and termination solder shall be confined to the height of the bobbin standoffs so that when the finished assembly is mounted on the printed wire board, the assembly sits on the standoffs and does not exceed a total maximum height of 0.500".
4	Starting at pin 3, wind 17 turns of Winding "N4" (3-4) bi-filar, using Item 4 (Wire, Magnet 28AWG) spread evenly across Item 2 (Bobbin) in a single layer crossing the finish backover to pin 4. No other crossovers allowed. See view below:  Note: - Wire from the winding window to the pins shall be contained in the bobbin slots provided for such wires. Wire wraps on the pins and termination solder shall be confined to the height of the bobbin standoffs so that when the finished assembly is mounted on the printed wire board, the assembly sits on the standoffs and does not exceed a total maximum height of 0.500".
5	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10" overlap.

P515	
Layers	Winding Instructions
6	Begin winding "N5" by starting at pin 9 and crossing over to the left side of the bobbin as pictured below. Wind 17 turns of Winding "N5" (9-10) bi-filar, using Item 5 (Wire, Magnet 30AWG) spread evenly across Item 2 (Bobbin) in a single layer finishing at pin 10, no other crossovers allowed.  BOTTOM VIEW Note: - Wire from the winding window to the pins shall be contained in the bobbin slots provided for such wires. Wire wraps on the pins and termination solder shall be confined to the height of the bobbin standoffs so that when the finished assembly is mounted on the printed wire board, the assembly sits on the standoffs and does not exceed a total maximum height of 0.500".
7	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10" overlap.
8	Wind 29 turns of Winding "N6" (1-2) single filar, using Item 5 (Wire, Magnet 30AWG) spread evenly across Item 2 (Bobbin) in a single layer starting at pin 1 and crossing back over per view below to pin 2, no other crossovers allowed.  BOTTOM VIEW Note: - Wire from the winding window to the pins shall be contained in the bobbin slots provided for such wires. Wire wraps on the pins and termination solder shall be confined to the height of the bobbin standoffs so that when the finished assembly is mounted on the printed wire board, the assembly sits on the standoffs and does not exceed a total maximum height of 0.500".
9	Final wrap with 1 layer minimum of Item 7 (Tape, Glass). Verify that all windings including leads, wire and tape are within the maximum dimensions of the bobbin winding window flanges and protrusions so that the finished height of the transformer when installed on the printed wire board will not exceed the maximum height of 0.500".

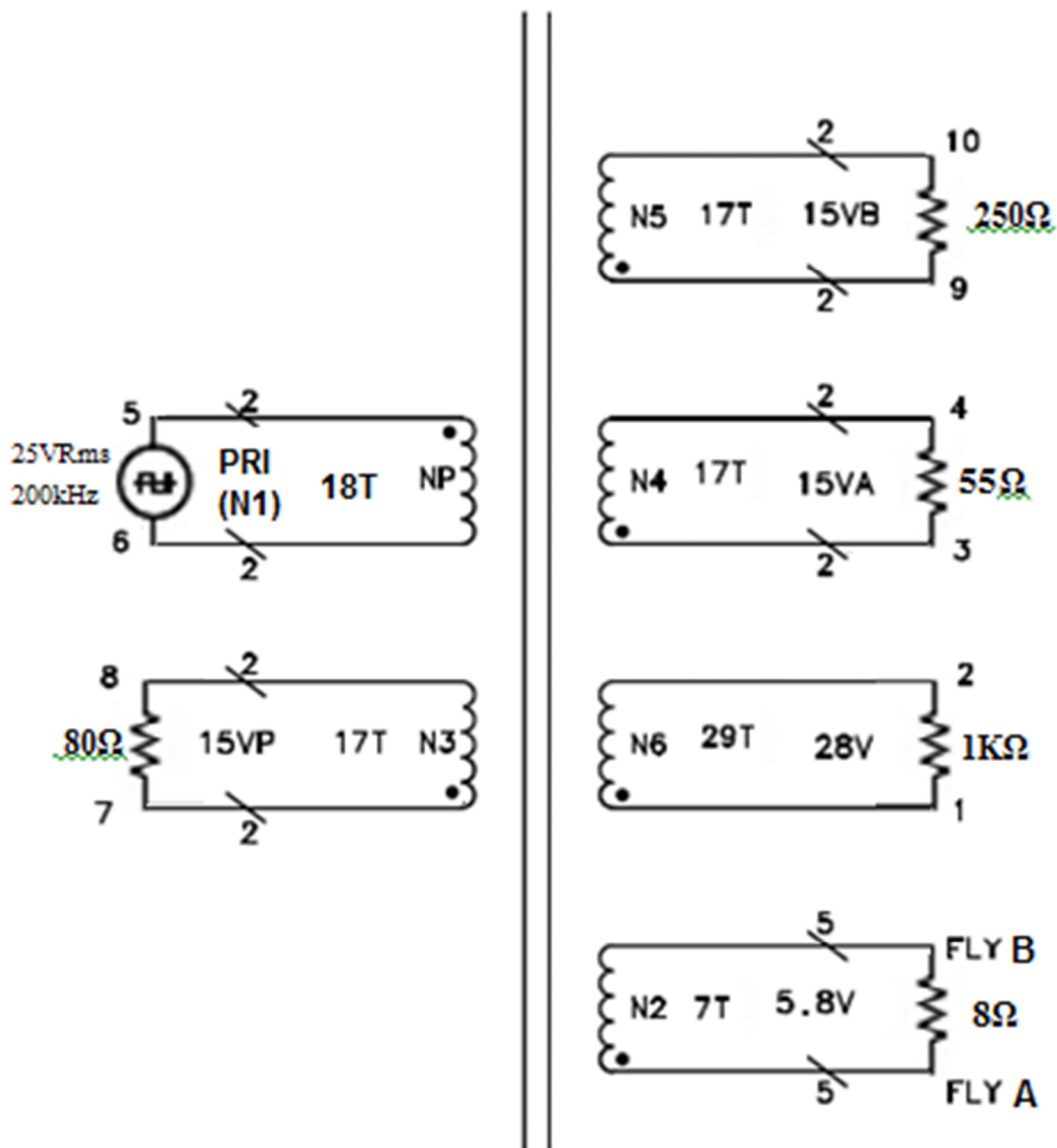
P515	
Layers	Winding Instructions
Winding Notes: 1. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnet wire in an oven set to 150°C $\pm 5^{\circ}\text{C}$ for 1 hour +10 minutes, -0 minutes. 2. All materials shall be per the P515 Materials List. 3. All insulation layers shall be spread around the perimeter to minimize buildup. 4. Lead breakout shall be as specified in the Outline Figure. 5. Final flying lead length shall be 1.50" ± 0.25". 6. Vacuum impregnate coil assembly using Item 9 (Impregnation). Height dimension is critical. Ensure no build up of impregnation exists on the coil assembly. 7. Apply Item 6 (Tape, Kapton) to notch in core top where flying leads exit. 8. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 9 (Core Cement). <u>Do not bond center leg</u>. Once cement cures ensure bobbin is not cemented to core. Ensure leads exit core per Outline figure. 9. Identify Pin 1 location with a white dot on top of core using Item 12 (Marking Ink). 10. Stake flying leads in core notch using Item 10 (RTV). 11. Apply two beads of Item 10 (RTV) at the top of the transformer between the core and the bobbin. Do not force the RTV into the narrow gap between the bobbin and core. RTV to be flush with top surface of core. Do not extend the RTV beads beyond the bobbin flange into the core corners. 12. All dimensions are in inches and include all materials used. 13. Mark each device per 3GPDE001 using Item 12 (Marking Ink). Tag each lead within 1/2 inch of core body.	



P515 Outline



P515 Schematic



P515 Burn-in/Life Test Circuit

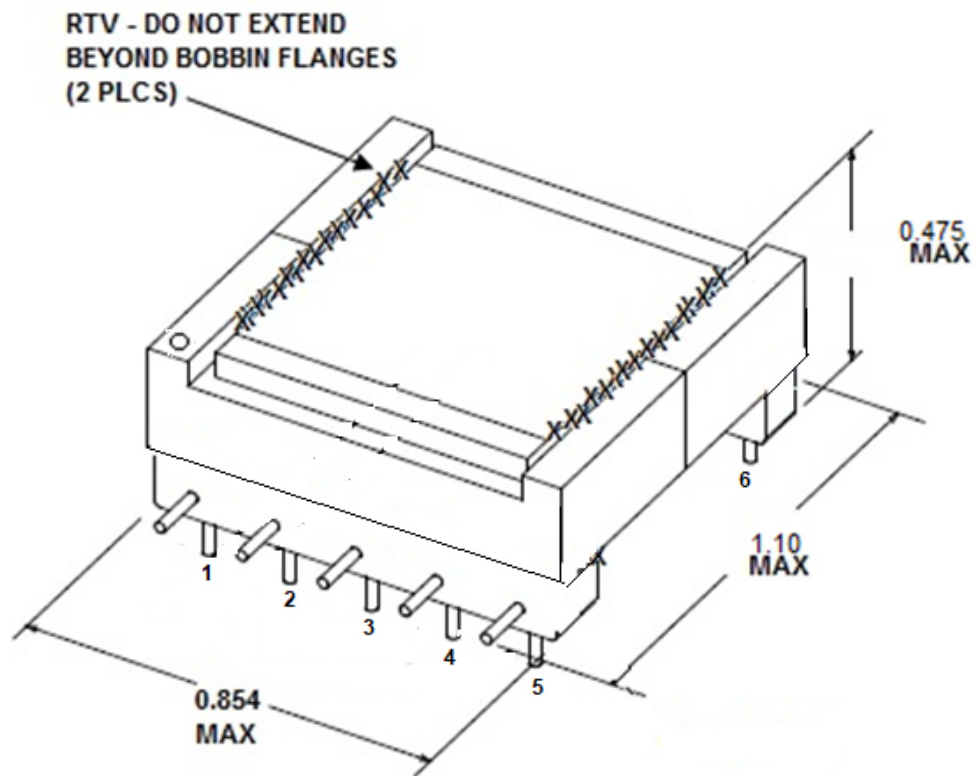
P516 Electrical Characteristics					
Characteristic	Limits	Limits	Limits	Limits	Limits
Winding Designation (start-finish)	N1 (1-2)	N2 (7-8)	N3 (9-10)	N4 (3-4)	N5 (5-6)
Turns	8	4	8	12	12
Wire (AWG) X Strands	28 X 3	26 X 3	28 X 3	26 X 1	26 X 1
Maximum Operating Current	1.11 Arms	1.65 Arms	1.04 Arms	0.47 Arms	0.16 Arms
Working Voltage (RMS)	175V	175V	175V	175V	175V
Dielectric Withstanding Voltage (1)	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (2)	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.
DC Resistance Max. (3)	21m Ω	7.35m Ω	21m Ω	60m Ω	60m Ω
Inductance (4)	9.03 μ H ($\pm$ 5%) (8)	2.26 μ H (REF)	9.03 μ H (REF)	20.30 μ H (REF)	20.30 μ H (REF)
Leakage Inductance (Max.) (5)	388 nH	Shorted	Shorted	Shorted	Shorted
Turns Ratio (6)	Reference	0.500 $\pm$ 2.0%	1.000 $\pm$ 2.0%	1.500 $\pm$ 2.0%	1.500 $\pm$ 2.0%
Inductance Delta Limits Percent (7)	$\pm$ 2%	N/A	N/A	N/A	N/A
Resistance Delta Limits Percent (7)	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%
Notes: 1. 500Vrms, 60 Hz, at 28 to 32 inches of Hg qualifies unit to 175V peak working voltage. 2. 500Vdc shall be applied between each winding and the core with all other windings grounded to the core. 3. Corrected to +20°C. 4. Measured at 1.0 Vrms Max, 200 KHz, 0 Adc and all other windings open. 5. Measured at 200 KHz. "Shorted" means all other windings shorted start to finish and shorted together. 6. Turns ratio data may be taken during the assembly process, prior to cementing cores. 7. Delta limits apply to burn-in and life test. 8. Grind center legs of cores symmetrical to achieve N1 inductance.					

P516 Materials List

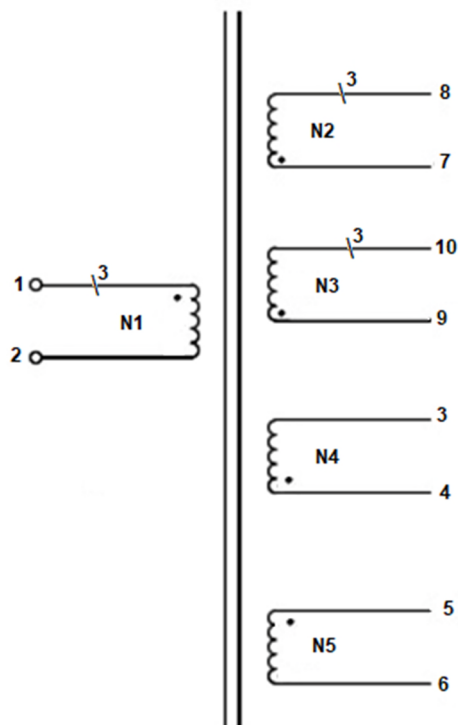
Item#	Material	Description	Source
1	Core (1)	EFD20/10/7-3F3	Ferroxcube
2	Bobbin	CPH-EFD20-1S-10PD-Z	Ferroxcube
3	Wire (2)	#28 AWG MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Wire (2)	#26 AWG MW035CHX000026* or MW016CHX000026*	Elektrisola, Essex, MWS, or REA
5	Not Used		
6	Tape, Kapton	P224 Kapton	Per approved Manufacturer's baseline
7	Tape, Glass	3M 79	Per approved Manufacturer's baseline
8	Impregnation	Scotchcast 280	Per approved Manufacturer's baseline
9	Core Glue (3)	RE2039/HD0243 or Epon 828/V140 or Epon 828/Epikure 3140	Per approved Manufacturer's baseline
10	Staking (Bobbin/Core)	RTV 142 or Dowsil6-1104	Per approved Manufacturer's baseline
11	Solder	SN60 or SN63	Qualified to J-STD-006
12	Ink/Catalyst	M-0-N or M-9-N/Cat 77 or B3 per A-A-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's baseline
13	Solder (Lead to Pin Terminations)	Sn10Pb88Ag2	Qualified to J-STD-006
Notes - Grind center legs of cores to achieve N1 inductance (Al = 141nH). - Approved magnet wire lubricant - * = "P" or "N". - Epon 828/V140 or Epon 828/Epikure 3140 Mix Ratio: 60:40 PBW.			

P516 Winding Instructions and Notes

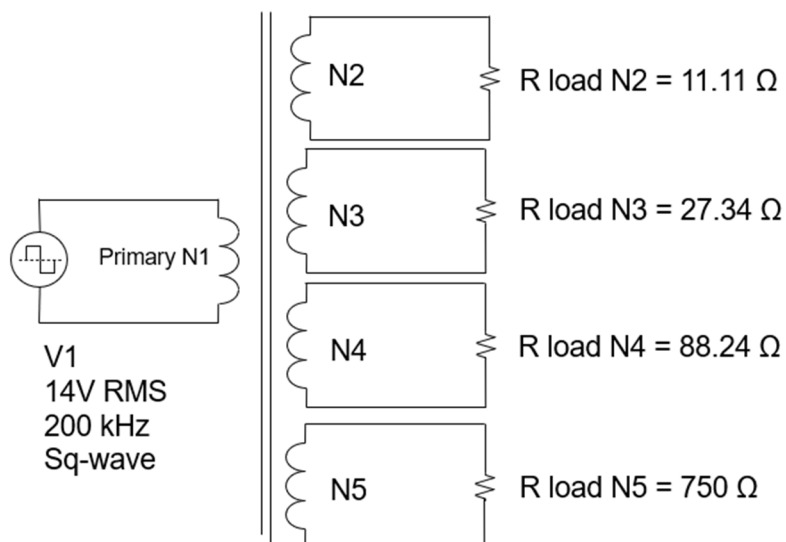
P516	
Layers	Winding Instructions
-	Observe polarity per schematic diagram.
-	Prior to winding, solder dip bobbin pins in their entirety twice using Item 11 (Solder).
-	Winding Item 2 (Bobbin) "Top Going".
1	Starting at pin 1, wind 8 turns of Winding "N1" (1-2) tri-filar, using Item 3 (Wire, Magnet 28 AWG) spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed.
2	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap. Return lead 2 to pin 2 termination side of bobbin. Cover pin 2 lead crossover with Item 6 (Kapton Tape).
3	Bring "N2" start leads from pin 7 termination point to pins 1-5 side of the bobbin and begin Winding 4 Turns of "N2" (7-8) tri-filar, using Item 4 (Wire, Magnet 26 AWG) spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed. Break out "N2" lead 8 to pin 8 termination.
4	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap.
5	Bring "N3" start leads from pin 9 termination point to pins 1-5 side of the bobbin and begin Winding 8 Turns of "N3" (9-10) tri-filar, using Item 3 (Wire, Magnet 28 AWG) spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed. Break out "N3" lead 10 to pin 10 termination.
6	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap.
7	Starting at pins 4 and 5 wind 12 turns of "N4" (4-3) and "N5" (5-6) in parallel both single filar using Item 4 (Wire, Magnet 26 AWG). Spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed. Break out "N5" lead 6 to pin 6 termination. Bring "N4" lead 3 back across the bobbin to terminate at pin 3.
8	Final wrap with 1 layer minimum of Item 7 (Tape, Glass).
Winding Notes: 1. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetics in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 2. All materials shall be per the P516 Materials List. 3. All insulation layers shall be spread around the perimeter to minimize buildup. 4. Terminate leads using Items 13 (Solder). 5. Vacuum impregnate coil assembly using Item 8 (Impregnation) in accordance with 3GPDE001. 6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 9 (Core Cement). <u>Do not bond center leg.</u> Once cement cures ensure bobbin is not cemented to core. 7. Identify Pin 1 location with a white dot on top of core using Item 12 (Marking Ink). 8. Apply two beads of Item 10 (RTV) at the top of the transformer between the core and the bobbin. Do not force the RTV into the narrow gap between the bobbin and core. RTV to be flush with top surface of core. Do not extend the RTV beads beyond the bobbin flange into the core corners. 9. All dimensions are in inches and include all materials used. 10. Mark each device per 3GPDE001 using Item 12 (Marking Ink). 11. Closed construction.	



P516 Outline



P516 Schematic



P516 Burn-in/Life Test Circuit

P517 Electrical Characteristics					
Characteristic	Limits	Limits	Limits	Limits	Limits
Winding Designation (start-finish)	N1 (A-B)	N2 (1,2 - 4,5)	N3 (9-10)	N4 (6-7)	N5 (3-8)
Turns	2	4	4	6	6
Wire (AWG) X Strands	Cu Foil	24 X 4	26 X 3	26 X 2	28 X 1
Maximum Operating Current	5.84 Arms	2.41 Arms	1.13 Arms	0.74 Arms	0.16 Arms
Working Voltage (RMS)	175V	175V	175V	175V	175V
Dielectric Withstanding Voltage (1)	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (2)	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.
DC Resistance Max. (3)	1.37 m Ω	4.19m Ω	8.97m Ω	20.16m Ω	63.7m Ω
Inductance (4)	1.1 μ H (REF)	4.4 μ H ($\pm$ 5%) (8)	4.4 μ H (REF)	9.9 μ H (REF)	9.9 μ H (REF)
Leakage Inductance (Max.) (5)	Shorted	206 nH	Shorted	Shorted	Shorted
Turns Ratio (6)	0.500 $\pm$ 2.0%	Reference	1.00 $\pm$ 2.0%	1.50 $\pm$ 2.0%	1.50 $\pm$ 2.0%
Inductance Delta Limits Percent (7)	N/A	$\pm$ 2%	N/A	N/A	N/A
Resistance Delta Limits Percent (7)	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%
Notes: 1. 500Vrms, 60 Hz, at 28 to 32 inches of Hg qualifies unit to 175V peak working voltage. 2. 500Vdc shall be applied between each winding and the core with all other windings grounded to the core. 3. Corrected to +20°C. 4. Measured at 1.0 Vrms Max, 200 KHz, 0 Adc and all other windings open. 5. Measured at 200 KHz. "Shorted" means all other windings shorted start to finish and shorted together. 6. Turns ratio data may be taken during the assembly process, prior to cementing cores. 7. Delta limits apply to burn-in and life test. 8. Grind center legs of cores symmetrical to achieve N1 inductance.					

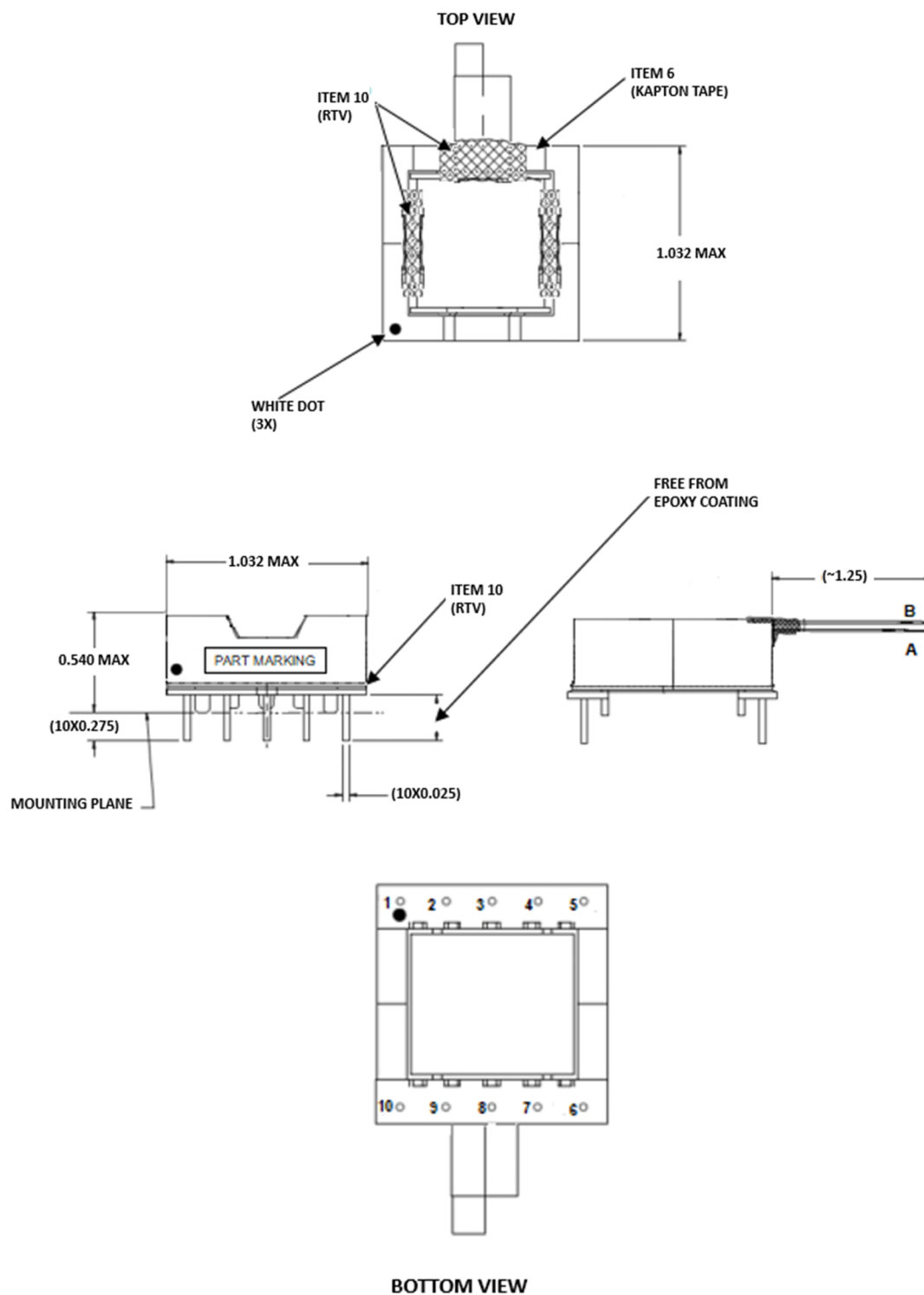
P517 Materials List

Item#	Material	Description	Source
1	Core (1)	EFD25/13/9-3F3	Ferroxcube
2	Bobbin	Custom Ultem Bobbin Figure #1 or #2 Detail	Per approved Manufacturer's baseline
3	Wire (2)	#24 AWG MW035CHX000024* or MW016CHX000024*	Elektrisola, Essex, MWS, or REA
4	Wire (2)	#26 AWG MW035CHX000026* or MW016CHX000026*	Elektrisola, Essex, MWS, or REA
5	Copper Foil	5 mil thick ETP copper foil, ASTM B152, Alloy 110	Per approved Manufacturer's baseline
6	Tape, Kapton	P224 Kapton	Per approved Manufacturer's baseline
7	Tape, Glass	3M 79	Per approved Manufacturer's baseline
8	Impregnation	Scotchcast 280	Per approved Manufacturer's baseline
9	Core Glue (4)	RE2039/HD0243 or Epon 828/V140 or Epon 828/Epikure 3140	Per approved Manufacturer's baseline
10	Staking (Bobbin/Core)	RTV 142 or Dowsil6-1104	Per approved Manufacturer's baseline
11	Solder	SN60 or SN63	Qualified to J-STD-006
12	Ink/Catalyst	M-0-N or M-9-N/Cat 77 or B3 per A-A-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's baseline
13	Wire (2)	#28 AWG MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
14	Bobbin Pins (5)	AA59551-H24S1S or QQW343S24S1S (#24 AWG Silver Plated Copper Wire)	Per approved Manufacturer's baseline
15	Bobbin Pin Cement (6)	Epon 828/V140/Cab-o-sil or Epon 828/Epikure 3140/Cab-o-sil	Per approved Manufacturer's baseline
Notes - Grind center legs of cores to achieve N2 inductance (AI = 275nH). - Approved magnet wire lubricant - * = "P" or "N". - See figure for copper foil approximate configuration. - Epon 828/V140 or Epon 828/Epikure 3140 Mix Ratio: 60:40 PBW. - Prior to cementing bobbin pins into bobbin, bobbin pins will be solder dipped twice in their entirety using item 9. - Epon 828/V140/Cab-o-sil or Epon 828/Epikure 3140/Cab-o-sil Mix Ratio: 60:40:4 PBW			

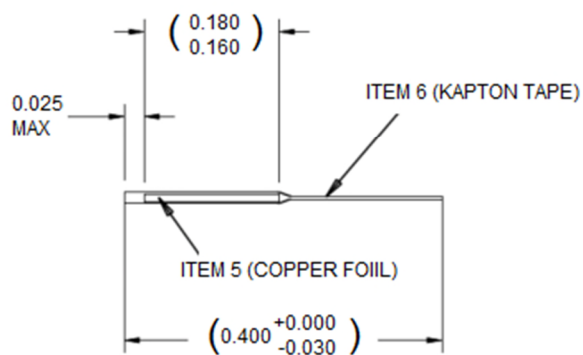
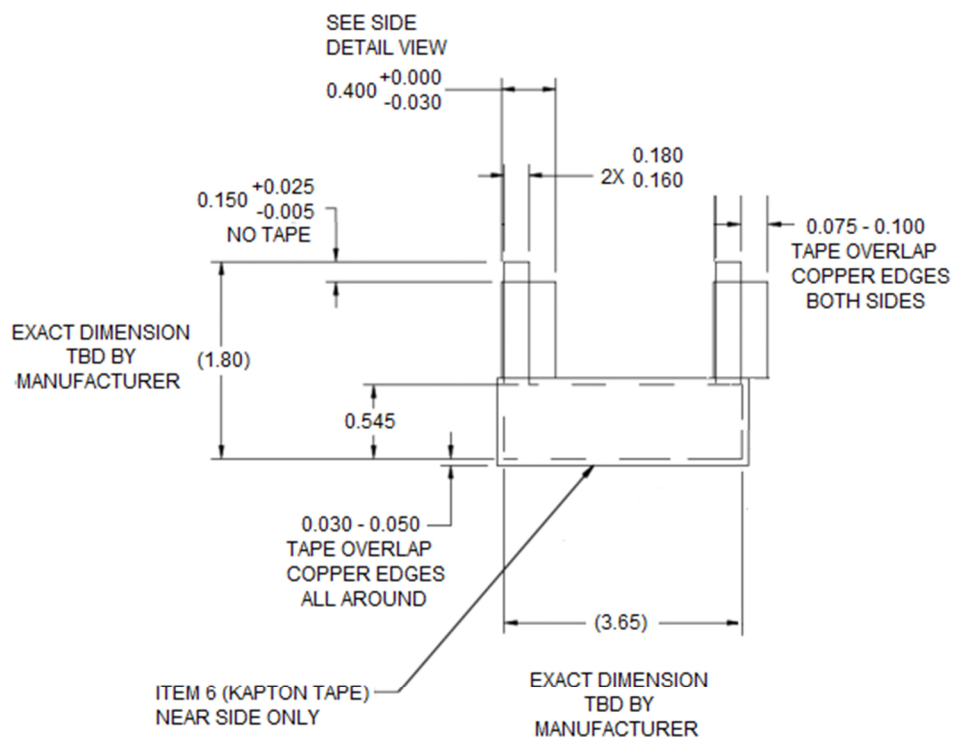
P517 Winding Instructions and Notes

P517	
Layers	Winding Instructions
-	Observe polarity per schematic diagram.
-	Prior to cementing Item 14 (Bobbin Pins) into Item 2 (Bobbin), solder dip bobbin pins in their entirety twice using Item 11 (Solder).
	Cement Item 14 (Bobbin Pins), into Item 2 (Bobbin), using Item 15 (Bobbin Pin Cement). Ensure Bobbin pins are of sufficient length to meet end item pin dimensions. Ensure bobbin pin cut out is completely filled with cement. ULTEM BOBBIN – DO NOT CLEAN WITH ACETONE OR VERTREL.
-	Winding Item 2 (Bobbin) “Top Going”.
-	Prepare Copper Foil winding “N1” using Item 5 (Copper Foil) and Item 6 (Kapton Tape) per the P517 Copper Foil Figure. All edges of the copper foil winding shall be deburred prior to applying Item 6. The exact length dimension shall be determined by the manufacturer during the initial winding process to ensure the copper foil winding contains the correct number of turns and the flying leads break out in the location specified on the P517 outline figure.
1	Start winding “N1” on top, opposite pin exits. Place flying lead “A” in bobbin notch on pins 6 thru 10 side, wind 2 turns of Winding “N1” (A-B) prepared copper foil lead with bare side of copper foil facing down onto bobbin. Place flying lead “B” in bobbin notch on pins 6 thru 10 side.
2	Secure copper foil winding with 1 layer of Item 6 (Kapton Tape) with a minimum 0.15 inch overlap.
3	Start on pins 1-5 side of the bobbin, begin winding “N2” (1,2 –4,5) start leads 1, 2 winding 4 turns of “N2” four filar, using Item 3 (Wire, Magnet 24 AWG) spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed. Bring “N2” finish leads 4, 5 back to pins 1 to 5 side of bobbin to pins 4, 5 for termination.
4	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap.
5	Starting on pins 6-10 side of the bobbin, begin winding “N4” (6-7) bi-filar using Item 4 (Wire, Magnet 26 AWG) start leads from pin 6 termination point and wind one turn in a single layer, no crossovers allowed. Hold winding. Starting on pins 6-10 side of the bobbin, begin Winding “N3” (9-10) tri-filar using Item 4 (Wire, Magnet 26 AWG) start leads from pin 9 termination point, joining with N4 as a multi-filar five stranding. Wind 4 turns spread evenly across Item 2 (bobbin) in a single layer, no crossovers allowed. Break out the tri-filar “N3” strands and bring back to pins 6-10 side of bobbin, terminate the “N3” finish leads on pin 10. Continue winding one more turn of “N4” bi-filar, completing a total of six turns for “N4” winding. Bring “N4” finish leads back to pins 6 to 10 side of bobbin and terminate on pin 7.
6	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap.
7	Start on pins 1-5 side of the bobbin, begin winding “N5” (3-8) single filar using Item 13 (Wire, Magnet 28 AWG) starting at pin 3. Wind 6 turns spread evenly across Item 2 (bobbin) in a single layer, no crossovers allowed. Bring finish lead to pin 8 for termination.
8	Final wrap with 1 layer minimum of Item 7 (Tape, Glass).

P517	
Layers	Winding Instructions
Winding Notes: 1. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnetics in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 2. All materials shall be per the P517 Materials List. 3. All insulation layers shall be spread around the perimeter to minimize buildup. 4. Terminate leads using Item 11 (Solder). For "N1" (1, 2 -4, 5) winding, terminate two wires to each pin as depicted in the schematic. 5. Vacuum impregnate coil assembly using Item 8 (Impregnation) in accordance with 3GPDE001. 6. Place Item 6 (Kapton Tape) in core slot where copper foil flying leads exit the bobbin to provide additional isolation from the core. 7. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 9 (Core Cement). <u>Do not bond center leg.</u> Once cement cures ensure bobbin is not cemented to core. 8. Identify Pin 1 location with a white dot on top of core using Item 12 (Marking Ink). 9. Apply two beads of Item 10 (RTV) at the top of the transformer between the core and the bobbin. Do not force the RTV into the narrow gap between the bobbin and core. RTV to be flush with top surface of core. Do not extend the RTV beads beyond the bobbin flange into the core corners. Fill bottom core to bobbin interface with Item 10 (RTV) 80 % fill minimum. 10. Stake copper foil flying leads using Item 10 (RTV) in core notch minimizing meniscus where flying leads extend out form the core. 11. All dimensions are in inches and include all materials used. Pin length from bottom of standoff is 0.275" 12. Mark each device per 3GPDE001 using Item 12 (Marking Ink). 13. Closed construction.	



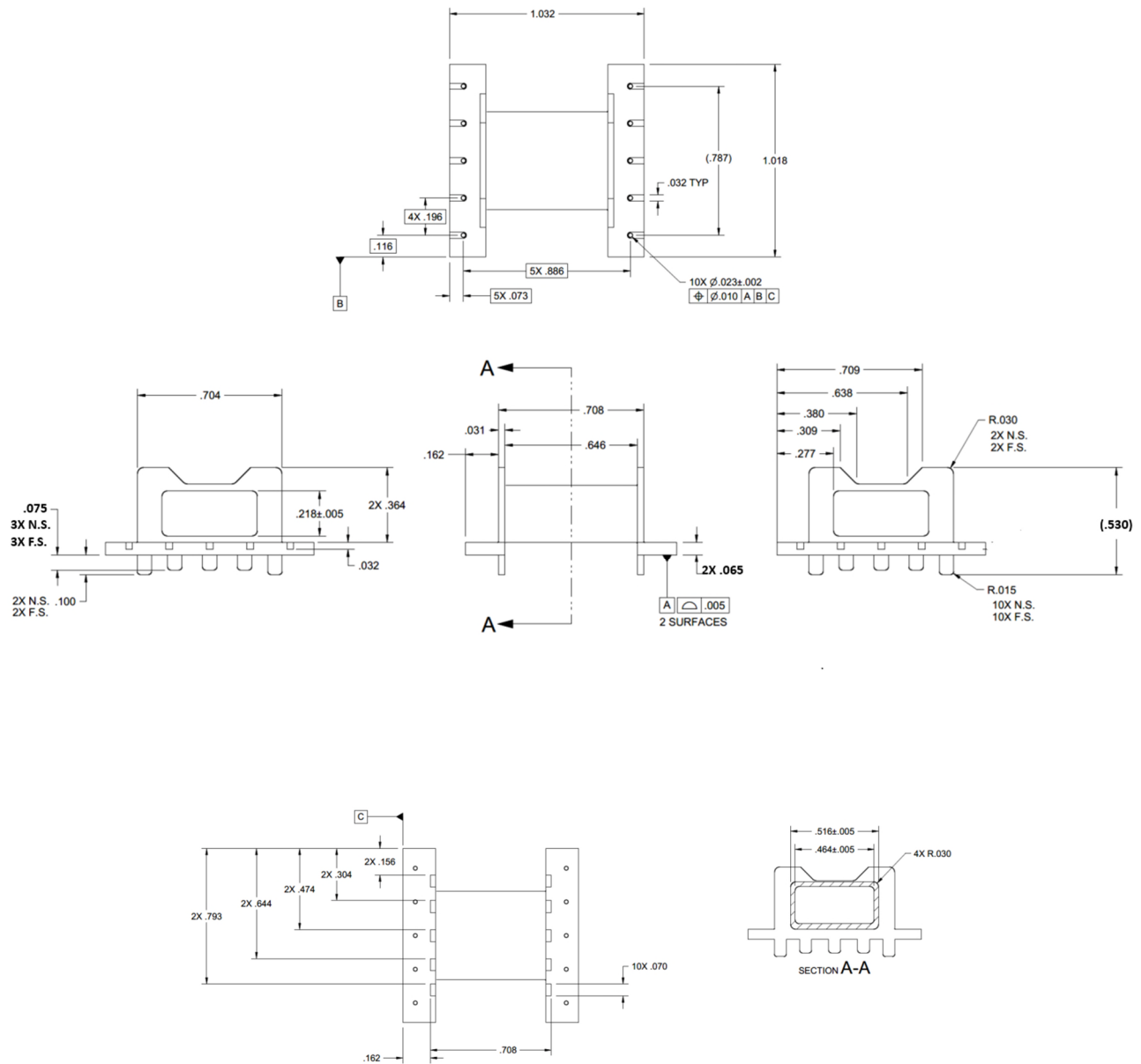
P517 Outline



SIDE DETAIL VIEW

SCALE: NONE

P517 COPPER FOIL

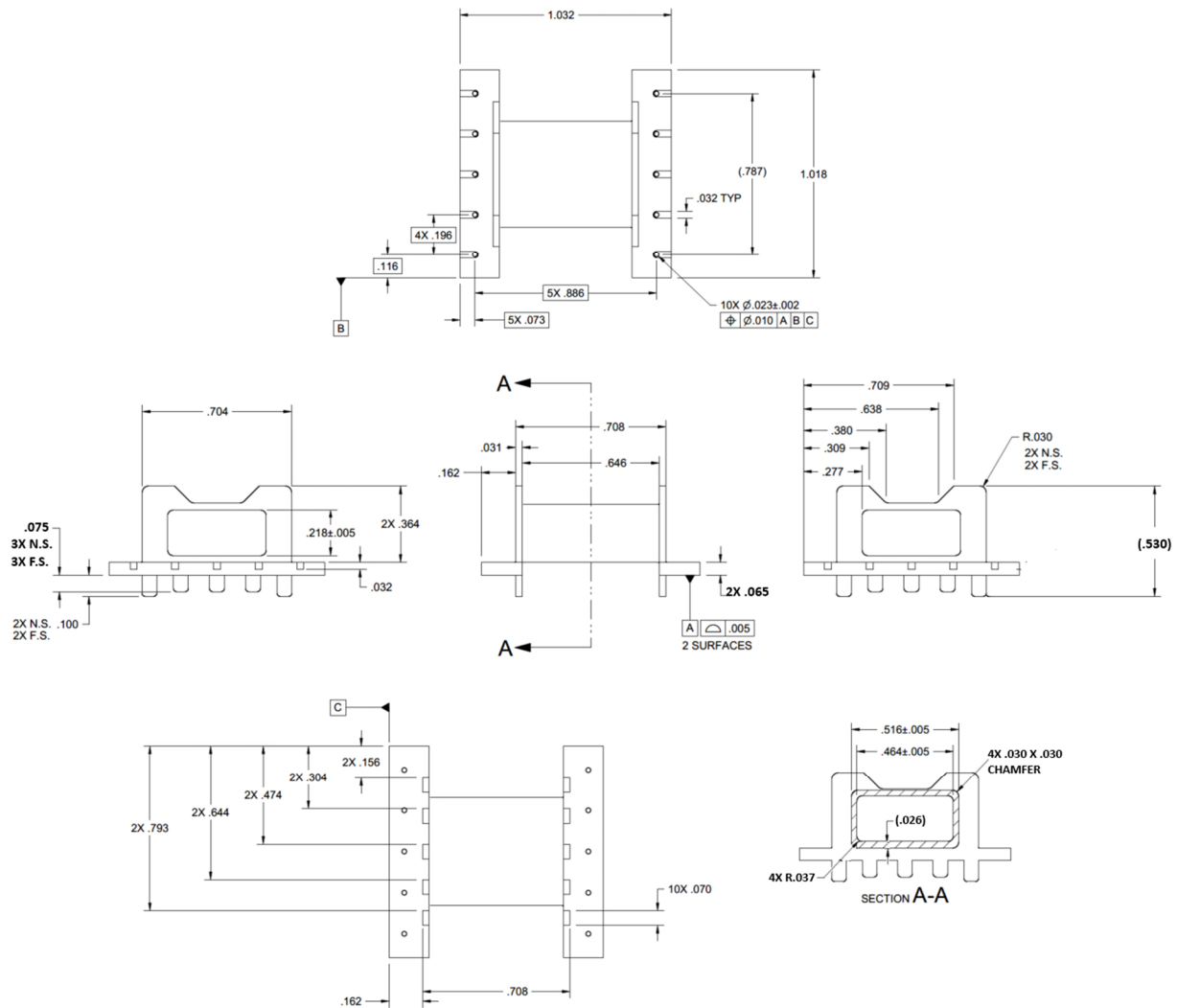


Bobbin Annealing Requirements

After final machining, Bobbin shall be annealed to 300°F at a maximum heating rate of 20°F per hour. Hold oven temperature at 300°F for 1 ½ hours. Increase oven temperature to 400°F at a maximum heating rate of 20°F per hour. Hold oven temperature at 400°F for 2 ½ hours. Oven cool to 150°F over 25 hours with a maximum cool rate of 10°F per hour. Turn off oven and allow to cool to room temperature with oven door closed before removing parts.

Tolerance (unless otherwise specified) - 3 decimal places: +/- .010

Custom Ultem Bobbin Figure #1

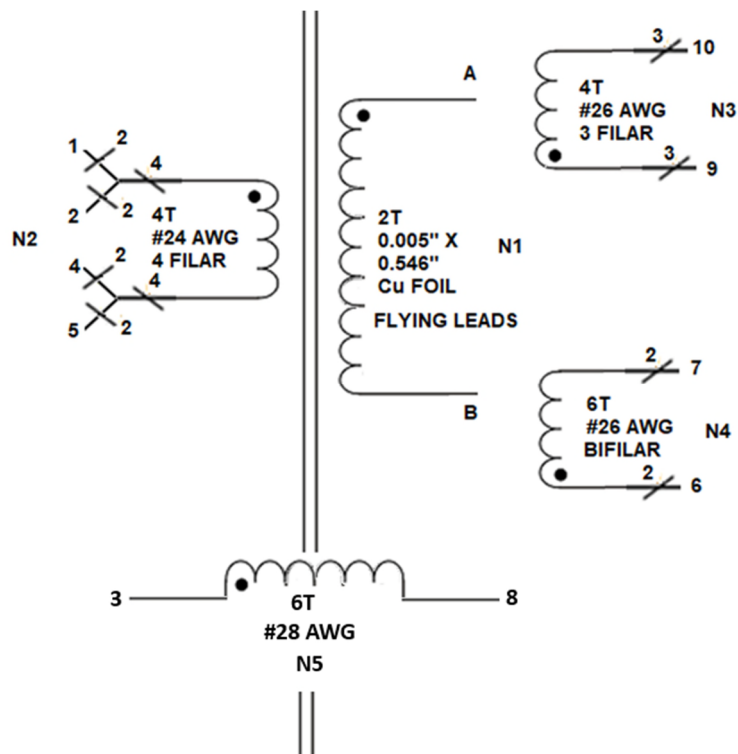


Bobbin Annealing Requirements

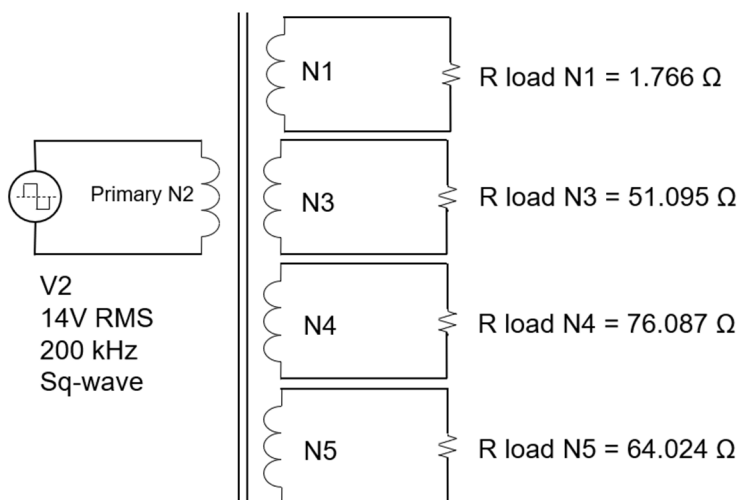
After final machining, Bobbin shall be annealed to 300°F at a maximum heating rate of 20°F per hour. Hold oven temperature at 300°F for 1 ½ hours. Increase oven temperature to 400°F at a maximum heating rate of 20°F per hour. Hold oven temperature at 400°F for 2 ½ hours. Oven cool to 150°F over 25 hours with a maximum cool rate of 10°F per hour. Turn off oven and allow to cool to room temperature with oven door closed before removing parts.

Tolerance (unless otherwise specified) - 3 decimal places: +/- .010

Custom Ultem Bobbin Figure #2



P517 Schematic



P517 Burn-in/Life Test Circuit

P518 Electrical Characteristics						
Characteristic	Limits	Limits	Limits	Limits	Limits	Limits
Winding Designation (start-finish)	N1 (A-B)	N2 (7-8)	N3 (9-10)	N4 (4-3)	N5 (5-6)	N6 (1-2)
Turns	8	4	8	12	12	12
Wire (AWG) X Strands	26 X 3	26 X 3	28 X 3	26 X 1	26 X 1	26 X 1
Maximum Operating Current	1.11 Arms	1.65 Arms	1.04 Arms	0.47 Arms	0.16 Arms	0.11 Arms
Working Voltage (RMS)	175V	175V	175V	175V	175V	175V
Dielectric Withstanding Voltage (1)	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
Insulation Resistance (2)	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.	10,000 Meg Ω min.
DC Resistance Max. (3)	15m Ω	8m Ω	21m Ω	60m Ω	60m Ω	60m Ω
Inductance (4)	9.03 μ H ($\pm$ 5%) (8)	2.26 μ H (REF)	9.03 μ H (REF)	20.30 μ H (REF)	20.30 μ H (REF)	20.30 μ H (REF)
Leakage Inductance (Max.) (5)	388 nH	Shorted	Shorted	Shorted	Shorted	Shorted
Turns Ratio (6)	Reference	0.500 $\pm$ 2.0%	1.000 $\pm$ 2.0%	1.500 $\pm$ 2.0%	1.500 $\pm$ 2.0%	1.500 $\pm$ 2.0%
Inductance Delta Limits Percent (7)	$\pm$ 2%	N/A	N/A	N/A	N/A	N/A
Resistance Delta Limits Percent (7)	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%	+3%/-5%
Notes: 500Vrms, 60 Hz, at 28 to 32 inches of Hg qualifies unit to 175V peak working voltage. 500Vdc shall be applied between each winding and the core with all other windings grounded to the core. - Corrected to +20°C. - Measured at 1.0 Vrms Max, 200 KHz, 0 Adc and all other windings open. - Measured at 200 KHz. "Shorted" means all other windings shorted start to finish and shorted together. - Turns ratio data may be taken during the assembly process, prior to cementing cores. - Delta limits apply to burn-in and life test. - Grind center legs of cores symmetrical to achieve N1 inductance.						

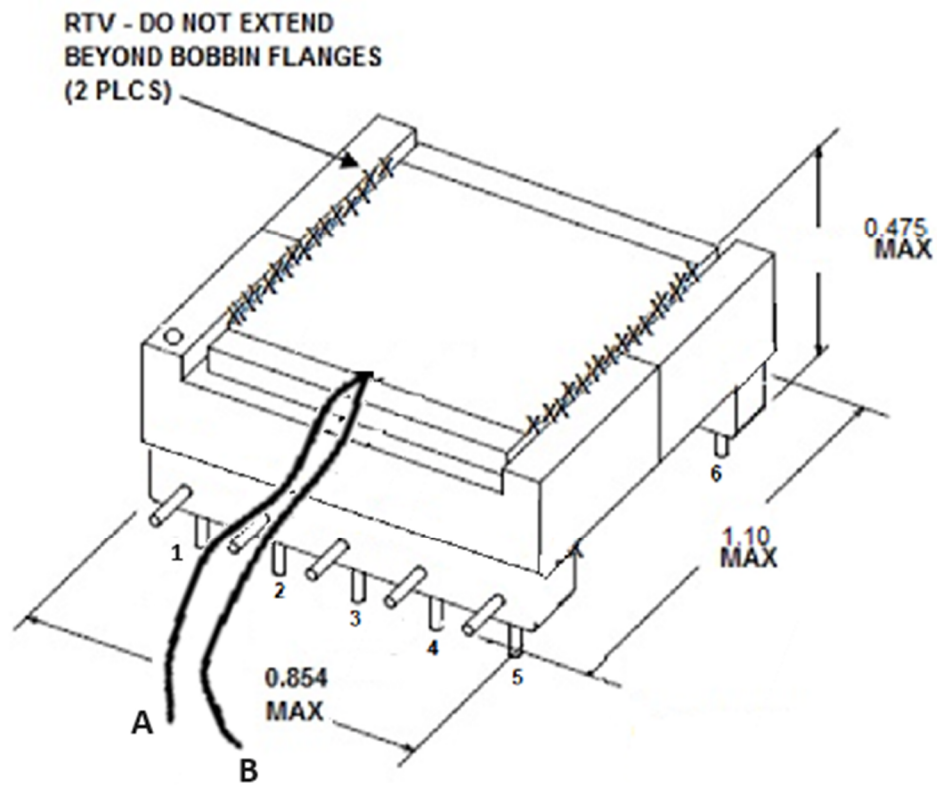
P518 Materials List

Item#	Material	Description	Source
1	Core (1)	EFD20/10/7-3F3	Ferroxcube
2	Bobbin	Custom Ultem Bobbin, EFD20	Per approved Manufacturer's baseline
3	Wire (2)	#28 AWG MW035CHX000028* or MW016CHX000028*	Elektrisola, Essex, MWS, or REA
4	Wire (2)	#26 AWG MW035CHX000026* or MW016CHX000026*	Elektrisola, Essex, MWS, or REA
5	Not Used		
6	Tape, Kapton	P224 Kapton	Per approved Manufacturer's baseline
7	Tape, Glass	3M 79	Per approved Manufacturer's baseline
8	Impregnation	Scotchcast 280	Per approved Manufacturer's baseline
9	Core Glue (3)	RE2039/HD0243 or Epon 828/V140 or Epon 828/Epikure 3140	Per approved Manufacturer's baseline
10	Staking (Bobbin/Core)	RTV 142 or Dowsil6-1104	Per approved Manufacturer's baseline
11	Solder	SN60 or SN63	Qualified to J-STD-006
12	Ink/Catalyst	M-0-N or M-9-N/Cat 77 or B3 per A-A-56032 (wht or blk) or M43553 (wht or blk). (Use Best Contrasting)	Per approved Manufacturer's baseline
13	Solder (Lead to Pin Terminations)	Sn10Pb88Ag2	Qualified to J-STD-006
14	Bobbin Pins (4)	AA59551-H24S1S or QQW343S24S1S (#24 AWG Silver Plated Copper Wire)	Per approved Manufacturer's baseline
15	Bobbin Pin Cement (5)	Epon 828/V140/Cab-o-sil or Epon 828/Epikure 3140/Cab-o-sil	Per approved Manufacturer's baseline
Notes - Grind center legs of cores to achieve N1 inductance (Al = 141nH). - Approved magnet wire lubricant - * = "P" or "N". - Epon 828/V140 or Epon 828/Epikure 3140 Mix Ratio: 60:40 PBW. - Prior to cementing bobbin pins into bobbin, bobbin pins will be solder dipped twice in their entirety using item 9. - Epon 828/V140/Cab-o-sil or Epon 828/Epikure 3140/Cab-o-sil Mix Ratio: 60:40:4 PBW			

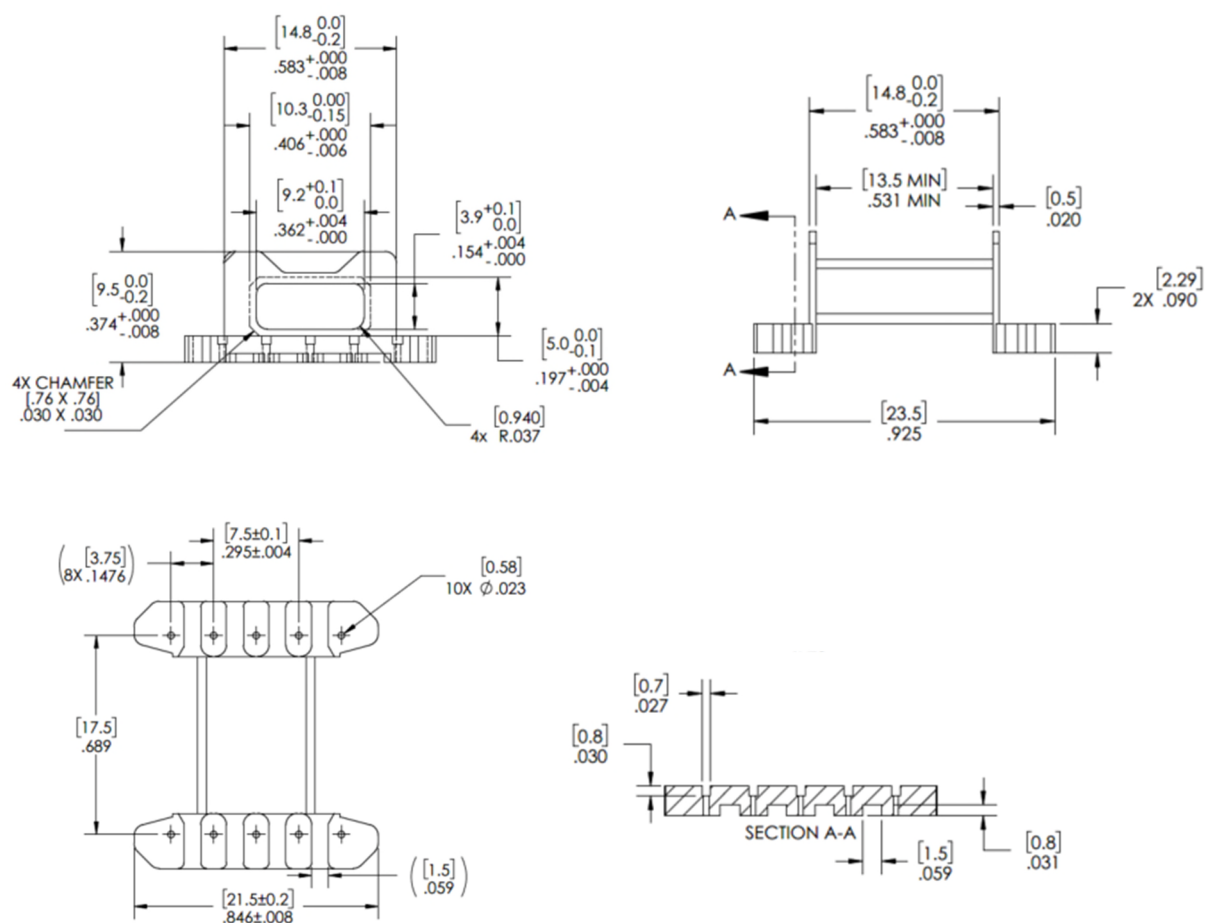
P518 Winding Instructions and Notes

P518	
Layers	Winding Instructions
-	Observe polarity per schematic diagram.
-	Prior to cementing Item 14 (Bobbin Pins) into Item 2 (Bobbin), solder dip bobbin pins in their entirety twice using Item 11 (Solder).
	Cement Item 14 (Bobbin Pins), into Item 2 (Bobbin), using Item 15 (Bobbin Pin Cement). Ensure Bobbin pins are of sufficient length to meet end item pin dimensions. Ensure bobbin pin cut out is completely filled with cement. ULTEM BOBBIN – DO NOT CLEAN WITH ACETONE OR VERTREL.
-	Winding Item 2 (Bobbin) "Top Going".
1	Starting on top, opposite side of pins on pin 1 side of bobbin at bobbin notch, wind 8 turns of Winding "N1" (A-B) tri-filar, using Item 3 (Wire, Magnet 28 AWG) spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed during winding. Once complete, bring "B" across winding to exit out through top bobbin notch.
2	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap. Return lead 2 to pin 2 termination side of bobbin. Cover pin 2 lead crossover with Item 6 (Kapton Tape).
3	Starting at Pin 4 wind 12 turns of "N4" (4-3) single filar using Item 4 (Wire, Magnet 26 AWG) using only half of the bobbin winding area. Bring "N4" lead 3 back across the winding to terminate at pin 3. Bring "N2" start leads from pin 7 termination point to pins 1-5 side of the bobbin and begin Winding 4 Turns of "N2" (7-8) tri-filar, using Item 4 (Wire, Magnet 26 AWG) spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed. Break out "N2" lead 8 to pin 8 termination.
4	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap.
5	Bring "N3" start leads from pin 9 termination point to pins 1-5 side of the bobbin and begin Winding 8 Turns of "N3" (9-10) tri-filar, using Item 3 (Wire, Magnet 28 AWG) spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed. Break out "N3" lead 10 to pin 10 termination.
6	Wind 1 layer of Item 6 (Kapton Tape) with a 0.15 ±0.10 inch overlap.
7	Starting at pins 2 and 5 wind 12 turns of "N6" (2-1) and "N5" (5-6) in parallel both single filar using Item 4 (Wire, Magnet 26 AWG). Spread evenly across Item 2 (Bobbin) in a single layer, no crossovers allowed. Break out "N5" lead 6 to pin 6 termination. Bring "N6" lead 1 back across the bobbin to terminate at pin 1.
8	Final wrap with 1 layer minimum of Item 7 (Tape, Glass).

P518	
Layers	Winding Instructions
Winding Notes: 1. Prior to allowing the wound magnet wire to come in contact with any cleaning solvents or solder flux, it shall be stressed relieved by heat treatment. Heat treat by placing the wound magnet wire in an oven set to 150°C ±5°C for 1 hour +10 minutes, -0 minutes. 2. All materials shall be per the P518 Materials List. 3. All insulation layers shall be spread around the perimeter to minimize buildup. 4. Terminate leads using Item 13 (Solder). 5. Vacuum impregnate coil assembly using Item 8 (Impregnation) in accordance with 3GPDE001. 6. Place coil into Item 1 (Core). Cement Item 1 (Core) together using Item 9 (Core Cement). <u>Do not bond center leg.</u> Once cement cures ensure bobbin is not cemented to core. 7. Identify Pin 1 location with a white dot on top of core using Item 12 (Marking Ink). 8. Apply two beads of Item 10 (RTV) at the top of the transformer between the core and the bobbin. Do not force the RTV into the narrow gap between the bobbin and core. RTV to be flush with top surface of core. Do not extend the RTV beads beyond the bobbin flange into the core corners. 9. All dimensions are in inches and include all materials used. Device Outline dimensions are per the P518 figure. The self leads from edge of core shall be 1.5 ±0.25 inches and the bobbin bottom pin length is 0.305 inches. 10. Mark each device per 3GPDE001 using Item 12 (Marking Ink). 11. Closed construction.	

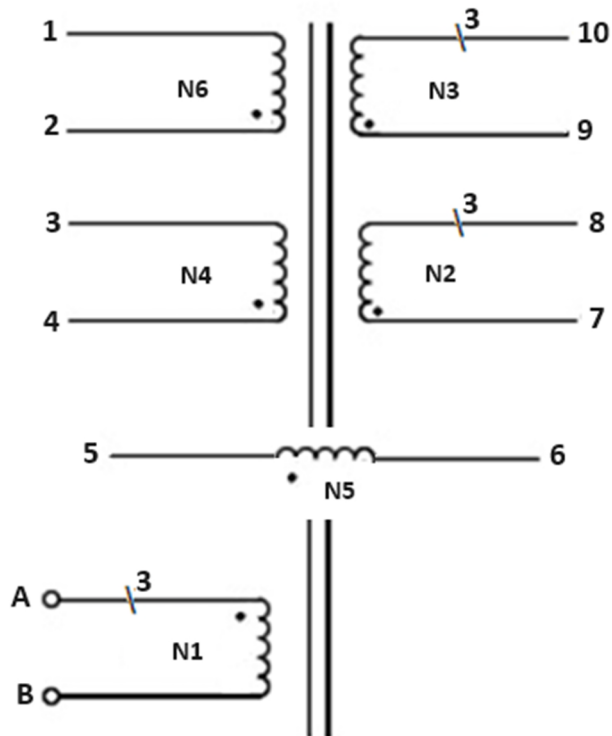


P518 Outline

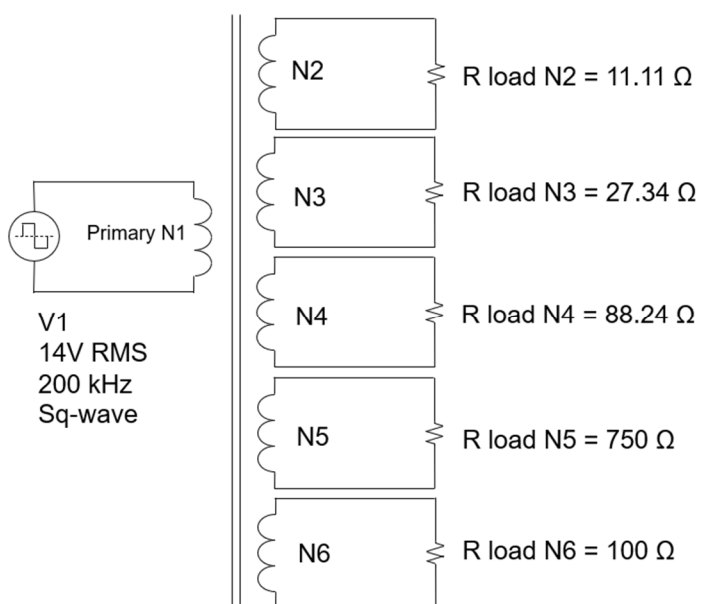


Note: Values in [] are in millimeters. All others are in inches.

P518 EFD20 Custom Bobbin



P518 Schematic



P518 Burn-in/Life Test Circuit